



# Release Notes

## MOBILE GATEWAY/CLOUD MOBILE GATEWAY

DN1000279072

Issue 01

7750 SERVICE ROUTER  
MOBILE GATEWAY/CLOUD MOBILE GATEWAY

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Release 26.7.R1

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# 1 About this document

This document describes the software delivery and is published for each software update. It specifies new and changed features together with any known problems or restrictions.

**Disclaimer:** CMG VNF deployments are not supported in this release.

## 1.1 Version history

Changes between document issues are cumulative. Therefore, the latest document issue contains all changes made to the previous issues.

### Version history

| Date       | Issue number | Description                                                                                                      |
|------------|--------------|------------------------------------------------------------------------------------------------------------------|
| July 2026  | 01           | 7750 SR MG and CMG 26.7.R1 Release Notes                                                                         |
| July 2026  | 01           | 7750 SR MG and CMG 26.3.R5 Release Notes                                                                         |
| June 2026  | 01           | 7750 SR MG and CMG 26.3.R4-1 Release Notes                                                                       |
| May 2026   | 01           | 7750 SR MG and CMG 26.3.R4 Release Notes<br>New appendix: Configuring manually the network interface card and PC |
| May 2026   | 01           | 7750 SR MG and CMG 26.3.R3 Release Notes                                                                         |
| April 2026 | 01           | 7750 SR MG and CMG 26.3.R2 Release Notes                                                                         |
| March 2026 | 01           | 7750 SR MG and CMG 26.3.R1 Release Notes<br>New chapter: Supported CMG appliances: Firmware on HPE hardware      |

## 2 Software identification details

### 2.1 Release content

The files from the following table are included in the software packages. The software packages can be accessed from the Nokia Support Portal:

<https://online.networks.nokia.com/SWD/>.

**NOTE:** CMG delivers perpetual Trust Wildcard licenses with all CMG features enabled for both CMG VNFs and CMG CNFs. To avoid service impact, in CMG CNF Intra ISSU, it is recommended to convert existing commercial licenses to Trust Wildcard licenses. For new CMG deployments, it is recommended to use the CMG licenses available in the Support Portal. These licenses do not support CMG Trusted Wireless Access Gateway and CMG Home Agent Gateway. For detailed instructions, contact your Nokia technical support representative.

Artifacts for CMG VNF: CMG (Cloud Mobile Gateway) > X.Y release > Rn>

| Artifacts for CMG product variant in Nokia support portal: folder location | Filename                                                                                                                                                                                                                                                            | Description                                                                                        |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| CMG-VNF                                                                    | Nokia-CMG-DB-VM-X.Y.Rn                                                                                                                                                                                                                                              | DB-VM image for CMG N:K redundancy topology                                                        |
| CMG-VNF                                                                    | Nokia-CMG-LTMT-X.Y.Rn<br>Note: LTMT files are included in the package of each major release. Nokia recommends using the latest LTMT because LTMT is backward compatible.<br>Use the 26.7.R1 version of LTMT with any CMG/CMM version from release 24.7 and onwards. | Software image for Lightweight Real-time Monitoring and Troubleshooting (LTMT) server installation |
| CMG-VNF                                                                    | Nokia-CMG-MIBs-X.Y.Rn                                                                                                                                                                                                                                               | CMG MIBs                                                                                           |
| CMG-VNF                                                                    | Nokia-CMG-YANG-X.Y.Rn                                                                                                                                                                                                                                               | CMG YANG models                                                                                    |

| Artifacts for CMG product variant in Nokia support portal: folder location | Filename                                       | Description                                                                                                                                                                                                         |
|----------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMG-VNF                                                                    | Nokia-CMG-OVA-X.Y.Rn                           | Open Virtualization Format (OVA) software image for automated CMG vApp on-boarding; From release 22.8.R1 onwards, the Virtual Machine Disk (VMDK) software image for manual CMG VM creation is included in the file |
| CMG-VNF                                                                    | Nokia-CMG-QCOW-2-X.Y.Rn                        | CMG Image QCOW2 file                                                                                                                                                                                                |
| CMG-VNF                                                                    | Nokia-CMG-SHA256-checksum-X.Y.Rn               | SHA256 checksums of all the files in X.Y release>Rn folder                                                                                                                                                          |
| CMG-VNF                                                                    | Nokia_External_Customer_license_All_LI_on.zip  | This is a commercial Trust Wildcard CMG license that enables all CMG features. The LI is enabled in this license.                                                                                                   |
| CMG-VNF                                                                    | Nokia_External_Customer_license_All_LI_on.pdf  | This is a supplement to Nokia_External_Customer_license_All_LI_on.zip that describes the license agreement                                                                                                          |
| CMG-VNF                                                                    | Nokia_External_Customer_license_All_LI_off.zip | This is a commercial Trust Wildcard CMG license that enables all CMG features. The LI is disabled in this license.                                                                                                  |
| CMG-VNF                                                                    | Nokia_External_Customer_license_All_LI_off.pdf | This is a supplement to Nokia_External_Customer_license_All_LI_off.zip that describes the license agreement                                                                                                         |

After downloading the software package, verify the software content by comparing the checksum of the downloaded package with the checksum provided in the tool.

Artifacts for CMG CNF: CMG (Cloud Mobile Gateway) > CMG-CNF > X.Y release > Rn>

| Artifacts for CMG product variant in Nokia support portal: folder location | Filename                                                                                                                                                                                                                                                          | Description                                                                          |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| CMG-CNF                                                                    | Nokia-CMG-DBCNF-X.Y.Rn                                                                                                                                                                                                                                            | Docker images for DB CNF (for N:K redundancy support)                                |
| CMG-CNF                                                                    | Nokia-CMG-LMG-X.Y.Rn                                                                                                                                                                                                                                              | Docker images for CMG CNF (for example, LMG)                                         |
| CMG-CNF                                                                    | Nokia-CMG-LMG-DBCNF-SHA256-checksum-X.Y.Rn                                                                                                                                                                                                                        | SHA256 checksums of all the files in X.Y release>Rn> folder                          |
| CMG-CNF                                                                    | Nokia-CMG-YANG-X.Y.Rn                                                                                                                                                                                                                                             | CMG YANG models                                                                      |
| CMG-CNF                                                                    | Nokia-CMG-CNF-OPERATOR-X.Y.Rn                                                                                                                                                                                                                                     | CMG CNF K8s Operator images and Helm charts                                          |
| CMG-CNF                                                                    | Nokia-CMG-LMGHELMCHARTS-X.Y.Rn                                                                                                                                                                                                                                    | CMG CNF (CMG and CDB) Helm Charts                                                    |
| CMG-CNF                                                                    | Nokia-CMG-LOGSPLIT-X.Y.Rn                                                                                                                                                                                                                                         | Provides splitting of event logs to FM and general logs                              |
| CMG-CNF                                                                    | Nokia-CMG-LTMT-NCOM-NSD-CNF-Xx.Yy.Rx                                                                                                                                                                                                                              | NSD package for LCM operations of CMG LTMT CNF with NCOM                             |
| CMG-CNF                                                                    | Nokia-CMG-LTMT-NCOM-NSD-input-CNF- Xx.Yy.Rx                                                                                                                                                                                                                       | Input files with modifiable attributes for deploying LTMT for cMG/cMM CNFs with NCOM |
| CMG-CNF                                                                    | Nokia-CMG-LTMT-image-signing-X.Y.Rn                                                                                                                                                                                                                               | Signatures of LTMT image                                                             |
| CMG-CNF                                                                    | Nokia-CMG-LTMT-CNF-X.Y.Rn Note: LTMT files are included in the package of each major release. Nokia recommends using the latest LTMT because LTMT is backward compatible. Use the 26.7.R1 version of LTMT with any CMG/CMM version from release 24.7 and onwards. | Docker images for the CNF Lightweight Monitoring Tool                                |

| Artifacts for CMG product variant in Nokia support portal: folder location | Filename                                                                                                 | Description                                                                                                        |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| CMG-CNF                                                                    | Nokia-CMG-NCOM-NSD-CNF-X.Y.Rn<br>The NCOM files are included in the package of each major release.       | NSD packages for LCM operations of CMG CNFs with NCOM                                                              |
| CMG-CNF                                                                    | Nokia-CMG-NCOM-NSD-input-CNF-X.Y.Rn<br>The NCOM files are included in the package of each major release. | Input files with modifiable attributes for deploying CMG CNFs with NCOM                                            |
| CMG-CNF                                                                    | Nokia-CMG-K8SCTL-X.Y.Rn                                                                                  | Provides Istio K8s service for HTTP2 traffic load balancing                                                        |
| CMG-CNF                                                                    | Nokia-CMG-image-signing-X.Y.Rn.zip                                                                       | Signatures of CMG images                                                                                           |
| CMG-CNF                                                                    | Nokia-CMG-NCOM-ANSIBLE-cmg_bof_path_update-X.Y.Rn                                                        | Updates the CMG BoF primary-configuration path as part of post deployment action using NCOM                        |
| CMG-CNF                                                                    | Nokia-CMG-NCOM-ANSIBLE-cmg_post_upgrade_pod_reset-X.Y.Rn                                                 | Deletes the CMG PODs from Kubernetes cluster as part of post upgrade (OSSU) action using NCOM                      |
| CMG-CNF                                                                    | Nokia-CMG-CertMgmt-X.Y.Rn.zip                                                                            | Automatic certificate manager                                                                                      |
| CMG-CNF                                                                    | Nokia_External_Customer_license_All_LI_on.zip                                                            | This is a commercial Trust Wildcard CMG license that enables all CMG features. The LI is enabled in this license.  |
| CMG-CNF                                                                    | Nokia_External_Customer_license_All_LI_on.pdf                                                            | This is a supplement to Nokia_External_Customer_license_All_LI_on.zip that describes the license agreement         |
| CMG-CNF                                                                    | Nokia_External_Customer_license_All_LI_off.zip                                                           | This is a commercial Trust Wildcard CMG license that enables all CMG features. The LI is disabled in this license. |
| CMG-CNF                                                                    | Nokia_External_Customer_license_All_LI_off.pdf                                                           | This is a supplement to Nokia_External_Customer_license_All_LI_off.zip that describes the license agreement        |

Artifacts for CMG-a: CMG (Cloud Mobile Gateway) CMG-CNF > X.Y release > Rn>

| Artifacts for CMG product variant in<br>Nokia support portal:<br>folder location | Filename                        | Description                                    |
|----------------------------------------------------------------------------------|---------------------------------|------------------------------------------------|
| CMG-a                                                                            | Nokia-CMG-a-X.Y.Rn              | CMG-a software image file                      |
| CMG-a                                                                            | Nokia-CMG-a-MD5-checksum-X.Y.Rn | MD5 Checksum for the CMG-a software image file |

Artifacts for CMG CBAM templates: CMG (Cloud Mobile Gateway) > CBAM > vX.Y.Z.U-n  
version >

| Artifacts for CMG product variant in<br>Nokia support portal:<br>folder location | Filename                                                    | Description                                                             |
|----------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|
| CBAM                                                                             | cMg_Cbam_Sriov_Ltmt_Package_vX.Y.Z.U                        | CMG CBAM Templates for LTMT Openstack                                   |
| CBAM                                                                             | cMg_Cbam_Sriov_MgRedirect_NkRedundancy_Package_vX.Y.Z.U     | CMG CBAM Templates for LB-Less N:K redundancy topology SR-IOV Openstack |
| CBAM                                                                             | cMg_Cbam_Sriov_MgRedirect_Package_vX.Y.Z.U                  | CMG CBAM Templates for LB-Less SR-IOV Openstack                         |
| CBAM                                                                             | cMg_cbam_Sriov_MgRedirect_No nRedGroup_Package_vX.Y.Z.UCM G | CBAM Templates for LB-Mode all active MG redundancy SR-IOV Openstack    |
| CBAM                                                                             | cMg_Cbam_Sriov_NkRedundancy_Package_vX.Y.Z.U                | CMG CBAM Templates for LB-Mode N:K redundancy topology SR-IOV Openstack |
| CBAM                                                                             | cMg_Cbam_Sriov_Package_vX.Y.Z.U                             | CMG CBAM Templates for LB-Mode SR-IOV Openstack                         |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_Ltmt_Package_vX.Y.Z.U                      | CMG CBAM Templates for LTMT VMware                                      |

| Artifacts for CMG product variant in<br>Nokia support portal:<br>folder location | Filename                                                                               | Description                                                                                           |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| CBAM                                                                             | cMg_Cbam_SriovLtmthHighAvailab<br>ilityPackage_vX.Y.Z.U This<br>package is deprecated. | CMG CBAM Templates for<br>High Available LTMT<br>Openstack                                            |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_MgRedirect<br>_NkRedundancy_PackageA_vX.Y.<br>Z.U                     | CMG CBAM Templates for LB-<br>Less N:K redundancy topology<br>VMware (Grantful mode)                  |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_MgRedirect<br>_NkRedundancy_PackageB_vX.Y.<br>Z.U                     | CMG CBAM Templates for LB-<br>Less N:K redundancy topology<br>VMware (Grantless mode)                 |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_MgRedirect<br>_NkRedundancy_PackageC_vX.Y.<br>Z.U                     | CMG CBAM Templates for LB-<br>Less N:K redundancy topology<br>VMware (Grantless mode &<br>one OrgVDC) |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_MgRedirect<br>_Package_vX.Y.Z.U                                       | CMG CBAM Templates for LB-<br>Less VMware (Grantless<br>mode)                                         |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_NkRedunda<br>ncy_PackageA_vX.Y.Z.U                                    | CMG CBAM Templates for LB-<br>Mode N:K redundancy<br>topology VMware (Grantful<br>mode)               |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_NkRedunda<br>ncy_PackageB_vX.Y.Z.U                                    | CMG CBAM Templates for LB-<br>Mode N:K redundancy<br>topology VMware (Grantless<br>mode)              |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_NkRedunda<br>ncy_PackageC_vX.Y.Z.U                                    | CMG CBAM Templates for LB-<br>Mode N:K redundancy<br>topology VMware (Grantless<br>mode & one OrgVDC) |
| CBAM                                                                             | cMg_Cbam_Vmxnet3_Package_v<br>X.Y.Z.U                                                  | CMG CBAM Templates for LB-<br>Mode VMware (Grantless<br>mode)                                         |

The -n versions/folders contain CMG CBAM Templates for VMware with the updated OVA file due to the new CMG software image.

## 2.2 NE to Nokia solution mapping

The following tables list the relevant releases of the other products that are supported with CMG release.

Release-load matrix

| CMG Release | CBIS                                          | CBAM                                          | NMS     | VMware                             | NCS                                                         | NRHO                 | NCOM                           |
|-------------|-----------------------------------------------|-----------------------------------------------|---------|------------------------------------|-------------------------------------------------------------|----------------------|--------------------------------|
| CMG26.3     | Release 26.3 does not support VNF deployments | Release 26.3 does not support VNF deployments | 26R1-NM | TCP3.0 with EDP<br>TCP5.1 with EDP | NCS 22.12<br>NCS 23.10<br>NCS 24.7<br>NCS 24.11<br>NCS 25.7 | NCP 24.7<br>NCP 25.4 | NCOM 24 FP2 PP1<br>NCOM 24 FP3 |

| CMG Release | CBIS               | CBAM                          | NMS                                                                                 | VMware             | NCS                                                         | NRHO                 | NCOM                               | RH OCP                           |
|-------------|--------------------|-------------------------------|-------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------|----------------------|------------------------------------|----------------------------------|
| CMG26.7     | CBIS 24<br>CBIS 25 | CBAM 24<br>CBAM 25<br>CBAM 26 | MantaRay 25R2-NM + FP1820<br>MantaRay 25R3-NM + FP1820<br>MantaRay 26R1-NM + FP1820 | TCI 2.2<br>TCI 3.0 | NCS 24.7<br>NCS 24.11<br>NCS 25.7<br>NCS 25.11<br>NCS 26.7* | NCP 24.7<br>NCP 25.4 | NCOM 24 FP2 PP1<br>NCOM 24 FP3 PP1 | OCP 4.16<br>OCP 4.18<br>OCP 4.20 |

\* Used NCS 26.7 pre-C5 version

NCOM supports CMG Deployment, offline upgrade and Termination. The CMG Operator for CMG ISSU/Canary is not supported with NCOM.

TCI3.0 requires ≥CBAM23

After downloading the software package, verify the software content by comparing the checksum included in the software package with the checksum of the downloaded package.

For example:

#### SHA-256 checksum

```
#sha256sum <zip package name>\  
  
<checksum value> <zip package name>  
  
# cat <zip package name>.sha256\  
  
<checksum value> <zip package name>
```

The checksum values must be identical.

## 3 Environments

### 3.1 Supported hardware

#### 3.1.1 Supported 7750 SR chassis

| Nokia Part | Hardware name           |
|------------|-------------------------|
| 7750 SR-7  | 7750 SR-7 slot chassis  |
| 7750 SR-12 | 7750 SR-12 slot chassis |

#### 3.1.2 Supported SFM, CPM, MCM, CCM, IOM, IMM and ISM cards

| Nokia Part | CLI String (Card) | CLI String (MDA)           | Description                                |
|------------|-------------------|----------------------------|--------------------------------------------|
| 3HE13727AA | iom5-e            | —                          | IOM - 7750 SR IOM5-e<br>400G – CR          |
| 3HE12334AA | —                 | me12-100gb qsfp28          | MDA-e-XP - 7750 SR<br>12pt 100GE QSFP28    |
| 3HE07158AA | imm-2pac-fp3      | p6-10g-sfp<br>p6-10g-sfp   | 7750 12-pt 10GE FP3<br>SFP+ IMM –L3HQ      |
| 3HE07158BA | imm-2pac-fp3      | p6-10g-sfp<br>p6-10g-sfp   | 7750 12-port 10GE FP3<br>SFP+ IMM – L3BQ Q |
| 3HE07159AA | cpm5              | p1-100g-cfp                | 7750 SR 1-port 100GE<br>FP3 CFP IMM – L3HQ |
| 3HE07159BA | iom3-xp-c         | p1-100g-cfp                | 7750 SR 1-port 100GE<br>FP3 CFP IMM – L3BQ |
| 3HE07303AA | sfm5-12           | p1-100g-cfp<br>p1-100g-cfp | 7750 SR 2-port 100GE<br>FP3 CFP IMM – L3HQ |
| 3HE07303BA | sfm5-7            | p1-100g-cfp<br>p1-100g-cfp | 7750 SR-2-port 100GE<br>FP3 CFP IMM – L3BQ |

| Nokia Part | CLI String (Card) | CLI String (MDA)           | Description                                                        |
|------------|-------------------|----------------------------|--------------------------------------------------------------------|
| 3HE07304AA | ism-mg-2pac-fp3   | p3-40g-qsfp<br>p3-40g-qsfp | 7750 SR 6-port 40GE<br>FP3 QSFP IMM – L3HQ                         |
| 3HE08174AA | ism-mg-2pac-fp3   | p10-10g-sfp<br>p20-1ge-sfp | 7750 SR 10-port 10GE<br>SFP+ + 20-port GE SFP<br>FP3 IMM – L3HQ    |
| 3HE08174BA | imm-2pac-fp3      | p10-10g-sfp<br>p20-1ge-sfp | 7750 SR 10-port 10GE<br>SFP+<br>+ 20-port GE SFP FP3<br>IMM – L3BQ |
| 3HE08175AA | imm-2pac-fp3      | p3-40g-qsfp<br>p20-1ge-sfp | 7750 SR 3-port 40GE<br>QSFP<br>+ 20-port GE SFP FP3<br>IMM – L3HQ  |
| 3HE08175BA | imm-2pac-fp3      | p3-40g-qsfp<br>p20-1ge-sfp | 7750 SR 3-port 40GE<br>QSFP<br>+ 20-port GE SFP FP3<br>IMM – L3BQ  |
| 3HE08423AA | cpm5              | —                          | 7750 SR CPM5                                                       |
| 3HE08426AA | iom4-e            | —                          | 7750 SR IOM3-XP-C                                                  |
| 3HE08428AA | iom4-e            | —                          | 7750 SR SFM5-12                                                    |
| 3HE08429AA | iom4-e            | —                          | 7750 SR SFM5-7                                                     |
| 3HE09117AA | ism-mg-2pac-fp3   | p-isa2-ms<br>p-isa2-ms     | 7750 Multiservice ISM                                              |
| 3HE09118AA | ism-mg-2pac-fp3   | p-isa2-ms-e<br>p-isa2-ms-e | 7750 Multiservice ISM-<br>E (no encryption)                        |
| 3HE09192AA | ism-mg-2pac-fp3   | p-isa2-ms<br>p1-100g-cfp   | 7750 MS-ISA2 + 1-port<br>100GE CFP IMM – L3HQ                      |
| 3HE09192BA | iom4-e-b          | p-isa2-ms<br>p1-100g-cfp   | 7750 MS-ISA2 + 1-port<br>100GE CFP IMM – L3BQ                      |

| Nokia Part | CLI String (Card) | CLI String (MDA)         | Description                                            |
|------------|-------------------|--------------------------|--------------------------------------------------------|
| 3HE09193AA | iom4-e-b          | p-isa2-ms<br>p10-10g-sfp | 7750 MS-ISA2 + 10pt<br>10GE SFP+ IMM –<br>L3HQ         |
| 3HE09193BA | iom4-e-b          | p-isa2-ms<br>p10-10g-sfp | 7750 MS-ISA2 + 10pt<br>10GE SFP+ IMM –L3BQ             |
| 3HE09648AA | iom4-e-hs         | —                        | IOM – 7750 SR IOM4-e<br>L3HQ                           |
| 3HE09648BA | iom4-e-hs         | —                        | IOM – 7750 SR IOM4-e<br>L3BQ                           |
| 3HE09648CA | ism-mg-2pac-fp3   | —                        | IOM – 7750 SR IOM4-e<br>L2HQ                           |
| 3HE10013AA | ism-mg-2pac-fp3   | —                        | 7750 ISM-MG-C<br>Multiservice ISM-E (no<br>encryption) |
| 3HE10556AA | imm-2pac-fp3      | —                        | 7750 ISM-MG-C<br>Multiservice ISM                      |
| 3HE10557AA | imm-2pac-fp3      | —                        | 7750 ISM-MG-C<br>Multiservice ISM-E (no<br>encryption) |
| 3HE10717AA | imm-2pac-fp3      | —                        | IOM - 7750 SR IOM4-e-<br>B L3HQ (Polaris-B)            |
| 3HE10717BA | imm-2pac-fp3      | —                        | 7750 SR IOM4-eB<br>L3BQ<br>Accepts up to two<br>MDA-e  |
| 3HE10717CA | cpm5              | —                        | SR IOM - 7750 SR<br>IOM4-eB L2HQ                       |
| 3HE11351AA | iom3-xp-c         | —                        | 7750 SR IOM4-eHS<br>L3HQ<br>Accepts up to two<br>MDA-e |

| Nokia Part | CLI String (Card) | CLI String (MDA) | Description                                      |
|------------|-------------------|------------------|--------------------------------------------------|
| 3HE11351CA | sfm5-12           | —                | 7750 SR IOM4-eHS L2HQ<br>Accepts up to two MDA-e |
| 3HE15879AA | sfm5-7            | —                | 7750 ISM-MG-C Multiservice ISM                   |
| 3HE15880AA | ism-mg-2pac-fp3   | —                | 7750 ISM-MG-C Multiservice ISM-E (no encryption) |

Interfaces support existing Layer 1, Layer 2, and Layer 3 Ethernet parameters as supported in SR OS Release 20.10.R10, including 802.1q VLANs, 802.3ad LAG, 802.1p, ECMP and network port QoS.

### 3.1.3 Supported MDAs, CMAs and ISA Cards

| Nokia Part | CLI String (Card) | CLI String (MDA) | Description                         |
|------------|-------------------|------------------|-------------------------------------|
| 3HE03611AA | m10-1gb-xp-sfp    | —                | 7750 SR 10-port GE SFP MDA-XP       |
| 3HE03612AA | m20-1gb-xp-sfp    | —                | 7750 SR 20-port GE SFP MDA-XP       |
| 3HE03685AA | m2-10gb-xp-xfp    | —                | 7750 SR 2-port 10GBASE – XFP MDA-XP |
| 3HE03686AA | m4-10gb-xp-xfp    | —                | 7750 SR 4-port 10GBASE – XFP MDA-XP |
| 3HE04274AA | m1-10gb-xp-xfp    | —                | 7750 SR 1-port 10GBASE – XFP MDA-XP |
| 3HE09649AA | me10-10gb-sfp+    | —                | MDA-e 10-port 10 GE SFP+            |

| Nokia Part | CLI String (Card) | CLI String (MDA) | Description                                                                                             |
|------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------|
| 3HE09881AA | me1-100gb-cfp2    | —                | MDA-e 1-port 100 GE CFP2<br>Accepts one CFP2 100GBase Optic Module                                      |
| 3HE10427AA | me-isa2-ms        | —                | ISA - 7750 SR MS-ISA2<br>Only Ethernet MDA-e cards are supported in IOM4-e-hs. ISAs are not supported   |
| 3HE10428AA | me-isa2-ms-e      | —                | ISA - 7750 SR MS-ISA2-E<br>Only Ethernet MDA-e cards are supported in IOM4-e-hs. ISAs are not supported |
| 3HE10642AA | me40-1gb-csfp     | —                | MDA-e 20-port GE SFP/40-port GE cSFP                                                                    |
| 3HE11030AA | me2-100gb-cfp4    | —                | MDA-e 2-port 100GE CFP4<br>Accepts two CFP4 100GBase Optic Module                                       |
| 3HE11031AA | me2-100gb-qsfp28  | —                | MDA-e 2-port 100GE QSFP28                                                                               |

### 3.1.4 Supported CMG appliances: CMG-a RM19 hardware

| Nokia Model             | Nokia Part | Release | Description                                       |
|-------------------------|------------|---------|---------------------------------------------------|
| CMG-a2 System Bundle DC | 3HE16618AA | 11.0r8  | Cloud Mobile Gateway appliance, CMG-a2 DC variant |
| CMG-a2 System Bundle AC | 3HE16619AA | 11.0r8  | Cloud Mobile Gateway appliance, CMG-a2 AC variant |

| Nokia Model             | Nokia Part | Release | Description                                       |
|-------------------------|------------|---------|---------------------------------------------------|
| CMG-a8 System Bundle DC | 3HE16620AA | 11.0r8  | Cloud Mobile Gateway appliance, CMG-a8 DC variant |
| CMG-a8 System Bundle AC | 3HE16621AA | 11.0r8  | Cloud Mobile Gateway appliance, CMG-a8 AC variant |

### 3.1.5 Supported CMG appliances: CMG-a on HPE Gen10+ hardware

| Nokia Model                            | Nokia Part | Release | Description                                       |
|----------------------------------------|------------|---------|---------------------------------------------------|
| HPE BNDL - CMG-a2 R23 System Bundle AC | 3HZ23315GA | 12.0r9  | Cloud Mobile Gateway appliance, CMG-a2 AC variant |
| HPE BNDL - CMG-a2 R23 System Bundle DC | 3HZ23316GA | 12.0r9  | Cloud Mobile Gateway appliance, CMG-a2 DC variant |
| HPE BNDL - CMG-a6 R23 System Bundle AC | 3HZ23317GA | 12.0r9  | Cloud Mobile Gateway appliance, CMG-a6 AC variant |
| HPE BNDL - CMG-a6 R23 System Bundle DC | 3HZ23318GA | 12.0r9  | Cloud Mobile Gateway appliance, CMG-a6 DC variant |

### 3.1.6 Supported CMG appliances: CMG-a on HPE Gen11 DL325 hardware

| Nokia Model                        | Nokia Part    | Release | Description                                       |
|------------------------------------|---------------|---------|---------------------------------------------------|
| BNDL g11 - CMG-a2 System Bundle AC | PCOEMHPHW5435 | 23.11   | Cloud Mobile Gateway appliance, CMG-a2 AC variant |
| BNDL g11 - CMG-a2 System Bundle DC | PCOEMHPHW5436 | 23.11   | Cloud Mobile Gateway appliance, CMG-a2 DC variant |

| Nokia Model                           | Nokia Part    | Release | Description                                             |
|---------------------------------------|---------------|---------|---------------------------------------------------------|
| BNDL g11 - CMG-a6<br>System Bundle AC | PCOEMHPHW5437 | 24.3    | Cloud Mobile Gateway<br>appliance, CMG-a6 AC<br>variant |
| BNDL g11 - CMG-a6<br>System Bundle DC | PCOEMHPHW5438 | 24.3    | Cloud Mobile Gateway<br>appliance, CMG-a6 DC<br>variant |

The HPE Gen11 CMG-a6 is not supported in release 23.11.R1.

### 3.1.7 Supported CMG appliances: CMG-a on HPE Gen11 DL345 hardware

| Nokia Model                                           | Nokia Part     | Release | Description                                             |
|-------------------------------------------------------|----------------|---------|---------------------------------------------------------|
| BNDL g11 CMG-a2<br>DL345 96c Sys Bundle<br>AC         | PCOEMHPHW5456X | 24.7    | Cloud Mobile Gateway<br>appliance, CMG-a2 AC<br>variant |
| BNDL g11 CMG-a2<br>DL345 96c Sys Bundle<br>DC         | PCOEMHPHW5457X | 24.7    | Cloud Mobile Gateway<br>appliance, CMG-a2 DC<br>variant |
| BNDL g11 - CMG-a2<br>DL345 96c V2 System<br>Bundle AC | PCOEMHPHW5507X | 26.3    | Cloud Mobile Gateway<br>appliance, CMG-a2 AC<br>variant |
| BNDL g11 - CMG-a2<br>DL345 96c V2 System<br>Bundle DC | PCOEMHPHW5508X | 26.3    | Cloud Mobile Gateway<br>appliance, CMG-a2 DC<br>variant |

### 3.1.8 Supported CMG appliances: CMG-a on HPE Gen12 DL345 hardware

| Nokia Model                                         | Nokia Part     | Release | Description                                             |
|-----------------------------------------------------|----------------|---------|---------------------------------------------------------|
| BNDL g12 - CMG-a2<br>DL345 192c System<br>Bundle AC | PCOEMHPHW5493X | 26.3    | Cloud Mobile Gateway<br>appliance, CMG-a2 AC<br>variant |

| Nokia Model                                         | Nokia Part     | Release | Description                                             |
|-----------------------------------------------------|----------------|---------|---------------------------------------------------------|
| BNDL g12 - CMG-a2<br>DL345 192c System<br>Bundle DC | PCOEMHPHW5494X | 26.3    | Cloud Mobile Gateway<br>appliance, CMG-a2 DC<br>variant |

### 3.1.9 [NEW] Supported CMG appliances: Firmware on HPE and DELL hardware

| CMG Variant                  | Server Model     | BIOS       | FPGA           | iLO                              | NIC Model(s)             | NIC Firmware           | IXR                               |
|------------------------------|------------------|------------|----------------|----------------------------------|--------------------------|------------------------|-----------------------------------|
| CMG-a2 64c                   | HPE DL325 Gen10+ | BIOS 4.00  | Not applicable | iLO5 3.18 (Rec)/2.72, 2.33 (Sup) | MCX65310/ MCX623106AS    | 20.43.8002/ 20.43.8002 | Not applicable                    |
| CMG-a6 64c                   | HPE DL325 Gen10+ | BIOS 4.00  | Not applicable | iLO5 3.18 (Rec)/2.72, 2.33 (Sup) | MCX65310/ MCX623106AS    | 20.43.8002/ 22.46.3048 | 7220 IXR-D3L – SR Linux vR25.10.3 |
| CMG-a2 64c                   | HPE DL325 Gen11  | BIOS 3.00  | Not applicable | iLO6 1.74 (Rec)/1.66 (Sup)       | MCX623436AC/ MCX623106AS | 22.41.1000/ 22.46.3048 | Not applicable                    |
| CMG-a6 64c                   | HPE DL325 Gen11  | BIOS 3.00  | Not applicable | iLO6 1.74 (Rec)/1.66 (Sup)       | MCX623436AC/ MCX623106AS | 22.41.1000/ 22.46.3048 | 7220 IXR-D3L – SR Linux vR25.10.3 |
| CMG-a2 96c (CPU EPYC 9654P)  | HPE DL345 Gen11  | BIOS 3.00  | Not applicable | iLO6 1.74 (Rec)/1.66 (Sup)       | MCX623106AS              | 22.46.3048             | Not applicable                    |
| CMG-a2 96c (CPU EPYC 9655P)  | HPE DL345 Gen11  | BIOS 3.00  | Not applicable | iLO6 1.74 (Rec)/1.66 (Sup)       | MCX623106AS              | 22.46.3048             | Not applicable                    |
| CMG-a2 192c (CPU EPYC 9965P) | HPE DL345 Gen12  | BIOS 1.50  | Not applicable | iLO7 1.17.00                     | MCX623106AS/ MCX75310AAS | 22.46.3048/ 28.48.1000 | Not applicable                    |
| CKD-c1                       | DELL XR8720t     | BIOS 1.1.3 | FPGA 1.2.3     | iDRAC 1.30.13.10                 | Not applicable           | Not applicable         | Not applicable                    |

### 3.1.10 Supported CMG deployments

The table shows the supported infrastructure and the LB architecture supported/MG redundancy of CMG deployments.

#### Supported CMG deployments

| Nokia Model          | Nokia Part      | Release                                                                                  |
|----------------------|-----------------|------------------------------------------------------------------------------------------|
| combined SGW/PGW     | VNF, CMG-a      | LB mode/ 1:1, all active<br>LB Less/1:1, all active, N:K<br>Hybrid/1:1, all active, N:K  |
| combined SGW-C/PGW-C | VNF,CNF,CMG-a   | LB mode/ 1:1, all active, N:K                                                            |
| combined SGW-U/PGW-U | VNF,CNF,CMG-a   | LB mode/ 1:1, all active<br>LB Less/1:1, all active, N:K<br>Hybrid/ 1:1, all active, N:K |
| SMF                  | VNF,CNF,CMG-a   | LB mode/ 1:1, all active, N:K                                                            |
| UPF                  | VNF,CNF,CMG-a   | LB mode/ 1:1, all active<br>LB Less/1:1, all active, N:K<br>Hybrid/ 1:1, all active, N:K |
| ePDG                 | VNF, CMG-a, CNF | LB mode: 1:1/all active                                                                  |
| TWAG                 | VNF             | LB mode: 1:1/all active                                                                  |

The following restrictions apply on CMG-a.

1. CMG-a on HPE servers must be used only for LB-less or hybrid mode.
2. N:K is supported only on CMG-a8 on the RM19 servers for the following models:
  - combined SGW-C/PGW-C
  - combined SGW-U/PGW-U
  - SMF
  - UPF
3. The all-active on CUPS UP deployments functionality is not recommended for LB-less deployments.
4. Combined SMF/GW-C or combined UPF/GW-U are supported on all VNF/CNF and CMG-a.

### 3.1.11 Supported Local High Availability Redundancy methods

1:1 method — the MG-VMs or pods are deployed in active and standby pairs

N:K method — an N number of active MG-VMs or pods are backed up by a K number of standby MG-VMs or pods; this method requires a set of DB-VMs or pods that store the state of the active MGs which is retrieved from the standby MGs when they become active

Both methods provide service continuation (subscriber signaling and User Plane state) in case an active MG-VM or pod fails.

Nokia strongly recommends that all live or production grade deployments incorporate one of the two above-mentioned High Availability redundancy methods.

While Nokia allows customers to deploy CMG with all MGs configured as active (all-active MGs deployment); this configuration must only be used in trial or lab environments.

This non-redundant configuration is referred to as N+ configuration and does not provide service continuation in case an active MG-VM or pod fails. Any subscriber sessions and User Plane forwarding bound to the failed MG are lost and must be reestablished by peer NFs or the UEs themselves.

CMG operators who choose the all-active MGs deployment must be aware of the following:

- The CMG binds subscribers and forwarding status to individual MG-VMs or pods; failure of an MG-VM or pod results in the loss of those subscriber sessions (signaling and User Plane forwarding state).

- The partial failure recovery of some CMG functions, for example Sx and GTP-C, does allow the CMG to perform bulk deletions of lost states; however, there may be cases where loss of subscriber signaling and forwarding state is "silent" (that is, peer NFs may not be aware of the loss of state by the CMG).

- Loss of a large amount of signaling state can result in a surge of signaling (storm) as that state is reestablished by the peer NFs.

- The CMG intra instance In Service Software Upgrade procedure relies on the N:K redundancy method and is therefore not possible in all-active MGs deployments.

If the operators still decide to deploy CMG using the all-active MGs deployment in a live environment, Nokia will provide support. However, in this case, operators must assume the risk of outages, such as those described in this section and agree not to subject Nokia to any escalations or penalties for outages that are due to the all-active MGs deployment.

## 4 Installation

### 4.1 Installation instructions

For installation instructions, see the following guides in the CMG documentation set:

CMG Installation Guide (DN1000133942)

CMG CNF Installation Guide (DN1000133986)

CMG-a Installation and System Management Guide (DN1000133953)

The CMG documentation set is available in Nokia Discovery Center at <https://doc.networks.nokia.com>.

### 4.2 Uploading an OVA vApp template with vCloud Director version 9.7

VMware introduced NSX-T next to NSX-V in vCloud NFV 3.2. However, the vCloud Director 9.7 (part of NFV 3.2) does not support network creation in NSX-T. To upload an OVA vApp template using vCloud Director version 9.7 the operator must perform the following:

1. Open the accompanied OVF file of the cMg CBAM package template and identify all the network names as defined in the file under section

`<ovf:NetworkSection>`.

Example

```
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt10ECP">
  <ovf:Description>The LB3ExternalPortGroup 10 network</ovf:Description>
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt11ECP">
  <ovf:Description>The LB3ExternalPortGroup 11 network</ovf:Description>
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt12ECP">
  <ovf:Description>The LB3ExternalPortGroup 12 network</ovf:Description>
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt13ECP">
  <ovf:Description>The LB4ExternalPortGroup 13 network</ovf:Description>
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt14ECP">
  <ovf:Description>The LB4ExternalPortGroup 14 network</ovf:Description>
</ovf:Network>
<ovf:Network ovf:name="loadBalancerExt15ECP">
```

## 5 Upgrade

### 5.1 CMG CBAM Templates for release 26.7.RX

The following table lists the available versions of the CMG CBAM templates and the verified combinations with CMG software releases and the CBAM releases. The CMG software build can be upgraded without CMG CBAM template package updates. For the CMG upgrade procedure, refer to the CMG Installation Guide.

Verified CBAM CMG templates with CMG SW and CBAM releases

| CMG releases | CBAM CMG templates | CBAM release     |
|--------------|--------------------|------------------|
| 26.7         | 26.7.1.1           | CBAM 26, CBAM 25 |
| 26.7         | 25.7.1.3           | CBAM 25          |

The following table lists the latest available versions.

Latest available CMG CBAM templates

| Nokia Model   | Nokia Part | Release | Description                                             | Release Date |
|---------------|------------|---------|---------------------------------------------------------|--------------|
| SR-IOV cMG    | cMG 26.7R1 | CBAM 26 | cMgCbamSriovPackage_v26.7.1.1.zip                       | 2026-07-31   |
| SR-IOV MG+    | cMG 26.7R1 | CBAM 26 | cMgCbamSriovMgRedirectPackage_v26.7.1.1.zip             | 2026-07-31   |
| SR-IOV NK     | cMG 26.7R1 | CBAM 26 | cMgCbamSriovNkRedundancyPackage_v26.7.1.1.zip           | 2026-07-31   |
| SR-IOV MG+ NK | cMG 26.7R1 | CBAM 26 | cMgCbamSriovMgRedirectNkRedundancyPackage_v26.7.1.1.zip | 2026-07-31   |
| SR-IOV N+     | cMG 26.7R1 | CBAM 26 | cMgCbamSriovMgRedirectNonRedGrp_v26.7.1.1.zip           | 2026-07-31   |
|               |            |         |                                                         |              |
| Vmxnet3 cMG   | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3Package_v26.7.1.1.zip                     | 2026-07-31   |

| Nokia Model    | Nokia Part | Release | Description                                                | Release Date |
|----------------|------------|---------|------------------------------------------------------------|--------------|
| Vmxnet3 MG+    | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3MgRedirectPackage_v26.7.1.1.zip              | 2026-07-31   |
| Vmxnet3 NK     | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3NkRedundancyPackageA_v26.7.1.1.zip           | 2026-07-31   |
| Vmxnet3 NK     | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3NkRedundancyPackageB_v26.7.1.1.zip           | 2026-07-31   |
| Vmxnet3 NK     | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3NkRedundancyPackageC_v26.7.1.1.zip           | 2026-07-31   |
| Vmxnet3 MG+ NK | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3MgRedirectNkRedundancyPackageA_v26.7.1.1.zip | 2026-07-31   |
| Vmxnet3 MG+ NK | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3MgRedirectNkRedundancyPackageB_v26.7.1.1.zip | 2026-07-31   |
| Vmxnet3 MG+ NK | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3MgRedirectNkRedundancyPackageC_v26.7.1.1.zip | 2026-07-31   |

The following table lists the available versions of the CMG CBAM templates and the compatible combinations with CMG software releases and the CBAM releases which are not verified by the RnD. These combinations are recommended to be verified by Customer Projects before they are live in Customers' network.

#### Compatible CBAM CMG templates with CMG SW and CBAM releases

| CMG releases | CBAM CMG templates | CBAM release   |
|--------------|--------------------|----------------|
| 26.7         | 26.7.1.1           | CBAM24         |
| 26.7         | 25.7.1.3           | CBAM26, CBAM24 |

- The new 26.7.1.1 CMG CBAM OpenStack Template version does not support AA Signature update.
- The CMG CBAM Templates for VMWare for the remaining future releases do not support SOL00x v3.3.1 specifications. The supported features are the same with 24.11.1.1 templates version. The OVA file is updated with 26.7.Rx image.

## Verified VNFD package upgrade paths

| VNFD package upgrade |
|----------------------|
| 25.3.1.x to 25.7.1.x |
| 25.7.1.x to 26.7.1.1 |

## 5.2 LTMT CBAM Templates for release 26.X.RX

The following table shows the latest LTMT CBAM templates.

### Latest available LTMT CBAM templates

| Nokia Model  | Nokia Part | Release | Description                             | Release Date |
|--------------|------------|---------|-----------------------------------------|--------------|
| SR-IOV LTMT  | cMG 26.7R1 | CBAM 26 | cMgCbamSriovLtmtPackage_v26.7.1.1.zip   | 2026-07-31   |
| Vmxnet3 LTMT | cMG 26.7R1 | CBAM 26 | cMgCbamVmxnet3LtmtPackage_v26.7.1.1.zip | 2026-07-31   |

The LTMT CBAM Template for HA LTMT is deprecated. To deploy LTMT HA with CBAM, contact your Nokia technical support representative.

## 5.3 CUPS upgrade

Nokia strongly recommends upgrading first the UPFs and immediately after the SMFs to the same software versions.

New functionality with impact in both SMF and UPF is allowed to be enabled only when both NEs are in the same software version which is the one where the new functionality is implemented or a later one.

To avoid any issues in Customers' live networks, contact your Nokia technical support representative to create together a secure upgrade path for your NEs.

## 5.4 Intra-CNF in-service upgrade procedure (Intra-CNF ISSU)

This section provides information about the CNF Intra-ISSU software upgrade paths.

For ISSU, the upgrade paths are specified by the CR24a model with a caveat that the target release build must be built after the source release build.

Contact your Nokia representative before planning for ISSU upgrades.

Since release 24.11, Intra-ISSU is supported for both CMG N:K and 1:1 redundancy models.

The supported paths to 26.7.R1 are as follows:

- 24.7.R6-R20 to 26.7.R1
- 25.7.R1-R13 to 26.7.R1 (25.7.R14 to 26.7.R1 is not supported)
- 25.11.R1-R9 to 26.7.R1
- 26.3.R1-R5 to 26.7.R1

Intra ISSU is supported for CMGs as SGW-C, SGW/PGW-C, SMF, SGW-U, SGW/PGW-U, UPF, and ePDG. From release 26.3.R1, ePDG CNF ISSU is supported.

The Intra-CNF ISSU is currently not supported for I-SMF- V-SMF, and N3IWF.

For more details about the upgrade procedure, refer to the *7750 SR MG and CMG Software Upgrade Guide*.

## 5.5 Inter-Chassis ISSU (ICR ISSU)

This section provides information about the release upgrade support using Major ICR ISSU.

### **Major ISSU as CMG 1:1 SGW-C, S/PGW-C and/or SMF and SGW-U, S/PGW-U and/or UPF**

- 24.7.R6-R20 to 26.7.R1
- 25.7.R1-R13 to 26.7.R1 (25.7.R14 to 26.7.R1 is not supported)
- 25.11.R1-R9 to 26.7.R1
- 26.3.R1-R5 to 26.7.R1

#### **Notes**

- ICR ISSU in CMG as SGW/PWG-U and/or UPF without GTP-U LB Function (LB-less) does not support the MC-redirection tunnel mechanism.
- ICR ISSU in CMG as I-SMF/V-SMF is not supported.

### **CMG ePDG with 1:1 redundancy**

- 25.11.R1-R9 to 26.7.R1
- 26.3.R1-R5 to 26.7.R1

For more details about the upgrade procedure, refer to the *7750 SR MG and CMG Software Upgrade Guide*.

## 5.6 CMG-a software upgrade for CMG-a2, CMG-a8, and CMG-a6 deployments

Upgrading to release 25.7.R1 results to upgrading RHEL version to 9.6. Due to this upgrade, the use of the `cmg rollback` command does not work. To revert to the previous release, SW re-installation and configuration restoration is required.

Upgrading to release 25.3.R1 from a previous release with the `cmg upgrade` command is not supported due to the upgrade of RHEL version to 9.4. To upgrade to release 25.3.R1, a clean install is required.

## 5.7 CMG-a upgrade paths

| RHEL | CMG-A SW release | Main release CMG upgrade path | Upgrade support    | Rollback support |
|------|------------------|-------------------------------|--------------------|------------------|
| 9.2  | CMG24.7          | CMG24.7.1 to CMG24.7.x        | YES                | YES              |
|      |                  | CMG24.7.1.x to CMG24.11.1     | YES                | YES              |
|      | CMG24.11         | CMG24.11.1 to CMG24.11.x      | YES                | YES              |
|      |                  | CMG24.11.x to CMG25.3         | NO (clean install) | NO               |
| 9.4  | CMG25.3          | CMG25.3.1 to CMG25.3.x        | YES                | YES              |
|      |                  | CMG25.3.x to CMG25.7.1        | YES                | NO               |
| 9.6  | CMG25.7          | CMG25.7.1 to CMG25.7.13       | YES                | YES              |
|      |                  | CMG25.7.x to CMG25.7.14       | YES                | NO               |
|      |                  | CMG25.7.13 to CMG25.11.x      | YES                | YES              |
|      |                  | CMG25.7.14 to CMG25.11.x      | NO (clean install) | NO               |

| RHEL | CMG-A SW release | Main release CMG upgrade path | Upgrade support | Rollback support |
|------|------------------|-------------------------------|-----------------|------------------|
|      | CMG25.11         | CMG25.11.1 to CMG25.11.x      | YES             | YES              |
|      |                  | CMG25.11.x to CMG26.3.1       | YES             | NO               |
|      | CMG26.3          | CMG26.3.1 to CMG26.3.x        | YES             | YES              |
|      |                  | CMG26.3.x to CMG26.7.1        | YES             | NO               |
|      | CMG26.7          | CMG26.7.1 to CMG26.7.x        | YES             | YES              |

## 6 Configuration

### 6.1 Configuring NetConf interface

A set of System-Provisioned Configuration (SPC) objects is provisioned automatically by SR OS. The SPC objects can be created at bootup time. Also, they can be created or removed dynamically based on configuration. The dynamically created SPC objects are typically children objects created as a side effect of the creation of their parent object.

Some of these objects can be deleted (removed) by a user (deletable SPC objects), and some SPC objects cannot be deleted (non-deletable SPC objects).

The ND-SPC objects are not displayed in model-driven interfaces as part of the configuration unless a user explicitly creates the object. This explicit creation of ND-SPC objects is only supported when operating in model-driven configuration mode; it is not supported in mixed configuration mode.

When a user explicitly creates an ND-SPC object, the SR OS remembers that it was explicitly created and displays it as part of the configuration. This can be useful for NetConf clients and tools that perform offline validation of the configuration against the SR OS YANG models and to resolve leafrefs that point to NDSPC objects.

There are several auto-created objects in CMG of which the following are the most referenced as per the SR documentation:

```
configure log log-id 101
configure service customer "1"
configure qos sap-egress queue 1
configure qos sap-ingress queue (1,11)
```

### 6.2 Configuring IXR 7220 SRL in CMG-a6

For switch A, follow the steps.

Enter the CLI in prompt.

```
sr_cli
```

Create a backup file in /tmp/SW-A-running-backup.json.

```
save file /tmp/filename
```

Save the file SW-A-running-backup.json to a local storage.

Run the following set of commands.

```
enter candidate
/interface ethernet-1/{9..14,17..22} subinterface 20 type bridged description
"CdbxSwA" admin-state enable vlan encaps single-tagged vlan-id 20
/interface lag1 subinterface 20 type bridged description "CdbxSwA" admin-state
enable vlan encaps single-tagged vlan-id 20
/network-instance mac-vrf-20 type mac-vrf admin-state enable description
"CdbxSwA" interface ethernet-1/{9..14,17..22}.20
/network-instance mac-vrf-20 type mac-vrf admin-state enable description
"CdbxSwA" interface lag1.20
/network-instance mac-vrf-20 bridge-table mac-duplication admin-state disable
commit save
```

For switch B, follow the same steps.

Enter the CLI in prompt.

`sr_cli`

Create a backup file in `/tmp/SW-B-running-backup.json`.

`save file /tmp/filename`

Save the file `SW-B-running-backup.json` to a local storage.

Run the following set of commands.

```
enter candidate
/interface ethernet-1/{9..14,17..22} subinterface 20 type bridged description
"CdbxSwB" admin-state enable vlan encaps single-tagged vlan-id 20
/interface lag1 subinterface 20 type bridged description "CdbxSwB" admin-state
enable vlan encaps single-tagged vlan-id 20
/network-instance mac-vrf-20 type mac-vrf admin-state enable description
"CdbxSwB" interface ethernet-1/{9..14,17..22}.20
/network-instance mac-vrf-20 type mac-vrf admin-state enable description
"CdbxSwB" interface lag1.20
/network-instance mac-vrf-20 bridge-table mac-duplication admin-state disable
commit save
```

## 6.3 Selecting GTP peer type

### 6.3.1 Peer type

A foreign peer is an S8 or Gp peer and a non-foreign peer is an S5 or Gn peer, depending on whether it is a GTPv2 peer or a GTPv1 peer respectively. When a peer is tagged as foreign, it remains so and any new attach requests from it that contradict this are dropped. Similarly, when a peer is tagged as non-foreign, any new roamer attach requests from it will be dropped. A GTP peer is uniquely defined by the triplet (local\_routing\_instance, local\_address, remote\_address) and can be either foreign or non-foreign, but not both. This means:

- if an SGW uses the same local address for S5 and S8, then the remote PGW S5 peer address must be different from the remote S8 peer address.
- if a PGW uses the same local address for S5 and S8, then the remote SGW S5 peer address must be different from the remote S8 peer address.
- if a GGSN uses the same local address for Gn and Gp, then the remote SGSN Gn peer address must be different from the remote Gp peer address.

The SGW and PGW-GGSN select the peer type based on the content of the incoming packets. The peer list selection is made after the peer type is determined.

- Peer-type selection on the SGW
- If the MCC/MNC portion (OI) of the APN string in the S11 CS Request does not match the locally configured MCC/MNC values in the EPC-NODE or any of the locally configured PLMN-IDs under the `home-plmn-list` command, then the PGW peer is treated as foreign and the S8 reference-point interface will be used, if configured.
- Peer-type selection on the PGW/GGSN
- If, in the first attach request received from a GTP peer, either the Serving Network IE (LTE) or the Routing Area Identity IE (2G/3G) does not match the locally configured MCC/MNC in the EPC-NODE or any of the locally configured PLMN-IDs under the `home-plmn-list` command, then the peer is tagged as foreign. Otherwise, the peer is tagged as non-foreign.

### 6.3.2 Configuring GTP profile for reference points

If the GTP profile of a reference point is not explicitly configured, the following rules determine which GTP profile is selected.

- If the reference point is explicitly configured but no GTP profile is defined, the system will use the GTP profile that is defined for the default reference point of the same type. For example, a specific reference point S5 test-1 will use the GTP profile defined for the reference point S5 default. If no GTP profile is defined for the default reference point of the same type, the system will use the default GTP profile of the PDN.
- If the reference point is not explicitly configured, the system will use the default GTP profile of the PDN.

### 6.3.3 Selecting the peer-list

The peer list for a reference point is selected based on the following:

If S5 and S8 reference point interfaces are different:

- peer lists configured for the respective reference-points are selected.

If S5 and S8 reference-point interfaces are NOT different:

- if the S5 peer list is configured, it is selected for use on both reference points.
- if the S5 peer-list is NOT configured, no peer list is selected for use on both reference points.

If a peer list is assigned to S8 but not S5, the S8 peer-list is NOT used in this case where S5/S8 share an interface.

- If Gn and Gp reference-point interfaces are different, the peer lists configured for the respective reference-points are selected.

If Gn and Gp reference-point interfaces are not different.

- If the Gn peer list is configured, it is selected for use on both reference points.
- If the Gn peer- list is not configured, no peer list is selected for use on both reference points.

If a peer list is assigned to Gp but not to Gn, the Gp peer list will not be used when the Gn/Gp share an interface.

## 6.4 Configuring DB VM cloud-init in VMware LB mode NK package

To provide DB VM cloud-init configuration in VMware LB mode NK package, the following steps are required.

The released template package includes an ISO file that consists of two other files (an ovf-env.xml and a user-data file). The operator must customize these files and generate a new ISO file.

From a high-level perspective, the following steps must be performed:

Extract the contents of the ISO file.

Edit the user-data file which includes the DB VM cloud-init configuration, for example:

- users:

- - name: cmguser (name of the new user)
- plain\_text\_passwd: 'password' (password of the new user).

These credentials must also be defined in dbUsername/dbPassword extensions afterwards.

Generate a new keypair in case you want to have access to DB VM through ssh key:

- ssh\_authorized\_keys: ssh-rsa  
AAAAB3NzaC1yc2EAAAABIwAAAQEA04mOVCdi128upOSk97ZUPtkN  
7co/tWMe68e4zdN5JJtByTRmRk59GDwAGinofa0Z32Uqn3XEefpoU  
gQpY8dPb6sIEsdeXPnV5VT3h0oHnb6SBRNuqtLUTu38luPEM1HYPu  
wyOFDVFfHY0NxckYCPegWvvyJTCap+T4hkcRz5yYWRQkVWScF5/EI  
JtkAwps2RaB10myzNmj/VpNSTOTggCNPpy1hBfUTa2RiN4IbYlVv2m  
3Un1YPI3ZI7lkWffc1JwMual2NjroVDjoFI5yIX6cTcPAAWwUBnBrKrYj  
f0baNxOahnNjkef+sIJZu9xOfFDIpus+QnMnNlqXx/HdAnQ==  
cmguser@localhost

On the eth0 device, the corresponding management interface properties (gateway IP, primary DNS/secondary DNS IP, netmask, and fixed IP address) are set.

On the eth1 device, the external network interface properties (netmask and fixed IP) are set.

In NUM\_INSTANCES=2, define the number of Redis Server instances. Based on the value set, also create new JSON blocks respectively if needed and continue incrementing port numbers one by one.

In the addr attribute, the database [S:> external:S] network IP address must be defined. To secure the Redis database connections, Redis must bind to the loopback IP address, that is, 127.0.0.1:

```
"redis_servers": [  
    {  
        "addr": "127.0.0.1",  
        "port": 6379  
    },  
    {  
        "addr": "127.0.0.1",
```

```
        "port": 6380
    }
],
```

Set the number of user and control workers:

```
"user_workers": 5,
"ctrl_workers": 5,
```

Encode the contents of the user-data file as base64:

```
base64 -w 0 <path to user-data file>
```

Paste the base64 string in ovf-env.xml:

```
<Property oe:key="user-data" oe:value=''<base64_string>'>/>
```

Set the new password in ovf-env.xml:

```
<Property oe:key="password" oe:value="<new_password>">/>
```

Use an ISO generation tool (for example, genisoimage and mkisofs) for creating the new ISO file:

```
e.g genisoimage -o <name of the ISO file to be generated> -r
<complete path to the folder>
```

Upload the new ISO in the vCD catalog in the media or other section. Before deployment, modify the VNF extension 'dbMediaName' and set the name of the catalog item that contains that ISO.

## 6.5 LTMT integration to MantaRay

| Parameter        | Description                                    | Default Value    |
|------------------|------------------------------------------------|------------------|
| snmpAgentEnabled | whether SNMP Agent is enabled                  | FALSE            |
| snmpVersion      | the SNMP version to be used; valid value: 2, 3 | 2                |
| snmpManagerIP    | the IP address or hostname of the SNMP manager | no default value |
| snmpManagerPort  | the port of the SNMP manager                   | 162              |
| snmpv2Community  | community string for the SNMP v2               | public           |

| Parameter            | Description                                        | Default Value    |
|----------------------|----------------------------------------------------|------------------|
| snmpv3User           | username for the SNMP v3                           | no default value |
| snmpv3SecurityLevel  | security level of the SNMP v3                      | authPriv         |
| snmpv3AuthPassPhrase | password for authentication                        | no default value |
| snmpv3AuthProto      | protocol for authentication; valid value: sha, md5 | no default value |
| snmpv3PrivPassPhrase | password for privacy                               | no default value |
| snmpv3EngineID       | unique ID of the SNMP v3 engine                    | no default value |

The `snmpv3SecurityLevel` parameter can be set with one of the following values:

- `authpriv`: enables both authentication and privacy; `snmpv3AuthPassPhrase` and `snmpv3PrivPassPhrase` must be configured
- `authnopriv`: enables authentication only; the `snmpv3AuthPassPhrase` must be configured
- `noauthnopriv`: both authentication and privacy are disabled

For the `snmpv3EngineID` parameter, the `engineID` is unique for each instance of SNMP agent; for example: `0x80001F8880739BBE4396F8896800000000`.

- If enabled, the LTMT uses the configured ID.
- If disabled, the LTMT generates a new ID.

For the LTMT, Nokia recommends using the static mapping:

- csv file generated: `.1.3.6.1.2.1.88.2.0.1` – minor
- storage usage high: `.1.3.6.1.2.1.88.2.0.2` – major
- storage usage low: `.1.3.6.1.2.1.88.2.0.3` – info
- ltmt restart: `.1.3.6.1.6.3.1.1.5.2` – major

# 7 MD-CLI non-backward compatibility

Release 26.7.R1 compatible changes

| Old format | New format |
|------------|------------|
| N/A        | N/A        |

## 8 New, changed, and deleted functionalities

This chapter supplies the list of new, changed, and deleted functionalities in this release.

### 8.1 Release 26.7.R1

#### New functionalities

| Key     | 5G                                                                   |
|---------|----------------------------------------------------------------------|
| MG-4300 | Weight-based UPF selection via static definition in CPF              |
| MG-4770 | Change of EPS interworking status                                    |
| MG-4801 | Increased CUPS UP selection policy scale scenarios                   |
| MG-4879 | DDN handling enhancements/configuration options                      |
| MG-4902 | Support for combination of LTE traffic using Rf and 5G SA using Nchf |
| MG-4989 | NF-id-list limit to be increased to at least 50                      |

| Key     | Application assurance                  |
|---------|----------------------------------------|
| MG-4290 | Host/URL rewrite support in HTTP proxy |

| Key     | CMG appliance                                            |
|---------|----------------------------------------------------------|
| MG-4658 | Install NEDR agent on CMG-a servers                      |
| MG-4923 | BIOS/ILO/NIC based on SPP with IXR                       |
| MG-4929 | IPSec from RHEL level                                    |
| MG-4933 | CNF appliance based on CKD-CMG-c1 edge (limited support) |

| Key     | CMG appliance           |
|---------|-------------------------|
| MG-4952 | CMG-c1 hosting DNS64/44 |

| Key     | CMG                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------|
| N/A     | CMG CNFs support for continuous integration with Nokia Continuous Delivery solution                    |
| N/A     | CMG CNF: Disable auto-mounting ServiceAccountToken for pods that do not need API server access         |
| N/A     | system-pki directory encryption                                                                        |
| MG-3600 | Renewal, update, and revocation of TLS certificates for CMG (limited support)                          |
| MG-4588 | CMG supports MantaRay NM fast pass                                                                     |
| MG-4856 | Different origin host within the same interface/Diameter peer config in CMG                            |
| MG-4783 | APN/DNN configuration per APN-NI only                                                                  |
| MG-4814 | CMG support for flexible, interface-specific APN/DNN handling with optional mapping and format control |
| MG-4904 | Optimization of CMG PoD alignment on CCDs in AMD CPU architecture                                      |
| MG-4936 | CMG OAuth 2.0 for 5G service authorization – phase 1 extended                                          |
| MG-5025 | CaaS ISSU leftovers for CMG1:1                                                                         |

| Key     | ePDG                                                     |
|---------|----------------------------------------------------------|
| MG-3027 | ePDG PGW selection based on UE-Usage-Type                |
| MG-4515 | CMG support of GCM encryption/cipher algorithms for ePDG |

| Key     | LTMT                                       |
|---------|--------------------------------------------|
| N/A     | LTMT UE QUERY and Workflows enhancements   |
| MG-4735 | LTMT support on RHOCP 4.18 with IPv6 stack |
| MG-5114 | CCAAS LTMT 25.7 (VNF) support on VMware    |

| Key     | KPI-KCI                                                                                 |
|---------|-----------------------------------------------------------------------------------------|
| MG-4921 | N2 cause and failure counters                                                           |
| MG-4953 | KPIs for GTPv2-C Message Request Retransmits and Drops                                  |
| MG-4954 | SBI service statistics KPIs with HTTP with error status granularity                     |
| MG-4995 | Enhancements in reporting of Echo Requests/Responses (KPIPathManagementS1U/N3) from RAN |

| Key     | PCMD                  |
|---------|-----------------------|
| MG-4978 | CMG PCMD enhancements |

## Changed functionalities

| Key     | Description                                                                           |
|---------|---------------------------------------------------------------------------------------|
| MG-3762 | ICR support for L2TP (PPP regeneration) sessions<br>Available for production networks |

#### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

## 8.2 Release 26.3.R5

#### New functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

#### Changed functionalities

| Key            | LTMT           |
|----------------|----------------|
| Not applicable | Not applicable |

#### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

## 8.3 Release 26.3.R4-1

#### New functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

#### Changed functionalities

| Key            | LTMT           |
|----------------|----------------|
| Not applicable | Not applicable |

#### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

### 8.4 Release 26.3.R4

#### New functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

#### Changed functionalities

| Key            | LTMT           |
|----------------|----------------|
| Not applicable | Not applicable |

#### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

### 8.5 Release 26.3.R3

#### New functionalities

| Key     | All mobile gateways                                      |
|---------|----------------------------------------------------------|
| MG-4126 | Increased UP connections per CMG CP instance             |
| MG-4706 | UL-CL+PSA insertion together with campus L-PSA insertion |

| Key     | CMG appliance                                     |
|---------|---------------------------------------------------|
| MG-5059 | Dispatcher task for IPsec out-of-sequence packets |

| Key     | CMG                              |
|---------|----------------------------------|
| MG-4533 | Enhancements in SCC capabilities |

| Key     | RADIUS                                                         |
|---------|----------------------------------------------------------------|
| MG-2718 | Support for up to 20 RADIUS Framed-Route and Framed-IPv6-Route |

#### Changed functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

#### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

## 8.6 Release 26.3.R2

#### New functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

#### Changed functionalities

| Key            | LTMT           |
|----------------|----------------|
| Not applicable | Not applicable |

## Deleted functionalities

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>The feature, Detection of the number of devices connected to a tethering device (MG-3417), is identified with limitations that prevent the system from accurately counting the number of devices. Also, no feasible path can improve the accuracy to an acceptable level.</p> <p>As a result, this feature must not be used in a live network environment.</p> <p>Support for this feature will be deprecated and removed from upcoming CMG releases.</p> |

## 8.7 Release 26.3.R1

### New functionalities

| Key     | 5G                                        |
|---------|-------------------------------------------|
| MG-4716 | SBA peer failure handling improvements    |
| MG-4898 | SMF use of NRF query parameters — phase 2 |

| Key     | Application assurance                                       |
|---------|-------------------------------------------------------------|
| MG-3261 | Multi destination for session filter redirect method        |
| MG-4818 | Cflowd records for PCF created dedicated bearers / flows    |
| MG-4855 | CMG support for AES-CBC-256 encryption for HTTP redirection |

| Key     | Charging                                                                              |
|---------|---------------------------------------------------------------------------------------|
| MG-4510 | Reduction of TPS toward Gx by no sending QOS-Update in case of mobility from 4G to 3G |
| MG-4634 | Increase number of policy rules to 30000 (from 10000)                                 |

| Key     | CMG appliance                                       |
|---------|-----------------------------------------------------|
| MG-4494 | CMG-a2 DL345 192c (AMD Turin CPU)                   |
| MG-4928 | AIM based on service IP                             |
| MG-4976 | Use of TPM and secure boot for full disk encryption |
| MG-5027 | CMG-a2 96c v2                                       |

| Key     | CMG                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------|
| N/A     | CMG CNF support for continuous integration with Nokia Continuous Delivery (NCD 25.5) solution for 26.3 release |
| MG-4576 | Authenticated emergency calls in 4G                                                                            |
| MG-4585 | SSH strong keys added in CMG product                                                                           |
| MG-4737 | UPF IP address allocation enhancement — remaining call flows                                                   |

| Key     | ePDG                                                                      |
|---------|---------------------------------------------------------------------------|
| MG-4727 | Non-SIM tablet ePDG support                                               |
| MG-4755 | ePDG Intra ISSU is automated with CMG K8s Operator                        |
| MG-4847 | ePDG support of Diameter AVP "APN-Configuration" on Emergency call attach |
| MG-4887 | N40 n3gallocation in case of ePDG via SMF/PGW-C                           |

| Key     | KPI-KCI                                                |
|---------|--------------------------------------------------------|
| MG-4606 | KPIPathManagementS1U module enablement be configurable |

| Key     | LTMT                                                            |
|---------|-----------------------------------------------------------------|
| MG-4233 | LTMT integration in NCD and NCOM                                |
| MG-4598 | entraID for LTMT GUIs                                           |
| MG-4683 | LTMT LCM operations automation with NCOM Evo (Deploy/Terminate) |

| Key     | PCMD                         |
|---------|------------------------------|
| MG-4926 | CMG PCMD hardening — phase 2 |

### Changed functionalities

| Key     | Description                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MG-2489 | UP-based UE IP address/prefix allocation<br>Added a limitation about UP-based IP address allocation for 2G/3G access that is not supported in the legacy Gn/Gp GPRS architecture |
| MG-4200 | Support encryption and integrity protection in non-SBA 5G interfaces - N3, N6 - IPSec<br>Available for production networks                                                       |
| MG-4583 | P-CSCF discovery via DNS with fallback to static configuration<br>Available for production networks                                                                              |
| MG-4852 | Operator-defined 5QI NAS remapping<br>Available for production networks                                                                                                          |
| MG-4882 | Observability for Rating Group failures                                                                                                                                          |
| MG-4943 | 5GSA virtual DNN support (full support)<br>Available for production networks                                                                                                     |

### Deleted functionalities

| Key            | Description    |
|----------------|----------------|
| Not applicable | Not applicable |

## 9 Corrected issues

This chapter supplies the list of issues resolved in release 26.X.RX. Corrected issues from past releases are added in this section with the label **[NEW]**.

Severity is indicated as follows:

- 0 – Critical
- 1 – Failed
- 2 – Degraded
- 3 – Minor
- 4 – Enhancement

A resolution of a corrected issue is provided when available.

### 9.1 Corrected issues in release 26.7.R1

#### 9.1.1 5G

| Key               | Severity | Description                                                                                                                                                                                                                                                  |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [440304/04442424] | 3        | <b>Symptoms:</b> When HTTPS is used in N10 interface, the schema of nudm-uecm service is HTTP in the show mobile-gateway pdn pdu-session supi detail CLI command.                                                                                            |
| [444712/05305570] | 2        | <b>Symptoms:</b> When the policy decision and condition data error handling is indicated by PCF, the SMF sends a subsequent update that incorrectly reports failures after normal PCC rule removal.                                                          |
| [446126/05374605] | 2        | <b>Symptoms:</b> Under rare condition during 5G roaming and when CMG acts as V-SMF, CMG does not handle the PDU Session Modification from H-SMF in case the H-SMF vendor pushes a duplicate match-all rule for the session during default flow modification. |
| [446320/05557656] | 2        | <b>Symptoms:</b> User traffic is not blocked for the first service-delay-timer after receiving END_USER_SERVICE_REJECTED from PCF.                                                                                                                           |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446431/05602953] | 2        | <b>Symptoms:</b> Under rare condition when sbi-correlation-info-header is enabled under sba-server, active OAM may experience a memory leak when a handover is in progress but receives Create SM Context Request.                                                                                                                                                                           |
| [446731/05668946] | 2        | <b>Symptoms:</b> Under rare condition when mode-D and default fallback mode are enabled in the N11 interface, SMF may fail to send the EE Service Request for the location after the link flapping during the 5G call. As a result, the call is dropped.                                                                                                                                     |
| [447594]          | 2        | <b>Symptoms:</b> The udm-sdm-profile parameter get-sm-subscription-data exclude-plmn-id is used to exclude the serving PLMN ID as query parameter for roaming sessions. However, for api-version 2 the SDM subscription create still contains the PLMN ID, which may cause UDM to reject the request. If the PLMN ID is excluded from the GET request, it should also be excluded from POST. |

## 9.1.2 All mobile gateways

| Key                      | Severity | Description                                                                                                                                                                                                                                                                                  |
|--------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445447/05266570]        | 3        | <b>Symptoms:</b> In the context of a handover to 4G, in case the serving node provides an AMBR greater than 4294968 kbps and there is no reason to communicate with the PCRF, CMG sends an Update Bearer Request towards the serving node to incorrectly downgrade the AMBR to 4294968 kbps. |
| [445567/05448248/439935] | 3        | <b>Symptoms:</b> Under rare conditions, standby CPM may experience instability due to dispatcher module.                                                                                                                                                                                     |
| [446096/05568651]        | 2        | <b>Symptoms:</b> Under rare condition when APN-based UPF selection is enabled and default target profile is selected, SMF may fail to select last UPF peer in the entry list after back-to-back MG card instability.                                                                         |
| [446143/05572195]        | 2        | <b>Symptoms:</b> In rare scenarios, when RADIUS is used as NAT logging method, accounting requests may be missing UE information, such as IMSI, IMEISV, and MSISDN.                                                                                                                          |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446151/05564500] | 2        | <p><b>Symptoms:</b> When multiple UE session bonding support is enabled, CMG aggregates the AMBR values from all the UE sessions to calculate the multi-UE session AMBR. Therefore, CMG allows traffic to be exchanged in a rate higher than the expected one.</p> <p><b>Resolution:</b> Taking into account that all UE sessions of a multi-UE session share common AMBR values per traffic direction, CMG uses these common AMBR values instead of the aggregated ones to police the multi-UE session traffic.</p> |
| [446771/05652554] | 2        | <p><b>Symptoms:</b> Under certain conditions, SMF may send NRF NF-Management heartbeats at the operator-driven http2-profile connection-timer cadence instead of the NRF advertised heartbeat timer.</p>                                                                                                                                                                                                                                                                                                             |
| [446983/05677505] | 2        | <p><b>Symptoms:</b> If MMEs use both lowercase and uppercase format for the APN with the <code>hold-for-pdn</code> command configured for the ip-pool, CMG assigns a different IP when UE reattaches within the configured hold time.</p>                                                                                                                                                                                                                                                                            |
| [447161]          | 2        | <p><b>Symptoms:</b> Under rare condition when cam-optimization is enabled, MG card may experience instability issue in UPF, due to failing to install a UL-only dynamic rule with PortList.</p>                                                                                                                                                                                                                                                                                                                      |
| [447676/05757301] | 3        | <p><b>Symptoms:</b> When installing a policy rule base with ASO characteristic, the VS.plcyRuleBaseUsers counter is incremented twice for each session.</p>                                                                                                                                                                                                                                                                                                                                                          |

### 9.1.3 CMG

| Key               | Severity | Description                                                                                                                                     |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| [445703/05496817] | 2        | <p><b>Symptoms:</b> In rare scenario after running sometimes, MG card may experience instability due to signaling module memory corruption.</p> |
| [447076/05704464] | 2        | <p><b>Symptoms:</b> In rare scenarios, checks are required to prevent incorrect memory access.</p>                                              |

| Key                | Severity | Description                                                                                                                                                                                                                                                                   |
|--------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447414/05734562]  | 2        | <b>Symptoms:</b> During the CDR pull via SFTP, CMG reads the files and retains them in its Linux page cache for an extended period, even after the file transfer is completed.<br><br>After the fix, CMG will release the Linux page cache once the SFTP process is finished. |
| [447640/05802368]  | 2        | <b>Symptoms:</b> The PCMD function is invoked to update bearer information. During this process, the session is accessed, which turns out to be NULL. However, the function lacks a NULL guard check, resulting in a crash.                                                   |
| [447782/058333630] | 2        | <b>Symptoms:</b> The PCMD Audit function has resulted in a dangling pointer for Multi-PDN UEs. Accessing this pointer in subsequent procedures causes a crash.                                                                                                                |

#### 9.1.4 CUPS

| Key               | Severity | Description                                                                                                                                                                                                       |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446137/05561369] | 2        | <b>Symptoms:</b> Under rare conditions, when UE IP is non-compressible, the flow description is missing in the session report request from UPF after the application is detected because the 5-tuple is too long. |
| [446444/05618945] | 2        | <b>Symptoms:</b> Under rare condition after running in UPF ICR setup, the secondary node may have fewer sessions than the primary node because the synchronization may fail unexpectedly.                         |
| [446579/05648286] | 2        | <b>Symptoms:</b> Under rare condition during high subscriber load in the UPF, the IOM task may delay to respond to the PFCP Modify Request from the SMF side.                                                     |

#### 9.1.5 ISSU/ISSD

| Key      | Severity | Description                                                                                                                                                                                                                                                                                      |
|----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446184] | 2        | <b>Symptoms:</b> Total session loss observed during Intra CNF ISSU for SMF with Sx going over IPsec tunnel.<br><br>For manual operation instructions, see sections <i>Upgrading the LLBs (N:K)</i> and <i>Intra-CNF ISSU when Sx-N4 is over IPsec</i> in the <i>CMG Software Upgrade Guide</i> . |

## 9.1.6 KPI-KCI counters

| Key      | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445991] | 2        | <b>Symptoms:</b> After Wi-Fi to 5G handover, the smf5GcapableUEsIn4GSessions counter under KCIBearerManagementSmfCP-ISA is getting incorrectly increased.                                                                                                                                                                                                                                                                                                                                                                        |
| [446017] | 2        | <b>Symptoms:</b> If a 5G PDU session with dedicated QoS flow is created, a handover to 4G occurs, and then a MSCP switchover occurs, 4G counters numberOfIpv4PdnSessions, numberOfIpv6PdnSessions, numberOfIpv4Ipv6PdnSessions, and realApnDedicatedBearers under KCIBearerManagementApnCP-ISA may have incorrect value zero for that session.                                                                                                                                                                                   |
| [446018] | 2        | <b>Symptoms:</b> If a 4G PDN session is created and then an MSCP switchover occurs, counter numberOfAttachedSubscribers that under KCIBearerManagementApnCP-ISA and counter smf5GcapableUEsIn4GSessions under KCIBearerManagementSmfCP-ISA may have incorrect value zero for that session.                                                                                                                                                                                                                                       |
| [446216] | 1        | <b>Symptoms:</b> In a 5G PDU Session, the following 4G counters under the KCIBearerManagementApnCP-ISA may be incorrectly increased: <ul style="list-style-type: none"> <li>• numberOfAttachedSubscribers</li> <li>• numberOfBearers</li> <li>• numberOfHomers</li> <li>• numberOfIpv4LocalPoolSessions</li> <li>• numberOfIpv6LocalPoolSessions</li> <li>• numberOfIpv4v6LocalPoolSessions</li> <li>• realApnSessions</li> <li>• realApnIpv4Sessions</li> <li>• realApnIpv6Sessions</li> <li>• realApnIpv4v6Sessions</li> </ul> |
| [447283] | 2        | <b>Symptoms:</b> After 5G to Wi-Fi handover, counters numberOfIpv6WlanS2bSessions and realApnIpv6S2bSessions under KCIBearerManagementApn-ISA are not getting increased. At the same time, counters numberOfIpv6WlanS2aSessions, realApnIpv6S2aSessions are getting incorrectly increased.                                                                                                                                                                                                                                       |

| Key             | Severity | Description                                                                                                                                                                |
|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447325]        | 2        | <b>Symptoms:</b> After Wi-Fi to 5G handover, counters numberOfIpv6WlanS2bSessions and realApnIpv6S2bSessions under KCBearerManagementApn-ISA remain incorrectly increased. |
| [447430]        | 2        | <b>Symptoms:</b> After 5G to 4G handover and MG card switchover, counter numberOfSmfUEs under KCBearerManagementSmfCP-ISA is getting incorrectly increased.                |
| [447606/447353] | 2        | <b>Symptoms:</b> After 5G to Wi-Fi handover, counters numberOfSmfUEs and smf5GcapableUEsIn4GSessions under KCBearerManagementSmfCP-ISA are getting incorrectly increased.  |

## 9.1.7 PGW/GGSN

| Key                                | Severity | Description                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445336/05416159/445768/05383611]  | 2        | <b>Symptoms:</b> In non-CUPS IPv6 configuration, the devices behind the UE may experience reachability issues when the UE IPv6 address is allocated from a remote pool and an IPv6 framed route exists.                                                                                                                                                                        |
| [446472/05632294]                  | 3        | <b>Symptoms:</b> Under rare condition during attach when the session setup fails while waiting for response from UPF, SMF does not send the PFCP Delete Request to UPF after the session is released.                                                                                                                                                                          |
| [447425/05733734, 447456/05765268] | 2        | <b>Symptoms:</b> In the context of a Wi-Fi to 4G handover of a session with dedicated bearer(s), the serving node may send a Create Bearer Response carrying a piggybacked Modify Bearer Request. In such a case, CMG acting as PGW sends neither a Modify Bearer Response to the SGW nor a Delete Bearer Request to the ePDG, and the handover is not completed successfully. |
| [447630/05441828]                  | 2        | <b>Symptoms:</b> CMG does not track ULI changes in PGW CDRs, in case the ULI change is reported via a Change Notification Request.                                                                                                                                                                                                                                             |

## 9.2 Corrected issues in release 26.3.R5

### 9.2.1 5G

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445727]          | 2        | <b>Symptoms:</b> Incorrect PVI left for regular user when the WPS user leaves the call.                                                                                                                                                                                                                                                          |
| [445533]          | 3        | <b>Symptoms:</b> In rare cases where 5G to 4G handover merges with Br/QoS del, the Ded-Qos-Flow statistics are not updated properly.                                                                                                                                                                                                             |
| [446039]          | 2        | <b>Symptoms:</b> In rare cases, during a PCF deregistration, internal subscription cleanup on the SMF may remove the NRF subscription record without invalidating the matching PCF profile in its local cache. As a result, the SMF keeps sending N7 Policy-Create traffic to the deregistered PCF until the next NF Discovery clears the cache. |
| [446599/05620582] | 1        | <b>Symptoms:</b> In rare scenarios, when FAR(Forwarding Access Rule) is internally created or deleted for an SDF(Service Data Flow), inconsistencies are created on the standby card, leading to incorrect deletions of some of the sessions on failover of cards at UPFs.                                                                       |
| [446640]          | 1        | <b>Symptoms:</b> In rare scenario, when an incoming connection request is served during the PDN shutdown/no-shutdown, due to which the deletion (on PDN shutdown) did not clean up the server instance completely and later no-shut, causes to looking at the stale server entry and fail to perform (re)initialization as well.                 |
| [446320/05557656] | 2        | <b>Symptoms:</b> User traffic is not blocked for the first service-delay-timer after receiving END_USER_SERVICE_REJECTED from PCF.                                                                                                                                                                                                               |
| [446649]          | 2        | <b>Symptoms:</b> SMF sends 500 Internal Server in response to Service Request from UE post session deletion, due to stale sessions that are not cleaned up.                                                                                                                                                                                      |
| [446869]          | 1        | <b>Symptoms:</b> The command, <code>configure mobile-gateway profile nsmf-profile myNsmfProf sbi-srvc-operation create http-status-code xyz application-error</code> , can cause configuration loading failure.                                                                                                                                  |

## 9.2.2 All mobile gateways

| Key                                | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446096/05568651, 446009/05551077] | 2        | <b>Symptoms:</b> Under rare condition when APN-based UPF selection is enabled and default target profile is selected, SMF may fail to select last UPF peer in the entry list after back-to-back MG card instability.                                                                                                                                                                                                                                                                                      |
| [446214/05588058]                  | 1        | <b>Symptoms:</b> In a high subscriber churn scenario (frequent attach/detach/reattach), the UPF may become unstable due to DPI module exceptions.                                                                                                                                                                                                                                                                                                                                                         |
| [446151/05564500]                  | 2        | <b>Symptoms:</b> When multiple UE session bonding support is enabled, CMG aggregates the AMBR values from all the UE sessions to calculate the multi-UE session AMBR. Therefore, CMG allows traffic to be exchanged in a rate higher than the expected one.<br><b>Resolution:</b> Taking into account that all UE sessions of a multi-UE session share common AMBR values per traffic direction, CMG uses these common AMBR values instead of the aggregated ones to police the multi-UE session traffic. |
| [446351/05572126]                  | 2        | <b>Symptoms:</b> When a second PDN session is initiated using the same UE static IP from HSS, the existing (first) session is deleted and the new session is rejected. However, the cleanup of the first session occurs without notifying or invoking the policy and accounting systems, resulting in lingering Gy, Gx, and RADIUS sessions.                                                                                                                                                              |
| [446455]                           | 3        | <b>Symptoms:</b> IPSec tunnel recovery may require more than 20 seconds when active card is rebooted.                                                                                                                                                                                                                                                                                                                                                                                                     |
| [446501/05597204]                  | 1        | <b>Symptoms:</b> In LB-less UPF deployments, the GTPU endpoint IP (for example, S8u/S5u/N3/GNu/GPu) that is assigned to a UE session may not align with the MG group that serves this UE.                                                                                                                                                                                                                                                                                                                 |
| [446756]                           | 2        | <b>Symptoms:</b> While N26 handover to 5G is ongoing, MME sends Delete Session Request message without SI for the 4G session. As a result, stale 5G sessions remain on the system.                                                                                                                                                                                                                                                                                                                        |
| [444712/05305570]                  | 2        | <b>Symptoms:</b> When the policy decision and condition data error handling is indicated by PCF, the SMF sends a subsequent update that incorrectly reports failures after normal PCC rule removal.                                                                                                                                                                                                                                                                                                       |
| [446306/05589836]                  | 2        | <b>Symptoms:</b> DSCP marking for traffic on the Gx interface over SCTP may not follow the configured SMF IP-DSCP value.                                                                                                                                                                                                                                                                                                                                                                                  |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446471/05629684] | 2        | <p><b>Symptoms:</b> As per DTS440872 and DTS441359, a behavioral change was introduced in 25.7.R1 to ensure that the Delete Session Requests from invalid peers are explicitly rejected with cause 64, instead of being silently dropped.</p> <p><b>Resolution:</b> Added the configuration, configure mobile-gateway pdn &lt;gw-id&gt; supported-vendor custom-1, to suppress sending Context Not Found for requests from invalid peers, effectively allowing a revert to the legacy behavior if required.</p> |

### 9.2.3 CMG

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445703/05496817] | 2        | <p><b>Symptoms:</b> In rare scenario after running sometimes, MG card may experience instability due to signaling module memory corruption.</p>                                                                                                                                                                                                                                                                      |
| [446410/05613221] | 2        | <p><b>Symptoms:</b> In rare cases, SMF may not send memberkey:nodeFunctionality in NCHF Release message.</p>                                                                                                                                                                                                                                                                                                         |
| [446654]          | 1        | <p><b>Symptoms:</b> In relation to MG-4583 feature, the fallback mechanism may not work properly. In case of incorrect DNS responses or timeout, the information updated to the MG groups may also remove the PCSCF static address groups. This leaves an empty PCSCP server list. Attach responses do not include the PCO options for the PCSCF server address that prevents users to register for IMS service.</p> |
| [447076/05704464] | 2        | <p><b>Symptoms:</b> In rare scenarios, checks are required to prevent incorrect memory access.</p>                                                                                                                                                                                                                                                                                                                   |
| [447289/05727868] | 2        | <p><b>Symptoms:</b> For an HR PDU, PLMN-ID in SmfRegistration towards UDM has serving PLMN instead of the home PLMN.</p>                                                                                                                                                                                                                                                                                             |

## 9.2.4 CNF

| Key               | Severity | Description                                                                                                                                                                                                                                                        |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446127/05578183] | 2        | <b>Symptoms:</b> During ICR ISSU upgrade from 24.7RX to 25.7X , certain memory tables may get corrupted. As a result, the "discovered" flag is set (from no to yes) in http2client connections to NF Peers on the OAM causing a subsequent ICR switchover to fail. |
| [446557/05645775] | 0        | <b>Symptoms:</b> When both SMF ICR master and ICR slave reboot, Gy module does not purge only peer related data, but includes dynamic data that is associated with realm client configuration data.                                                                |

## 9.2.5 Charging

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446388/05112605] | 3        | <b>Symptoms:</b> When the ccfh-profile>cc-failure-handling command is configured with the result-code-3xxx retry-and-continue parameter, the UE attached with both primary and secondary OCS peers reply with CCA-I RC3004. After session-restore-retry timer expires, a new CCA-I is returned with MSCC RC4012 and FUI Redirect, CMG allows further UE traffic unlimitedly. |

## 9.2.6 CUPS

| Key               | Severity | Description                                                                                                                                                                                                   |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446137/05561369] | 2        | <b>Symptoms:</b> Under rare conditions, when UE IP is non-compressible, the flow description is missing in the session report request from UPF after the application detected due to the 5-tuple is too long. |
| [445734/05497414] | 1        | <b>Symptoms:</b> When the urr-id is used with a subscriber and, if it is removed and is attempted to be recreated, it may get stuck in cache.                                                                 |
| [446316/05477233] | 2        | <b>Symptoms:</b> PFCP-U tunnels (both APN PFCP-U and default/DD tunnels) do not come up after configuring a pdn no shutdown when the CMG is rebooted with the PDN in shutdown state.                          |

| Key               | Severity | Description                                                                                                                                                                                         |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446444/05618945] | 2        | <b>Symptoms:</b> Under rare condition after running sometimes in UPF ICR setup, the secondary node may have fewer sessions than the primary node because the synchronization may fail unexpectedly. |

### 9.2.7 ISSU/ISSD

| Key      | Severity | Description                                                                                                               |
|----------|----------|---------------------------------------------------------------------------------------------------------------------------|
| [446184] | 2        | <b>Symptoms:</b> CMG CNF ISSU: Total session loss observed during Intra CNF ISSU for SMF with Sx going over IPsec tunnel. |

### 9.2.8 Lawful interception

| Key               | Severity | Description                                                                                                                                                                   |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443034/05056319] | 2        | <b>Symptoms:</b> If a UPF ICR switchover occurs during the monitored UE is online when the UE detaches, the correlation session ID value in SMF LI target may not be cleared. |
| [447020]          | 1        | <b>Symptoms:</b> LI 3G/2G pdpCtxActivation encode function limits the content to 255 bytes. As a result, the IRI message is truncated and missing on the LI.                  |

### 9.2.9 PGW/GGSN

| Key                               | Severity | Description                                                                                                                                                                                               |
|-----------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445336/05416159/445768/05383611] | 2        | <b>Symptoms:</b> In non-CUPS IPv6 configuration, the devices behind the UE may experience reachability issues when the UE IPv6 address is allocated from a remote pool and there is an IPv6 framed route. |
| [445826/05474312]                 | 3        | <b>Symptoms:</b> CMG accepted message with corrupted IMSI, in certain scenarios and session is created with incorrect IMSI.                                                                               |
| [446472/05632294]                 | 3        | <b>Symptoms:</b> Under rare condition during attach when the session setup fails while waiting for response from UPF, SMF does not send the PFCP Delete Request to UPF after the session is released.     |

| Key      | Severity | Description                                                                                            |
|----------|----------|--------------------------------------------------------------------------------------------------------|
| [446484] | 2        | <b>Symptoms:</b> Interim-update is sent before a start message in case of Selective-RADIUS-accounting. |
| [446848] | 1        | <b>Symptoms:</b> Stale QFI is left on default bearer after WLAN<->LTE handover with ARP change.        |

## 9.3 Corrected issues in release 26.3.R4-1

### 9.3.1 5G

| Key      | Severity | Description                                                                                                                                                       |
|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447084] | 2        | FQDN-resolution-failed fallback for PCF/UDM/AMF/CHF caches requires further error handling to prevent inconsistencies that may lead to IOM-V-MG card instability. |

### 9.3.2 CMG

| Key               | Severity | Description                                                                                                                                                                                                             |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447033/05685499] | 1        | A narrow timing race between a subscriber detach and one or more concurrent in-flight attaches that share the same subscriber IP-prefix block, causes the UPF active CP to program the LB-VM downlink-forwarding entry. |

## 9.4 Corrected issues in release 26.3.R4

### 9.4.1 All mobile gateways

| Key             | Severity | Description                                                                                               |
|-----------------|----------|-----------------------------------------------------------------------------------------------------------|
| [446696/446834] | 1        | <b>Symptoms:</b> ICR and Intra-CNF ISSU may not be functional with some of the configurations in 26.3.R3. |

## 9.5 Corrected issues in release 26.3.R3

### 9.5.1 5G

| Key                | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [4445362/05391686] | 2        | <b>Symptoms:</b> Under rare condition during 5G to 4G handover, the voice call may drop while voice rule is created using a downlink-only dynamic rule; this creation collides with a handover to 4G.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [444863/05293803]  | 2        | <b>Symptoms:</b> The number of retransmissions for assign-ebi request may be less than the configured number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [445452/05444361]  | 1        | <b>Symptoms:</b> An internal timing issue during callflow failure may cause system instability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| [445513/05393805]  | 2        | <p><b>Symptoms:</b> When dns-snooping is configured, the snooped DNS responses may be incorrectly matched and charged under the installed policy rule with the arithmetically lowest precedence.</p> <p>When dns-snooping is configured, in case a policy-rule with UDP port 53 match criteria (protocol 17, remote-port eq 53) is installed, the DNS responses are expected to be charged under this rule. However, this is valid only for non-CUPS deployments.</p> <p>Also, in case there is no policy-rule with UDP port 53 match criteria installed, the DNS responses are expected to match to an internally created rule and be charged under rating-group 0. However, they are charged under the installed policy rule with the arithmetically lowest precedence, both in CUPS and non-CUPS deployments.</p> <p><b>Resolution:</b> In case a policy-rule with UDP port 53 match criteria is installed, the snooped DNS responses are charged under this rule. In case no such rule is installed, the DNS responses are charged under the fqdn-list dns-snooping rule that has the arithmetically lowest precedence.</p> |
| [446036/05551077]  | 2        | <b>Symptoms:</b> Exceeding internal JH transaction count in certain scenarios may cause system instability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [446126/05374605]  | 2        | <b>Symptoms:</b> Under rare condition during 5G roaming and when CMG acts as V-SMF, CMG does not handle the PDU Session Modification from H-SMF in case the H-SMF vendor pushes a duplicate match-all rule for the session during default flow modification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## 9.5.2 All mobile gateways

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443267/05094000] | 2        | <b>Symptoms:</b> BFD session cleanup on MG group shutdown is not clean.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| [443570/04778783] | 3        | <b>Symptoms:</b> SSH NetConf task may consume high CPU usage on standby CPM.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [444904/05360038] | 3        | <b>Symptoms:</b> Under rare condition during 4G to 3G handover with default QoS change, and when 3G SGSN Update PDP Context Request is received for an update, the dynamic policy rule may get corrupted, causing MG card instability.                                                                                                                                                                                                                                                                                     |
| [445447/05266570] | 3        | <b>Symptoms:</b> In the context of a handover to 4G, in case the serving node provides an AMBR greater than 4294968 kbps and there is no reason to communicate with the PCRF, CMG sends an Update Bearer Request towards the serving node to incorrectly downgrade the AMBR to 4294968 kbps.                                                                                                                                                                                                                               |
| [445627]          | 3        | <b>Symptoms:</b> After DNAI-based UPF reselection, PCO (Protocol Configuration Options) DNS values are looked-up using the DNN session instead of the APN/DNAI associated with the anchor UPF. The PFCP peer PCO configuration is keyed by peer IP APN; the anchor UPF has PCO under the DNAI APN. Lookup with DNN fails or does not match the anchor entry. Therefore, DNS is not taken from the UPF profile pco-option and remains from the PDN APN configuration (configure mobile-gateway pdn 1 apn <DNN> pco-option). |
| [445889]          | 1        | <b>Symptoms:</b> Server addresses are duplicated under redirect-server-policy obj when executing double CPM switchover.                                                                                                                                                                                                                                                                                                                                                                                                    |
| [446210/05576930] | 2        | <b>Symptoms:</b> After implementing mTLS for the SBI interface, TLS certs-profile are shown as invalidCertFile for client and server.                                                                                                                                                                                                                                                                                                                                                                                      |
| [446213/05577814] | 2        | <b>Symptoms:</b> Under rare conditions, the MG card may experience instability due to memory corruption.                                                                                                                                                                                                                                                                                                                                                                                                                   |

### 9.5.3 CMG

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444792/05343410] | 1        | <b>Symptoms:</b> Large TMD files may cause significant TS file size inflation, leading to potential storage issues and prolonged TS file processing times.                                                                                                                                                                                                                                                                                                                                                                                      |
| [446367]          | 2        | <b>Symptoms:</b> Parameterizing the cliConfig file name is not supported because it only uses <code>default mg.cfg</code> file name for day0 configuration as part of the CMG CNF deployment and upgrade.<br><b>Resolution:</b> Updated CMG Helm template files <code>Dut_ConfigMap.yaml</code> and <code>Card_ConfigMap.yaml</code> to parameterize the name of the configuration file that is executed on CMG CNF node as part of day0 configuration, by configuring the <code>config.configFile</code> <code>valyes.yaml</code> parameter.   |
| [446368]          | 2        | <b>Symptoms:</b> Configuring the TechSupport creation directory under <code>Dut_ConfigMap.yaml</code> is not supported as part of day0 configuration on CMG CNF.<br><b>Resolution:</b> Updated CMG Helm template file <code>Dut_ConfigMap.yaml</code> to parameterize location for TechSupport creation directory by setting the <code>techSupport.techSupportFileLoc</code> <code>values.yaml</code> parameter in a way that TechSupport files are created automatically under requested <code>cf1</code> directory when enabling techSupport. |
| [446370]          | 2        | <b>Symptoms:</b> Supporting to push LI mobile-gateway configuration via NetConf interface, with liuser and liprofile, created as part of Day0 configuration.<br><b>Resolution:</b> Updated CMG Helm template file <code>Dut_ConfigMap.yaml</code> by updating the liprofile to allow the <code>li li-mobile-gateway</code> command privileges. As a result, the LI mobile-gateway configuration push is supported via the NetConf interface with liuser and liprofile created as part of Day0 configuration.                                    |
| [446644]          | 2        | <b>Symptoms:</b> Configuring license as part of <code>values.yaml</code> parameter is not supported because it only configures license by reading <code>license.txt</code> file.<br><b>Resolution:</b> Updated CMG Helm template file <code>License_ConfigMap.yaml</code> to support license configuration via <code>values.yaml</code> by reading the <code>license</code> parameter to configure license on CMG CNF node.                                                                                                                     |

## 9.5.4 CNF

| Key      | Severity | Description                                                            |
|----------|----------|------------------------------------------------------------------------|
| [446049] | 2        | <b>Symptoms:</b> AA CPM does not update after CNF ISSU without reboot. |

## 9.5.5 Charging

| Key      | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445565] | 2        | <b>Symptoms:</b> When PCC is enabled and the RAT_CHANGE trigger is simultaneously enabled for both Nchf and Npcf interfaces during the LTE to WLAN handover, CMG accounting may miss the final usage snapshot for that transition and may not include all passed volumes in the immediate Nchf Update Request due to the RAT_CHANGE trigger. In this case, the missed volume snapshot is reported in the subsequent Ncfh Requests.                   |
| [445658] | 2        | <b>Symptoms:</b> For the MG-4887, with the keep-wlan-loc-info-on-partial-update option, when the Nchf USER_LOCATION_CHANGE trigger is not provided for the S2b/ePDG session and the n3gaLocation information is partially updated by Modify Bearer Request command (for example, UE local IP address + UE UDP Port only), then subsequent CHF events (for example, quota-exhausted CDR-U) may report the old IP address and port in n3gaLocation IE. |
| [445673] | 2        | <b>Symptoms:</b> For the MG-4360 and MG-4887, when the keep-wlan-loc-info-on-partial-update option is enabled and a new UE Local IP address is reported by either a Create Bearer Response or Update Bearer Response without an associated local port number, CMG may clear the previously stored port number (and the corresponding protocol identifier) and, therefore, may not report them in Nchf/Npcf interfaces n3gaLocation IE.               |

## 9.5.6 CUPS

| Key               | Severity | Description                                                                                                                                          |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444572/05306756] | 3        | <b>Symptoms:</b> UPF peer is selected even though it has been removed from up-peer-list (under the config>mobile>profile>pfcg>up-peer-list command). |

| Key               | Severity | Description                                                                                                                                                                                     |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445612/05482382] | 2        | <b>Symptoms:</b> Under rare condition when msm-stat-poll parameter is enabled, MG card may experience instability issue in UPF, during session deletion while cleaning quota.                   |
| [445846/05507601] | 3        | <b>Symptoms:</b> Under rare condition during attach, when the session setup fails while waiting the UPF response, SMF does not send the PFCP Delete Request to UPF after releasing the session. |
| [445932/05484558] | 2        | <b>Symptoms:</b> Under rare condition when handling exceptional packets in user plane, the MG card may experience instability.                                                                  |

### 9.5.7 ePDG

| Key      | Severity | Description                                                                                                            |
|----------|----------|------------------------------------------------------------------------------------------------------------------------|
| [445938] | 2        | <b>Symptoms:</b> The show mobile-gateway pdn pdu-session ran-gap-id ue-id detail command may cause system instability. |

### 9.5.8 KPI-KCI counters

| Key             | Severity | Description                                                                                                                                                                                                                                                                                   |
|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442231/445740] | 2        | <b>Symptoms:</b> In the KCI table (BearerManagementApn), numberOfPdnWithDedicatedBearers counter is incremented for 5G calls instead for 4G calls only.<br>Counter numberOfIpv6EutranPdnSessions is not cleared after handover to 5G in 4G-to-5G call scenarios and continues to be retained. |
| [445716/445206] | 1        | <b>Symptoms:</b> In a 5G PDU Session with dedicated QoS flow, 4G KCIs numberOfDedicatedBearers, realApnDedicatedBearers that belong to KCIBearerManagementApnCP-ISA are incorrectly increased.                                                                                                |

## 9.5.9 LTMT

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A               | N/A      | <p><b>Symptoms:</b> LTMT with local Timezone set, may apply incorrect filenames to selective Log CSV output files. In LTMT 26.3.R1 and LTMT 25.11.R2, Sel Log file names use local timezone time in the filenames.</p> <p>In LTMT, the names do not use local LTMT timezone correctly.</p>                                                                                             |
| [444392/05238941] | 2        | <p><b>Symptoms:</b> If LTTM VNF uses the TACAS+ LTMT feature to manage LTMT users centrally, a user with sudo rights cannot be created after SSH to LTTM-VM. Even though TACACS+ has created a user that belongs to both ltmtsupport and ltmtsudo authorization groups, the user cannot successfully execute sudo command inside LTMT-VM. This issue affects TACAS+ created users.</p> |

## 9.5.10 PGW/GGSN

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443769/04775380] | 2        | <p><b>Symptoms:</b> Under specific circumstances, when CMG receives two RAR messages for the same session from a PCRF at the same time, it may respond to a different PCRF for the second RAR, in case the alt-peer-rar command is configured under the config&gt;mobile&gt;pdn&gt;gx context and a diameter-peer-group-list is used for PCRF selection.</p> <p>During a session activation, a PCRF is initially selected. However, if the alt-peer-rar command is configured under the config&gt;mobile&gt;pdn&gt;gx context, CMG can also accept and handle messages from alternate PCRFs, during the session lifetime.</p> <p>In such a case, if CMG receives two RAR messages at the same time, it can send back a RAA with Result-Code: DIAMETER_OUT_OF_SPACE (4002) for the second RAR, to report that it currently cannot handle this message because a RAR is already in progress.</p> <p>In case the sender of these two RAR messages is an alternate PCRF, CMG may send the DIAMETER_OUT_OF_SPACE RAA for the second RAR message toward the initially selected PCRF, instead of the alternate PCRF.</p> <p>This issue occurs only if a diameter-peer-group-list is used for PCRF selection.</p> |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443806/05214077] | 2        | <b>Symptoms:</b> DNS redirect related messages require additional error handling.                                                                                                                                                                                                                                                                                                                                |
| [445502/05310166] | 2        | <b>Symptoms:</b> ACR-Interim may be sent with no SDCs when the offline time limit and the session idle timeout expire at the same time.                                                                                                                                                                                                                                                                          |
| [445701/05293627] | 2        | <b>Symptoms:</b> During an MG card restart, the PGW may not immediately and accurately determine whether the peer SGW is local or foreign at the time the card becomes operational. As a result, a locally connected peer may be incorrectly classified as FOREIGN on the MG card. When this misclassification occurs, CSRs destined for the MG card may be silently dropped, with no explicit error indication. |
| 445722/05468965   | 2        | <b>Symptoms:</b> When UP-based IP address allocation is enabled and hold-for-pdn is specified in the IP pool, LTE to Wi-Fi handover may fail with: Context Not Found.                                                                                                                                                                                                                                            |
| [445914/05526484] | 2        | <b>Symptoms:</b> When the session is in Assume positive, and the session restoration timer is enabled, CMG stores the events that occurred during that time. With multiple "Suspend" "Resume", and armed triggers of "RECOVERY_OF_BEARER, LOSS_OF_BEARER or both", the rule reports start building up causing system instability.                                                                                |
| [446056/05479332] | 2        | <b>Symptoms:</b> Generally, an offline-only RG is not affected by the Gy session going into CCFH. However, when the <i>Separate RGs for Offline</i> feature is enabled, the SDC must be marked when the session goes into CCFH and close the SDC containing the usage up to the point when the Gy session went into CCFH.                                                                                        |

## 9.6 Corrected issues in release 26.3.R2

### 9.6.1 5G

| Key      | Severity | Description                                                                                                                                                                                                                                                     |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443352] | 2        | <b>Symptoms:</b> The call-procedure-retry does not apply when the PGW sends a Create Bearer Request PCF-initiated for IMS call. As a result, the UE moves from LTE to Wi-Fi, and ePDG sends a Create Session Request before MME sends a Create Bearer Response. |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443907]          | 2        | <p><b>Symptoms:</b> Under rare conditions, the CNF LMG pod is unresponsive and shows high CPU utilization. As a result, service on this pod may be impacted.</p> <p><b>Resolution:</b> Changes implemented to prevent the scenarios that may cause this issue. Also, debugs are added to gather more information.</p>                                                                                                                                           |
| [445039]          | 3        | <p><b>Symptoms:</b> After system reboot or PDN reset, additional TCP sockets are created listening on 0.0.0.0 for SBI HTTP2 servers when a namf-comm parameter is configured without a system-interface.</p>                                                                                                                                                                                                                                                    |
| [445374/05363815] | 2        | <p><b>Symptoms:</b> When Model B is used for PCF communication and the initially selected PCF is unresponsive during the PDU session establishment procedure, failover is not executed to any of the other cached PCFs that is received from NRF during the discovery phase.</p> <p>Failover to cached PCF peers is currently supported only in case the initially selected PCF responds with an error code during the PDU session establishment procedure.</p> |
| [446048]          | 1        | <p><b>Symptoms:</b> Under certain JH message resend scenarios, memory inconsistency may occur leading to system instability.</p>                                                                                                                                                                                                                                                                                                                                |

## 9.6.2 All mobile gateways

| Key      | Severity | Description                                                                                                                                                                                                                                                                                                                                                                           |
|----------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446013] | 1        | <p><b>Symptoms</b> Concurrent call flows may cause PCMD inconsistency and system instability.</p>                                                                                                                                                                                                                                                                                     |
| [446044] | 0        | <p><b>Symptoms</b> The Multi destination for session filter redirect method (MG-3621) functionality is enabled by default. In certain cases, a DPI flow may enable the server-policy redirected feature. Subsequent fast-path processing treats those flows as server-policy redirected, corrupting statistics and potentially altering DNS and TCP behavior for unrelated flows.</p> |

### 9.6.3 KPI-KCI counters

| Key             | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445955]        | 1        | <p><b>Symptoms:</b> KPI counters from tables KPIBearerManagementSmfCP-ISA (edgeUpf and SSCMode related counters) and KPIBearerManagementSmfIUpfCP-ISA (I-UPF related counters) may be populated with zero value in few counters in alternative intervals.</p> <p>KPIBearerManagementSmfCP-ISA</p> <ul style="list-style-type: none"> <li>VS.pduSessEstablishWithSSCMode1</li> <li>VS.pduSessRelocationWithSSCMode1</li> <li>VS.edgeUpfPduEstablishSucc</li> <li>VS.edgeUpfPduEstablishFail</li> <li>VS.edgeUpfWithULCLInsertSucc</li> <li>VS.edgeUpfWithULCLInsertFail</li> </ul> <p>KPIBearerManagementSmfIUpfCP-ISA</p> <ul style="list-style-type: none"> <li>VS.pduSsnInsertIUpfSuccess</li> <li>VS.pduSsnInsertIUpfFail</li> <li>VS.pduSsnRemoveIUpfSuccess</li> <li>VS.pduSsnRemoveIUpfFail</li> <li>VS.pduSsnChangeIUpfSuccess</li> <li>VS.pduSsnChangeIUpfFail</li> <li>VS.pduSsnInsrtIUpfRmvISmfSuccess</li> <li>VS.pduSsnInsrtIUpfRmvISmfFail</li> <li>VS.pduSsnRmvIUpfInsrtISmfSuccess</li> <li>VS.pduSsnRmvIUpfInsrtISmfFail</li> </ul> |
| [442231/445740] | 2        | <p><b>Symptoms:</b> In the KCI table (BearerManagementApn), numberOfPdnWithDedicatedBearers counter is incremented for 5G calls instead for 4G calls only.</p> <p>Counter numberOfIpfv6EutranPdnSessions is not cleared after handover to 5G in 4G-to-5G call scenarios and continues to be retained.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## 9.7 Corrected issues in release 26.3.R1

### 9.7.1 5G

| Key                                    | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443244/04793211]                      | 2        | <p><b>Symptoms:</b> An additional retry to the retries configured in udm-sdm-profile, may appear.</p> <p><b>Resolution:</b> The attempts correspond to the configured number of retries in retry-count parameter on udm-sdm-profile.</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| [445164]                               | 1        | <p><b>Symptoms:</b> After establishing voice call, N3IWF is not receiving any ESP for RTP/RTCP from UE. The issue applies only to IPv6 PDN types in N3IWF.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [444043/05231167,<br>[444325/05256895] | 2        | <p><b>Symptoms:</b> Under rare condition, the standby MG card of the group may experience instability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| [440304/04442424]                      | 3        | <p><b>Symptoms:</b> When HTTPS is used in N10 interface, the schema of nudm-uecm service is HTTP in the show mobile-gateway pdn pdu-session supi detail CLI command.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [441959/04840970]                      | 2        | <p><b>Symptoms:</b> Under rare condition with I-Smf scenario when handling the invalid pduSessionCreate message from PSA-SMF/H-SMF, MG may experience instability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| [443315]                               | 2        | <p><b>Symptoms:</b> In case a PDU Session Establishment Request uses a quoted boundary parameter, the SMF rejects it and returns 400 Bad Request.</p> <p>According to the specifications, quotes can be used when defining the boundary. However, the SMF extracts the multipart boundary value from the content-type header without handling quotes. As a result, the surrounding quotes for quoted string values are not stripped. Therefore, the returned boundary may include quotes, causing mismatch with the multipart body's delimiter that is unquoted and exact. In this case, the SMF incorrectly rejects the request.</p> |
| [443608/05158016]                      | 3        | <p><b>Symptoms:</b> Under rare scenarios, due to stale entry build-up related to RGs in UPF, the system may experience instability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [443833/05060968]                      | 2        | <p><b>Symptoms:</b> When 5QI is greater than 15, SMF cannot include PPI in the N1N2MessageTransfer Request.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444052]          | 2        | <b>Symptoms:</b> Under rare scenario, when the regular non-emergency 5G session is created in an APN with emergency-profile configured, after an MG card switchover, the smfEmergencyPduSessions counter is increased for this session.                                                                                          |
| [444324/05277294] | 2        | <b>Symptoms:</b> Under rare condition, when 5G to 4G handover fails, the MG card may experience instability.                                                                                                                                                                                                                     |
| [444516/05252382] | 2        | <b>Symptoms:</b> If SMF has subscribed to UE location change event from AMF during PDU session establishment, and the AMF sends a UE location change notification before the PFCP session is established, the SMF proactively deletes the PDU session.                                                                           |
| [444706/05255784] | 3        | <b>Symptoms:</b> Under rare condition when mode-D is enabled and FQDN is used in smContextStatusUri, SMF may send the Status Notify message to invalid IP when the attach is rejected due to failure handling.                                                                                                                   |
| [444787/05347166] | 1        | <b>Symptoms:</b> SMF does not update nf-set-id in 3GPP-SBI-ROUTING-BINDING header in smpolicy update or delete.<br><b>Resolution:</b> The SMF includes the nf-set-id information received from the PCF within the 3gpp-sbi-routing-binding header when sending the request to the SCP.                                           |
| [444867/05300239] | 2        | <b>Symptoms:</b> The NF status is displayed as Registered under the tools dump mobile-gateway 1 nrf-client-cache nf-type udm command even if the UDM has deregistered.                                                                                                                                                           |
| [444946/05350327] | 3        | <b>Symptoms:</b> Under rare condition when handling policy update response with rules, the MG card may become non-responsive when there are many SDFs in the pended policy notify messages in parallel.                                                                                                                          |
| [445034]          | 3        | <b>Symptoms:</b> When Nchf Charging is used, the Charging Data Request message triggered upon Quota Holding Time expiration may not include usedUnitContainer information under certain conditions when service ID level reporting is in use.                                                                                    |
| [445071]          | 1        | <b>Symptoms:</b> When the udm-after-vdn command is enabled (MG-4943 feature) is enabled with apn-network-id type=public or without apn-network-id configured (virtual DNN sent by default), UDM notifications for session subscription changes fail with HTTP 404. This prevents applying new policies on the affected sessions. |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445278/05420431] | 3        | <b>Symptoms:</b> Under rare condition when the PDU does not exist in SMF but is in AMF, SMF responds with a 400 bad request toward the PFU Update Request from AMF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [445280/05394222] | 2        | <b>Symptoms:</b> During the SCP failover SMF adds one more leading slash in api-prefix while sending the message toward the other available SCP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [445351/05409452] | 2        | <b>Symptoms:</b> If the primary locally configured PCF sends a Create Response with a specific http-status-code, and the retry-and-terminate command is configured in the npcf-profile for this code, CMG may fail over to an unexpected PCF, instead of the next PCF in the same nf-id-list that is configured under the config>mobile>pdn>apn>pcf-selection context.<br>In case a specific nf-id-list is locally configured under the config>mobile>pdn>apn>pcf-selection context, this list must be used for PCF selection of the initial messages. Therefore, if a Create Response triggers a PCF failover, the next PCF must be selected from this list, based on the nf-priority of the nf-prof-id entries. However, under specific conditions, CMG may reselect a PCF that is configured under another nf-id-list in some cases. |
| [445356/05424753] | 3        | <b>Symptoms:</b> Under rare condition while freeing the location data during 5G to 4G (combined) handover, MG may experience instability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [445358]          | 1        | <b>Symptoms:</b> In a rare scenario, when several dynamic PCC rules have been installed on non-default QoS flow, and QoS or Charging data is modified for some of them while others remain the same, MG card restart may occur.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| [445579/05479189] | 2        | <b>Symptoms:</b> When mode-D and TLS is enabled in N40, SMF does not use the updated chfSet and FQDN for the subsequent request toward N40 after SCP changes the chfSet and FQDN in response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [445584/05442740] | 3        | <b>Symptoms:</b> When modifying ApiFullVersion of the Nsmf-pdusession service with CLI on the fly, SMF does not update the new version to NRF and the nf-profile under the tools dump context.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Key      | Severity | Description                                                                                                                                                                                                                                                                                                   |
|----------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445687] | 1        | <b>Symptoms:</b> In some cases, after MG switchover per APN level 4G sessions counter may be incorrectly populated (for example: ipv4/ipv6/ipv4v6 sessions, numberOfIpv4Ipv6PdnSessionsApn ). This is a counter issue and is auto-corrected when system runs through further handover call-flows (4G <-> 5G). |

## 9.7.2 All mobile gateways

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443940]          | 1        | <b>Symptoms:</b> In specific race conditions, when consecutive N26-based handovers fail, the SMF/PGW may not delete the existing session, leading to increased utilization of the IP pool.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| [443941/05207153] | 2        | <p><b>Symptoms:</b> Under specific conditions, MG may unexpectedly not send a RAT_CHANGE CCRU toward the PCRF in the context of a handover from 2G to 4G, although this event trigger has been previously armed by the PCRF.</p> <p>When a handover from 4G to 2G takes place, MG, after communicating with the PCRF, may be triggered to send an Update PDP context Request toward the SGSN. In such a case, if a Create Session Request for a handover back to 4G is received while waiting for the SGSN to respond back, MG may not send toward the PCRF a CCRU due to RAT_CHANGE that is expected to be sent, because this trigger is armed.</p> <p>This issue may appear in case the use-last-known-uli command is configured under the config&gt;mobile&gt;pdn&gt;charging context.</p> |
| [444102/05200622] | 2        | <b>Symptoms:</b> The GTP peer restart reason "unknown" in the SNMP trap and system log is inconsistent with the one defined in the MIB file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [444981]          | 2        | <b>Symptoms:</b> In rare scenario, when the Dynamic PCC rules are changed from bidirectional filter (no ports, using port list) to separate UPLINK and DOWNLINK filters where a local port is missing from UL filter, UPF may fail to update SDF, and the session is deleted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445238]          | 3        | <p><b>Symptoms:</b> When the charging-rule-unit under mobile-gateway&gt;profile&gt;policy-options MD-CLI command is enabled, an invalid charging method is selected for configurations involving a rating group and custom-online-sru. This incorrect configuration may cause validation errors.</p> <p><b>Resolution:</b> The system correctly implements the charging method for such charging-rule-unit configurations to be either online or both, preventing instability.</p> |
| [445177/05405510] | 1        | <p><b>Symptoms:</b> In specific scenarios, during 4G to 3G handover, changes in QoS/ARP may cause instability issues in the MG if obtaining the response from the UPF is delayed, along with the new updates for SGSN-initiated AMBR changes that are received prior to the response from the UPF.</p>                                                                                                                                                                             |
| [444253/05260814] | 3        | <p><b>Symptoms:</b> When fh-close-cdr-containers is enabled and the failure handling is triggered by negative response code from OCS during SMF request quota in first CCRU message, the closed PGW-CDR containers do not set dCCAContinueOngoingSession field correctly.</p>                                                                                                                                                                                                      |
| [445436]          | 1        | <p><b>Symptoms:</b> When QoS-Based Charging is enabled for 2G/3G/4G roaming sessions that use statically configured policy rules (Rulebase or static rule reference), the QBC containers are not included in Charging Data Request messages sent to CHF.</p>                                                                                                                                                                                                                       |
| [445789/05507053] | 3        | <p><b>Symptoms:</b> Under rare condition when session-timeout is configured as 24 hours, SMF may delete the session unexpectedly before session-timeout expires.</p>                                                                                                                                                                                                                                                                                                               |

### 9.7.3 Charging

| Key                | Severity | Description                                                                                                                 |
|--------------------|----------|-----------------------------------------------------------------------------------------------------------------------------|
| [444808, CMGN-394] | 2        | <p><b>Symptoms:</b> The roamingQBCInformation section is reported only in the CDReq-initial message for 2G/3G/4G users.</p> |

## 9.7.4 CMG appliance

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443662/05157332] | 2        | <p><b>Symptoms:</b> For CMG-a deployments, unnoticed mgmt user password expiration may disrupt system operation because licenses and system configuration cannot be retrieved. The default password expiration date is one year after a clean installation or upgrade.</p> <p><b>Resolution:</b> The expiration of mgmt user password is disabled by default for a clean installation or upgrade.</p> |

## 9.7.5 CMG

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442525/04835875] | 3        | <p><b>Symptoms:</b> In N:K deployments, call-insight trace may not capture the UL GTP packets.</p>                                                                                                                                                                                                                                                                                                                      |
| [443301]          | 4        | <p><b>Symptoms:</b> The DPDK memory pool monitoring is not supported to identify the reason for any DPDK memory resource exhaustion.</p> <p><b>Resolution:</b> Updated CMG Helm templates to DPDK memory pool monitoring via CMG Helm chart <code>values.yaml</code>.</p>                                                                                                                                               |
| [443796]          | 2        | <p><b>Symptoms:</b> Providing soft affinity between LOAM and LLB pods is not supported; only hard affinity support is supported between LOAM deployment and LLB statefulset pods to colocate on the same worker node.</p> <p><b>Resolution:</b> Updated CMG Helm templates to add support soft affinity between LOAM and LLB pods via CMG Helm chart <code>values.yaml</code>, to colocate them on the same worker.</p> |
| [443984/05223849] | 3        | <p><b>Symptoms:</b> Configuring the <code>admin save</code> command may fail for <code>nvsys.info</code> after multiple CPM switchovers because the CF information is missing from the file <code>uri</code>.</p>                                                                                                                                                                                                       |
| [444035]          | 1        | <p><b>Symptoms:</b> The repository for <code>td-agent-bit</code> that is used in the <code>fluentbit</code> sidecar, is obsolete.</p> <p><b>Resolution:</b> Updated <code>Dockerfile.fluentbit</code> to use the <code>fluent-bit</code> instead of the <code>td-agent-bit</code>, for getting the <code>fluent-bit</code> sidecar container.</p>                                                                       |

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444217/05255234] | 2        | <b>Symptoms:</b> Under rare condition after 4G combined session handover to 3G failure, the MG may experience instability due to memory corruption.                                                                                                                                                                                                                                                                                                                                                        |
| [444825]          | 2        | <b>Symptoms:</b> Under CMG Helm chart <code>StatefulSet.yaml</code> , incorrect labels are mapped for LOAM deployment pod, to support antiAffinity between LOAM and LLB pods.<br><b>Resolution:</b> CMG Helm templates add correct label under <code>antiAffinity</code> section of <code>StatefulSet.yaml</code> , supporting antiAffinity between LOAM and LLB pods.                                                                                                                                     |
| [444898]          | 2        | <b>Symptoms:</b> If the <code>config&gt;mobile&gt;pdn pdn&gt;apn apn&gt;ssc&gt;upf-reselection location</code> command is set for an existing 4G Idle session performing EPS to 5GS Mobility Registration to 5G, the SMF may not trigger PSA reselection.                                                                                                                                                                                                                                                  |
| [444504/CMGN-382] | 3        | <b>Symptoms:</b> In Kubernetes deployments, when a pod is terminated, a delay of up to three seconds may occur before the system fully recognizes the load balancer as unavailable. This may affect fail-over timing during planned or cause unplanned pod terminations.<br><b>Resolution:</b> The pod termination process immediately notifies the control plane when a load balancer is going down, eliminating the unnecessary delay and improving fail-over responsiveness in Kubernetes environments. |
| [445390]          | 2        | <b>Symptoms:</b> The PriorityClass resource can be configured only when enabling <code>priorityclass.values</code> under CMG Helm chart <code>values.yaml</code> .<br><b>Resolution:</b> Updated CMG helm templates to create the PriorityClass resource only after enabling <code>priorityclass.values</code> under CMG Helm chart <code>values.yaml</code> ; this creates PriorityClass resource with the configured value.                                                                              |
| [445412]          | 2        | <b>Symptoms:</b> When enabling external i/ports on MG card for ePDG CNF, the MG card may experience instability because the third MDA is configured as MG MDA instead of as m20-v.<br><b>Resolution:</b> The third MDA can be configured as m20-v under <code>smbios bootString</code> allowing the MG card to boot up correctly when the third MDA is enabled for ePDG/N3IWF deployment.                                                                                                                  |

## 9.7.6 CNF

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444165/05216052] | 3        | <b>Symptoms:</b> CMG CNF does not respond to the gNMI Subscription Request message.                                                                                                                                                                                                                                                                                                                                               |
| [444630/05327547] | 2        | <b>Symptoms:</b> SSH access from a CNF pod to the CNF OAM (BOF IP) may not work properly.                                                                                                                                                                                                                                                                                                                                         |
| [445554]          | 2        | <b>Symptoms:</b> When performing the <code>admin reboot</code> command, logs were not printed under <code>show log output</code> . As a result, no information on user details was available.<br><b>Resolution:</b> Added details on user and client that executed the <code>admin reboot</code> command under <code>show log output</code> . Also, <code>reserveMem</code> files are created under <code>/logs</code> directory. |

## 9.7.7 CUPS

| Key                | Severity | Description                                                                                                                                                                                                                                                                                                      |
|--------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442349, CMGN-259] | 3        | <b>Symptoms:</b> In the output of the SMF <code>show transport-statistics</code> command, URR IDs shown for policy rules do not match the URR IDs in PFCP Session Establishment signaling between UPF and SMF, and several policy rules with multiple AA charging groups appear with duplicated counter entries. |
| [444032/05226199]  | 2        | <b>Symptoms:</b> The force option with block on remote-ip-pool is not saved in the configuration. As a result, the configuration is rejected during system reboot.                                                                                                                                               |
| [444111/05240819]  | 2        | <b>Symptoms:</b> When the <code>show mobile-gateway pdn pdn-context default-session</code> commands or any derivative is executed, the MG cards may experience instability.                                                                                                                                      |
| [444418]           | 1        | <b>Symptoms:</b> In MG-4537, when both "radius via upf all" and duplicated accounting are enabled, RADIUS accounting on/off messages sent to the duplicated accounting server incorrectly include the APN-level NAS-ID instead of the UPF peer-specific NAS-ID.                                                  |
| [444795/05323946]  | 3        | <b>Symptoms:</b> In rare scenario, after all MG cards reboot due to external reason, UPF may report the PFCP load incorrectly, and the log enhancement receives more information about the failure scenario.                                                                                                     |

| Key                       | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [445486/05460693]         | 1        | <p><b>Symptoms:</b> The MG-VM may experience instability when receiving a Diameter code 4144 (DIAMETER_PENDING_TRANSACTION) from the PCRF without a Destination-Host AVP. This occurs when the CMG attempts to encode the missing Destination-Host AVP to re-transition the message.</p> <p><b>Workaround:</b> Add the following pending-transaction-timer/pending-transaction-retry-count with value 0 on the profile gx.</p> <pre>*A:Dut-C# admin display-config   match Nokia-PCF context all #----- --     mobile-gateway     profile         gx "Nokia-PCF"             fh-session-no-resp continue             fh-time-limit 60             pending-transaction-timer 0    &lt;-- ----- new line, to be added             pending-transaction-retry-count 0 &lt;----- new line, to be added             message ccr-i                 result-code 3002 continue             exit  *A:Dut-C#</pre> <p>If required, the customer can create a separate profile for the affected APN only, instead of using the Nokia-PCF profile that is used by other APNs.</p> |
| [445498/05460693, 437491] | 1        | <p><b>Symptoms:</b> Under rare conditions, synchronization may not be completed between CPM and MSCP regarding the functionalities of the PFCP peers. As a result, traffic endpoints of the sessions are not established properly, affecting the flow of the user-plane traffic.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 9.7.8 ePDG

| Key      | Severity | Description                                                                                                                                                       |
|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [430925] | 2        | <p><b>Symptoms:</b> Charging Data Request message may contain incorrect values of subscribedSessionAMBR after 4G to eDPG handover when CHF interface is used.</p> |

| Key               | Severity | Description                                                                                                                                                                                       |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443929/05189275] | 3        | <b>Symptoms:</b> CMG ePDG reports recurrent alarm messages in log after upgrade.                                                                                                                  |
| [444015/05213916] | 3        | <b>Symptoms:</b> The gateway check may incorrectly overlook the TWAG and ePDG. As a result, KCI Bearer Management Per Qci CP and KCI Bearer Management Per Qci CP counters are missing from ePDG. |

## 9.7.9 KPI-KCI counters

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444876]          | 1        | <b>Symptoms:</b> The KCI Dnn Nchf Fail Handling Continue counter under current Dnn Chf Fhc Sessions may temporarily show incorrect values after MG group switchover event. Existing sessions that have already moved from fail-open state to the 'failure handling continue' state are not reflected in the new active ones.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [444579/05272710] | 3        | <p><b>Symptoms:</b> CMG CNF may experience discrepancies between KPI-KCI and Prometheus metrics. This occurs when a client or peer uses different port numbers for different operations with the same peer, causing the sidecar to report multiple values (old and recent) for the same peer. Prometheus aggregates the data because the sidecar cannot identify the peer-port combination as a single entity. As a result, this may lead to double or triple counting.</p> <p><b>Resolution:</b> The port number must be removed from the label list for Prometheus metrics. This enables reporting per peer, rather than per peer per port. Therefore, only one entry is displayed for each peer, eliminating duplicate entries with different port numbers. This resolution applies to both LMG and LOAM.</p> |

## 9.7.10 ICR

| Key               | Severity | Description                                                                                                                                                                         |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442717/05004156] | 3        | <b>Symptoms:</b> Under rare condition in ICR setup while secondary node synchronizes with primary node during MC link flapping, the secondary node may experience card instability. |

### 9.7.11 Lawful interception

| Key      | Severity | Description                                                                                                               |
|----------|----------|---------------------------------------------------------------------------------------------------------------------------|
| [441704] | 1        | <b>Symptoms:</b> In case of non-CUPS deployment, the interception does not work for a s8hr-media target with bearer-id 0. |

### 9.7.12 LTMT

| Key | Severity | Description                                                                                                                                                                                                 |
|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | N/A      | <b>Symptoms:</b> When LTMT sends more than 2 billion (2147483648) PCMD records to selective log CSV files, LTMT stops writing. No more PCMD records/CSV file rows are written.                              |
| N/A | N/A      | <b>Symptoms:</b> When LTMT Workflow generates a large text file output (typically 100 MB) and links it from LTMT Main GUI Notification area, the user cannot open and download it by clicking to this link. |

### 9.7.13 PGW/GGSN

| Key               | Severity | Description                                                                                                                                                                                                                                                                                                         |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443367/05136587] | 1        | <b>Symptoms:</b> In case a proxy is used for GTPv1 messages, the first Create PDP Context Request is ignored, and an Echo request is initiated.                                                                                                                                                                     |
| [444362/05281836] | 2        | <b>Symptoms:</b> In certain re-attach scenario, GGSN/Gn builds two Gn/Gp-U traffic endpoints on the same realm (plus the SGi TE), causing malformed 3-TE (2 Gn + 1 SGi) Session Establishment Requests that the UPF rejects with PFCP cause 73 (rule creation/modification failure), and degrading PFCP SE success. |
| [444810]          | 2        | <b>Symptoms:</b> During roaming with LTE to Wi-Fi handover with ViLTE, QoS parameters may be handled differently between the visitor network and the home PCF, causing a call failure.                                                                                                                              |
| [444930]          | 2        | <b>Symptoms:</b> When using the info or info detail commands, the configuration status of the keep-wlan-loc-info-on-partial-update CLI command is not displayed in the configuration under the config>mobile>pdn>apn context.                                                                                       |

## 9.7.14 RADIUS

| Key               | Severity | Description                                                                     |
|-------------------|----------|---------------------------------------------------------------------------------|
| [445097/CMGN-403] | 2        | <b>Symptoms:</b> CMG does not perform accounting on Radius server when enabled. |

## 10 Deprecated functionality

This section describes the deprecated functionalities and the 7750 SR MG and CMG CLI commands that have been deprecated and/or replaced by new commands. Unless otherwise noted, deprecated commands are accepted during boot and execution, and are converted on upgrade. Also, the deprecated SR MG hardware is listed in this section.

Deprecated functionalities and commands in current release are added with the label **[NEW]**.

| Key               | Description                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| [438026]          | The <code>config&gt;mobile&gt;profile&gt;canary-test&gt;user-range&gt;ue-type</code> command has been deprecated.                             |
| [439228/04386717] | The <code>disable-skip-indicator</code> CLI parameter has been deprecated.                                                                    |
| [432726]          | The commands under the <code>config&gt;mobile&gt;profile&gt;qfi-mapping-profile&gt;static-mapping&gt;qfi</code> context have been deprecated. |

# 11 Limitations

This chapter supplies a list of known limitations in product design. Limitations from past releases are added in this section with the label **[NEW]**.

## 11.1 5G

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446671/05661457] | <p><b>Symptoms:</b> In rare circumstances, during the 5G to 4G handover, if the handover cancellation and the next handover CSReq is coming closely together, CMG may drop this CSReq because the cleanup of the resources is not completed.</p> <p><b>Workaround:</b> Configure a proper retry timer on MME side to let CMG handle the CSReq.</p>                                                                                                                                                                                                                            |
| [445461]          | <p><b>Symptoms:</b> When the UE is on a 5G VPLMN and charging ID is provided by the v-SMF, the gateway does not send Radius accounting messages.</p> <p><b>Workaround:</b> To support disconnect, LRDM must use charging ID to identify the session. This requires mapping the charging ID to the session ID. In the case of roaming when foreign charging ID is used, LRDM cannot make the association between the charging ID and the session ID. LRDM validates the charging ID when accounting request is received. If the validation fails, the request is rejected.</p> |
| [429814]          | <p><b>Symptoms:</b> Incorrect QFI ID is present in PFCP session report when DL data is received.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [430982]          | <p><b>Symptoms:</b> For the 5GC UPF support on 7750-MG (MG-3601) feature, the following applies: hardware UPF registration with NRF is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                             |
| [431401]          | <p><b>Symptoms:</b> The IPSec tunnel sync-up is not supported at SR level on CNF platform, therefore the ICR support for encryption and integrity protection over N4, Sx (PFCP, GTP-u) (MG-4040) feature does not currently support CNF configuration.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                            |
| [432085]          | <p><b>Symptoms:</b> In rare circumstances, when CPM switchover occurs on mobile gateway that is configured as "role primary" with the current operation role as slave, the IPSec tunnel on the gateway goes down. To avoid this, the mobile gateway configured with "role primary" must be made as master by performing a force switchover again to make primary as master and secondary as slave.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                |

| Key      | Description                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [430509] | <p><b>Symptoms:</b> In case of a PDU session with multiple QoS Flows, hardware UPF includes incorrect QFI ID and priority level in the PFCP Session Report message toward SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                  |
| [330589] | <p><b>Symptoms:</b> 5G systems using IP local pools of size /24 to /30 may experience attach failures if the number of MG groups in the SMF is similar or greater than the number of pools defined per DNN/APN.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> The Hold for PDN functionality does not work with virtual APNs.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                             |

## 11.2 Application assurance

| Key               | Description                                                                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A               | <p><b>[NEW] Symptoms:</b> HTTPS classification fails if the TLS packets arrive out-of-order and the SNI cannot be extracted from the Client Hello packet.</p> <p><b>Workaround:</b> Configure a session filter to drop out-of-order packets for the classification to work.</p> |
| N/A               | <p><b>[NEW] Symptoms:</b> TLS enrichment will not be performed if the Client Hello packet is segmented and arrives out of order.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                    |
| [429516/03044797] | <p><b>Symptoms:</b> In rare occasions when a session, bearer or RG is deleted MG may not have the related values to populate to the respective Cflowd record.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                       |
| N/A               | <p><b>Symptoms:</b> In CMG CNF deployments, the performance when TCP-O is enabled is lower than the performance in CMG VNF deployments.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                             |

| Key     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [91811] | <p><b>Symptoms:</b> The following features cannot be used when AA L7 classification is disabled:</p> <ul style="list-style-type: none"> <li>• HTTP/HTTPS header enrichment</li> <li>• application start/stop notification</li> <li>• HTTP redirect, HTTPS redirect, error-redirect, notification</li> <li>• DNS IP cache for rating group anti-fraud verification</li> <li>• ALG (flow wildcarding)</li> <li>• GTP/SCTP Firewall</li> <li>• URL filter (parental control via ICAP or local list)</li> <li>• tethering detection</li> </ul> <p>When more than one bearer per session is diverted to AA for analysis, only one bearer can have AA rules installed. The other bearers use 5-tuples.</p> <p>If Cflowd export from AA exceeds the rate that the CPM/CFM can process, Cflowd packets may be silently discarded.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A     | <p><b>Symptoms:</b> When deleting an application or an application group, statistics for the current accounting interval are lost. The workaround is to first remove all references to the application and application group, allow the accounting intervals to occur, and then delete the application or application group.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| N/A     | <p><b>Symptoms:</b> The CLI and SNMP commands relating to the following AA features are not supported:</p> <p>residential and VPN AA subscriber scale modes</p> <p>transit-ip-policy</p> <p>transit-prefix-policy</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### 11.3 All mobile gateways

| Key      | Description                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [438269] | <p><b>Symptoms:</b> Modifying the tapping-interface of the serving/active network-trace instance may not be effective depending on the changed mode.</p> <p><b>Workaround:</b> The tapping-interface can be modified when the network-trace instance is disabled.</p> |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [437461]          | <p><b>Symptoms:</b> CHASSIS #2104 tmnxEqLowSwitchFabricCap and CHASSIS #2105 tmnxEqLowSwitchFabricCapClear are a pair of events that reflect the status of the switch fabric capacity. The tmnxEqLowSwitchFabricCap is triggered when a change occurs, and tmnxEqLowSwitchFabricCapClear is triggered when it returns to normal. After CPM switchover, the state of the fabric is re-calculated and either tmnxEqLowSwitchFabricCapClear or tmnxEqLowSwitchFabricCap are generated based on the state of the fabric.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A               | <p><b>Symptoms:</b> The number of the supported HTTP2 client connection pairs is 1500.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| N/A               | <p><b>Symptoms:</b> In CMG, the limit for the UDMs is up to 255 peers.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [428508]          | <p><b>Symptoms:</b> When using the stat-rule-unit command inside the charging-rule-unit command, the configuration update of the charging-rule-unit with a new rating-group/service-id is not possible.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                              |
| [424142/02588630] | <p><b>Symptoms:</b> In N:K deployments, call-insight trace may not capture all packets in case of MG-VM switchover.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| [265413]          | <p><b>Symptoms:</b> In a dual-stack configuration environment, if gateway has S5 "default" configured with only IPv4 address, attach failures may occur. Nokia recommends having both IPv4 and IPv6 address configured on S5/S8/Gn/Gp "default" ref-point on a dual-stack config environment.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                        |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [373652/00840673] | <p><b>Symptoms:</b> The default charging group for Unreferenced Charging Groups Returned by AA (MG-2469/RFE-333679) does not apply to applications that meet all three conditions as follows:</p> <ul style="list-style-type: none"> <li>• they are detected on the first packet (so they match the AA inactive charging group on the first packet)</li> <li>• they are very short-lived (for example, one packet in each direction)</li> <li>• their RG gets in redirect status</li> </ul> <p>An example for this is DNS. The reason is that the DNS transaction is technically over in two packets. This is sooner than the time required by the internal mechanism to detect the RG in redirect status, and, therefore, the following packets are dropped.</p> <p><b>Workaround:</b> In such cases, the operator is recommended to configure valid AA rules matching DPI under the config&gt;mobile&gt;profile&gt;policy-options context. This limitation applies to both CUPS and non-CUPS deployments.</p> |
| [390839/01783591] | <p><b>Symptoms:</b> With the exception of SSG, the maximum number of default bearers per subscriber is eight.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [382798/01787040] | <p><b>Symptoms:</b> The maximum number of assigned filters per bearer is 15. This applies to both dedicated and default bearers.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| N/A               | <p><b>Symptoms:</b> The following limitations apply to the SGW with Local Breakout (MG-2771) feature.</p> <ul style="list-style-type: none"> <li>• Dual stack to single stack conversion is not correctly supported regarding the resource allocation in SGW. [359722]</li> <li>• Only one LBO bearer per session is supported</li> <li>• Multiple LBO policy rules with different QCI/ARP are not supported</li> <li>• Rating group on-the-fly change is not currently supported; it is allowed but with incorrect accounting</li> <li>• PGW-initiated dedicated bearer creation with same the QCI/ARP is not currently supported</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                           |
| N/A               | <p><b>Symptoms:</b> PCO fallback using configured user-name (PCO fallback using configured user-name (MG-2964)).</p> <p>The pco-fallback-acct command requires an APN shutdown/no shutdown for change to take effect.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [342743] | <p><b>Symptoms:</b> The value of the max-pdn-connections command under the config&gt;mobile&gt;pdn&gt;apn context must not be less than the available operational MG groups.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                              |
| [336750] | <p><b>Symptoms:</b> The following limitations apply to the number of rule bases that can be installed from the PCRF.</p> <ul style="list-style-type: none"> <li>• 40 policy rule bases per PDN session in total</li> <li>• 20 Charging-Rule-Base-Name AVPs in the Charging-Rule-Install AVP</li> <li>• 16 Charging-Rule-Install AVPs in a Gx message, such as CCAI</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [335620] | <p><b>Symptoms:</b> The configured value of the max-activations command may be exceeded on a short-term basis due to periodic internal reporting.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                         |
| [326044] | <p><b>Symptoms:</b> Installation of multiple rules with the same aa-charging-group command is supported only for predefined rules. The rules must have the same flow-gate-status command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                 |
| [305412] | <p><b>Symptoms:</b> Removing a Diameter peer from the PDN is allowed if it is part of an unassigned Diameter peer group list. However, this may have unpredicted results.</p> <p><b>Workaround:</b> A PDN shutdown is suggested before removing this peer.</p>                                                                                                                                                                        |
| N/A      | <p><b>Symptoms:</b> Result code 404 is returned if a query over T6c reference points refers to a session, where real and virtual APNs are using a different VRPN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                         |
| N/A      | <p><b>Symptoms:</b> Secondary RAT reports may be missing from the CDR for some sessions when received in DBResp.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                          |
| N/A      | <p><b>Symptoms:</b> For CUPS configuration, the feature Application Detection Control is extended to static PCC and optimization for VoIP is currently supported for lab trials only.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                     |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> A maximum of only four dedicated bearers can be activated per Diameter message. Requests for creation of more than four dedicated bearers in one message are rejected.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                  |
| N/A      | <p><b>Symptoms:</b> No dedicated bearers are triggered if the policy rule and policy rule base configuration using the <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;dedicated-bearers&gt;default-bearer-qci</code> command requires more than four immediate dedicated bearers.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                              |
| N/A      | <p><b>Symptoms:</b> For virtual APN selection using the IMEI-TAC + SVN of the IMEISV content in the MEI IE, the feature does not work if the MEI IE includes the IMEI and not the IMIESV.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                   |
| [273300] | <p><b>Symptoms:</b> It is not possible to configure a rule or rule base with an empty flow description for the <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;dedicated-bearers&gt;default-bearer-qci</code> command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                          |
| [275705] | <p><b>Symptoms:</b> The <code>show&gt;mobile&gt;pdn&gt;bearer-context</code> command does not show sessions with eight or more dedicated bearers on hardware platforms. Bearer information details can be accessed using the <code>show&gt;mobile&gt;pdn&gt;bearer-context imsi bearer</code> command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                      |
| [276454] | <p><b>Symptoms:</b> If the Locally Trigger Default Bearer Modification with QoS Update feature changes the default bearer QCI and/or ARP to apply or remove the QCI or ARP entry in the <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;dedicated-bearers&gt;default-bearer-qci</code> command for the APN session, the changes are not applied on the PCC rules or rule bases.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [269743] | <p><b>Symptoms:</b> For the dynamic PCC per RAT type feature, handovers between LTE and Trusted 3GPP Wi-Fi and Untrusted 3GPP Wi-Fi are not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                      |

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A              | <p><b>Symptoms:</b> When the disable-delayed-acct command is changed in the config&gt;mobile&gt;system context, some sessions may experience minor errors in charging data counts. Any delayed accounting reported after setting the disable-delayed-acct command is ignored, so some data flows with application identification still in progress may be undercharged. To avoid this possibility, make the configuration change when the gateway is in a shutdown state.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A              | <p><b>Symptoms:</b> While writing to the CF card, the ZFS compression feature may temporarily increase CPU usage by 10 to 15 percent.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                     |
| [229715]         | <p><b>Symptoms:</b> LI peer configurations must be deleted before deleting LI interfaces and VPRN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                        |
| [101485]         | <p>A ping from the UE to the Gi interface address of the PGW/GGSN is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                       |
| [111843]         | <p><b>Symptoms:</b> Interleaved TFT filter precedence across multiple PDP contexts is not supported if dynamic PCC is not enabled for the session.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                        |
| [136591]         | <p><b>Symptoms:</b> When the SGW receives a Resume message for a session in DDN-DAMP state, it does not change the state to IDLE.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                         |
| [123036]         | <p><b>Symptoms:</b> Modification of filters and/or precedence of match-all-traffic dynamic rule is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                         |
| [124287]         | <p><b>Symptoms:</b> Initiating a CPM and ISM-MG switchover at the same time is unsupported and may cause system instability. This limitation applies to both hardware as well as virtual systems.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                         |
| [124290, 124846] | <p><b>Symptoms:</b> Shutdown of the gateway during reconcile of the active and standby ISM-MG card is unsupported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                        |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| [130151] | <p><b>Symptoms:</b> Using the secondary address of an interface for MG signaling is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                              |
| [134557] | <p><b>Symptoms:</b> When duplicate policy rules and/or duplicate charging group names are active for a session, only one of them is applied to the traffic. The supported configuration is to have unique policy rule and charging group names.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                 |
| [136951] | <p><b>Symptoms:</b> If AA is configured and a persistent HTTP connection is used when the quota limit for the RG is reached HTTP redirection may not be invoked.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                |
| [146339] | <p><b>Symptoms:</b> In the BearerTrafficMSM KPI, the s1u5uDIIDropPackets measType counts uplink and downlink drops.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                             |
| [141559] | <p><b>Symptoms:</b> The maximum number of TFT filters that can be signaled for a dedicated bearer or secondary PDP context is 15. These TFT filters come from dynamic rules signaled by the PCRF and non-AA flow descriptions in policy rule units that apply.</p> <p><b>Workaround:</b> The total count of these TFT filters intended for the same dedicated bearer or secondary PDP context must not be more than 15.</p> |
| N/A      | <p><b>Symptoms:</b> For the IMS Dedicated Bearer Idle Timer Per QCI feature, configuration is in seconds per QCI but this feature has a one-minute granularity. For example, a value of 20 secs for a particular QCI is rounded to one minute and the timer is started.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                         |
| [147757] | <p><b>Symptoms:</b> Adding or removing the AA group configuration of an APN requires that no session has been active for that APN during the past ten minutes.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                  |
| [162975] | <p><b>Symptoms:</b> Dedicated bearers cannot be created if they contain a filter matching all the traffic.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                     |
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| [167209] | <p><b>Symptoms:</b> The Gx and S6b Diameter peers must be configured with IP addresses. There is currently no support for FQDN configuration for these two applications.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                            |
| [168341] | <p><b>Symptoms:</b> For the S5/Gn and S8/Gp interfaces, to apply a peer list to only one of the two reference points, the operator must assign an empty peer list to the other reference point. The same applies for multiple interfaces where peer list inheritance is active.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                     |
| [178901] | <p><b>Symptoms:</b> The ipv6-assigned-length optional parameter of an ipv6-prefix under ip-local-pool is not supported. The parameter is deprecated.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                |
| [182602] | <p><b>Symptoms:</b> On the S4 interface, in scenarios where the ULI contains SAI and RAI, only the SAI is delivered to the LIG.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                     |
| [182696] | <p><b>Symptoms:</b> NEMO control messages in the uplink direction, such as Registration Requests, are not accounted for in charging applications for the outer LTE/eHRPD/HRPD sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                             |
| [185096] | <p><b>Symptoms:</b> DNS-proxy redirect is not supported with fragmented packets.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                    |
| [185162] | <p><b>Symptoms:</b> DNS-proxy redirect rule base referenced by the Gx interface is not supported for Home Agent.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                    |
| [185688] | <p><b>Symptoms:</b> The Home Agent IP-in-IP Tunnel and Unicast Reverse Path Forwarding are two mutually exclusive features. They cannot operate under the same VPRN at the same time.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                               |
| [186821] | <p><b>Symptoms:</b> In the scenario, where a 100% LTE or 100% non-3GPP access traffic-hash profile is configured to force traffic onto a specific session based on an AA charging rule match, uplink statistics are counted against the bearer on the specified session, regardless of which session the uplink traffic arrived on.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                            |
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| [187279] | <p><b>Symptoms:</b> Executing a previously saved detailed configuration file (using the <code>admin&gt;save detail</code> command) may fail under certain circumstances.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                   |
| [187841] | <p><b>Symptoms:</b> Individual prefix rejection is not supported for a Mobile Router's Registration Request. A whole request is rejected if a prefix cannot be allocated.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                  |
| [187920] | <p><b>Symptoms:</b> Tuning of SCTP and Diameter retransmission parameters to the Mobile Core Network characteristics is needed so that on an SCTP path failure, the failure is detected and path switching occurs quickly and properly.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                    |
| [188825] | <p><b>Symptoms:</b> For systems that support DSL-bonding, if a traffic flow stops before the AA has fully classified the flow, the delayed accounting statistics may be counted on the incorrect session in both uplink/downlink directions. This only affects very short-lived flows.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                     |
| [188825] | <p><b>Symptoms:</b> If forcing traffic onto a specific session using a 100% LTE/non-3GPP hash-profile based on an AA Charging rule match, the initial few packets that pass before classification is complete are counted on the specified session regardless of the session the packets were initially hashed upon before classification.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [188954] | <p>Under a Pi reference-point configuration, the VPRN ID specified in the <code>ip-in-ip-tunnel vprn-id</code> command must match with the VPRN ID of the APN specified in the <code>peer-list peer-list-name</code> configuration.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                        |
| [191055] | <p><b>Symptoms:</b> Policy rule base change without shutting down the APN is not supported for a PRB that contains Application-Assurance policy rules.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                     |
| [191130] | <p><b>Symptoms:</b> When DNS packets with DSCP marking are redirected, preservation of the DSCP marking is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                  |

| Key      | Description                                                                                                                                                                                                                                                                                              |
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| [192235] | <p><b>Symptoms:</b> With the DNS-proxy redirect feature, Gy/Gx redirection for the session or SDF is not supported at the same time.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                         |
| [192571] | <p><b>Symptoms:</b> Gx redirection (flow gates status as HTTP redirect) on PCRF-referenced static DPI rules is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                |
| [197249] | <p><b>Symptoms:</b> The FQDN flow description feature is not supported for hybrid access bonding sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                   |
| [210361] | <p><b>Symptoms:</b> TFTP mode to fetch a certificate is not supported in the ePDG IPsec profile.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                             |
| [212677] | <p><b>Symptoms:</b> The traffic-hash command in the config&gt;mobile&gt;profile&gt;policy-options&gt;usage-monitoring-info&gt;monitoring-key context is a read-only field that defaults to defaultXds1HashProfile; no changes are permitted.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [213816] | <p><b>Symptoms:</b> The no ike-sa and no esp-sa commands in the config&gt;mobile&gt;profile&gt;ipsec-profile context do not remove the profile parameters. The no option was disabled in release 7.0.R6.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                     |
| [213968] | <p><b>Symptoms:</b> TCP-MSS adjust for ISM-MG-originated TCP connections is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                   |
| [219887] | <p>The PCMD is not currently reporting bulk records in the event of a path failure, mobile gateway shutdown, or clear APN command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                           |
| [220534] | <p><b>Symptoms:</b> Multiple PDN attaches to the same APN are rejected if DHCP is used to assign the IP address for the UE.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                  |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| [221254] | <p><b>Symptoms:</b> If a UE has multiple PDNs for the same APN with the IP hold for PDN option enabled, only the session that was deleted last for the APN retains the IP address hold entry.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                       |
| [221332] | <p><b>Symptoms:</b> Multiple PDNs to the same APN are not officially supported on S4 and S12 interfaces.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| [224561] | <p><b>Symptoms:</b> For TWAG systems, the AP peer MAC address for the <code>show&gt;mobile&gt;pdn&gt;ref-point-peers sww tunnel</code> command is only populated when the attachment includes DHCP Option-82.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                       |
| [226541] | <p><b>Symptoms:</b> The interface used for the SWw reference point must be in a different routing instance than the one used for the SWu and IP-in-IP MIPv4 reference points.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                       |
| [232161] | <p><b>Symptoms:</b> In a geo-redundant configuration, a policy rule/rule base is not removed from the configuration while sessions are referring to it.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                             |
| [231074] | <p><b>Symptoms:</b> The APN must be shut down while making changes to the <code>config&gt;mobile&gt;pdn&gt;apn&gt;ambr-policing</code> command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [236969] | <p><b>Symptoms:</b> For online charging to function properly in release 8.0, the system requires a Gy peer profile to be configured in at least one of the following contexts:</p> <pre>config&gt;mobile&gt;profile&gt;charging&gt;charging-profile config&gt;mobile&gt;pdn&gt;apn config&gt;mobile&gt;pdn&gt;gy</pre> <p>Pre-release 8.0 configurations that do not have a Gy peer profile specified in one of these contexts must be updated prior to upgrading to release 8.0. Failure to do so may result in online charging not being activated after upgrade.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [276152] | <p><b>Symptoms:</b> BOF save to compact flash drive formatted with ZFS file system is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [429394] | <p><b>Symptoms:</b> Installing or removing ADC dedicated bearer rules that share the same AA charging-group with a previously installed rule may not have any effect after an MSCP switchover.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                      |
| [268174] | <p><b>Symptoms:</b> If the PCRF installs the same rule or rule base name for a session that is already active for the session via <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;dedicated-bearers&gt;default-bearer-qci</code>, the PCRF-installed rule or rule base must be rejected, but it is not.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.4 Call trace

| Key      | Description                                                                                                                                                                                                                                                            |
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| [322032] | <p><b>Symptoms:</b> Inconsistencies between the counter user-plane messages and the total number of call-insight statistics have been noticed. The extra packets in the counter are router advertisements.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [309875] | <p><b>Symptoms:</b> Call trace is not supported with multiple-UE sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                 |
| N/A      | <p><b>Symptoms:</b> Call trace arguments ingress, egress, and statistics are not supported with NAT.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                       |
| N/A      | <p><b>Symptoms:</b> The path management messages Echo, Node Alive, and Redirection are not supported for Ga even if the Ga reference point is included in the CLI command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                 |
| N/A      | <p><b>Symptoms:</b> The Call Trace PCC Rule Metadata feature (MG-1742) is currently only supported for the combined CP-UP CMGs.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                            |

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| N/A              | <p><b>Symptoms:</b> Call trace UP statistics and PCAP file limitations</p> <p>The following limitations apply to per UE Call Trace for UP available for CMG and Cloud ePDG:</p> <ul style="list-style-type: none"> <li>• Debug data traffic statistics per UE are not supported for PMIP</li> <li>• Supports a maximum of 15 UEs for UP (release 10.0.R3 and later)</li> <li>• Supports a maximum of 16 UEs for UP (release 10.0.R1 to 10.0.R2)</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A              | <p><b>Symptoms:</b> For 7750 SR MG hardware-based systems, live-output must set in the all UEs call trace profile commands apart from the path-management.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                             |
| [232539, 234553] | <p><b>Symptoms:</b> Throughput of user-traffic call trace for the SGi interface is limited to 4Mbit/s for all UEs together or 270kbit/s per UE for 15 UEs. Packets exceeding this limit may fail to be captured.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                       |
| [238922]         | <p><b>Symptoms:</b> With Li Ulic v0 enabled for a UE, call trace is not supported on SGi reference point.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                              |

## 11.5 Charging

| Key      | Description                                                                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [436610] | <p><b>Symptoms:</b> For the MG-4523 feature, if OCS replies with FAILOVER_NOT_SUPPORTED, a failover is performed in case DRA is non-responsive. This allows performing a DRA failover keeping the same OCS.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.6 CMG

| Key | Description |
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| N/A | <p><b>Symptoms:</b> The current CMG release uses an SSH version that supports legacy host keys. When you try to establish an SSH connection using a recent SSH client version, the system responds with the following error:</p> <p>Unable to negotiate keys with no matching host key type found. Their offer: ssh-rsa,ssh-dss.</p> <p><b>Workaround:</b> Update the SSH client configuration to allow connecting to CMG SSH server.</p> <ol style="list-style-type: none"> <li>1. Create the following files. <ul style="list-style-type: none"> <li>• /etc/ssh/ssh-legacy.conf <pre>.include /etc/ssl/openssl.cnf [openssl_init] alg_section = evp_properties [evp_properties] rh-allow-sha1-signatures = yes</pre> </li> <li>• /etc/ssh/ssh_config <pre>Host *     KexAlgorithms +diffie-hellman-group14-sha1     HostKeyAlgorithms +ssh-rsa     PubkeyAcceptedKeyTypes +ssh-rsa     RequiredRSASize 1024</pre> </li> </ul> </li> <li>2. Execute the following command. <pre>export OPENSSL_CONF= /etc/ssh/ssh-legacy.conf</pre> </li> </ol> <p>This step can be applied either to the host running the SSH client or to the LOAM pod. If these modifications are applied to the LOAM pod, the change is not persistent.</p> <p>To make the change persistent, modify the <code>Deployment.yaml</code> file.</p> <pre>- /bin/sh - -c - &gt;-     echo -e ".include /etc/ssl/openssl.cnf" &gt;&gt; /etc/ssh/ssh-legacy.conf;     echo -e "[openssl_init]" &gt;&gt; /etc/ssh/ssh-legacy.conf;     echo -e "alg_section = evp_properties" &gt;&gt; /etc/ssh/ssh-legacy.conf;     echo -e "[evp_properties]" &gt;&gt; /etc/ssh/ssh-legacy.conf;     echo -e "rh-allow-sha1-signatures = yes" &gt;&gt; /etc/ssh/ssh-legacy.conf;     echo -e "Host *" &gt;&gt; /etc/ssh/ssh_config;     echo -e "    KexAlgorithms +diffie-hellman-group14-sha1" &gt;&gt; /etc/ssh/ssh_config;     echo -e "    HostKeyAlgorithms +ssh-rsa" &gt;&gt; /etc/ssh/ssh_config;     echo -e "    PubkeyAcceptedKeyTypes +ssh-rsa" &gt;&gt; /etc/ssh/ssh_config;     echo -e "    RequiredRSASize 1024" &gt;&gt; /etc/ssh/ssh_config; {{- if \$loamMgmtDict }} {{- if \$dualStackEnable }}</pre> |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | You can include these commands as part of the LOAM pod initContainer and set the environment variable under LOAM deployments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [441445] | <p><b>Symptoms:</b> VNFD package upgrade is blocked from 24.x.y.z to 25.3.1.2 and from 24.x.y.z to 25.7.1.1 with Openstack SrioVMgRedirect packages.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| N/A      | <p><b>Symptoms:</b> For the MG-2489 feature, the UP-based IP address allocation does not work with gx-before-aaa.</p> <p><b>Resolution:</b> The workaround call-flow where the PCRF/PCF transaction is performed before the AAA/RADIUS will be provided in a future release.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                       |
| [438722] | <p><b>Symptoms:</b> In case of on-the-fly modification of the <code>nf-id-list</code> command under the <code>sba-client</code> context, memory leak problems are caused.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                          |
| [437231] | <p><b>Symptoms:</b> CBAM upgrade LCM from any previous version to 24.7R1 (24.7.1.1 N:K OpenStack packages) fails at DB-VM health checks.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [436860] | <p><b>Symptoms:</b> SSH from CMG to DB-VM is not supported from release 24.7.R1 due to security enhanced in DB-VM when operating system is upgraded to RHEL 9.</p> <p><b>Workaround:</b> Nokia recommends using the way of directly SSH to DB VM or jumping from any other node in OAM network. Another workaround but not recommended by Nokia is to manually run the script in DB-VM <code>/usr/sbin/cmg-ssh-compatible.sh</code> as root user and reboot. However, the security of DB VM will be degraded.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                      |
| N/A      | <p><b>Symptoms:</b> For the <u>CMG support for network trace of all SBI signaling messages in an unencrypted format (MG-3497)</u>, the following limitations apply.</p> <ul style="list-style-type: none"> <li>vTap is supported only in CNF/VNF SMF and UPF. It is not supported in 7750-MG.</li> <li>vTap supports only communication with one packet broker.</li> </ul> <p>To establish communication with packet broker, only IP address can be configured. FQDN is not supported. Network Trace is automatically disabled when CPU is <math>\geq 80\%</math>.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| N/A | <p><b>Symptoms:</b> When the configuration mode is set to mixed, and an nf-prof-id is enabled with a UUID and a prioritized-address-list, attempting to remove the UUID or prioritized-address-list using the classic CLI may lead to a validation error. This can cause the Model-Drive CLI to lose access to the MCI database and become unavailable. [429863]</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                        |
| N/A | <p><b>Symptoms:</b> Changing the CMG management IP address (for SSH access to CMG CLI).</p> <p><b>Workaround:</b> Reinstall/upgrade the CMG where glusterfs storage class is used, perform the following steps:</p> <ol style="list-style-type: none"> <li>1. delete the <code>/mnt/glusterfs/smf-smf-cf1-a-volume/bof.cfg</code> and <code>/mnt/glusterfs/smf-smf-cf1-b-volume/bof.cfg</code> paths on any worker node of the CMG cluster, or</li> <li>2. delete the <code>cf1:/bof.cfg</code> path inside loam-a and loam-b pods</li> </ol> <p>Then, trigger a new Helm installation to instantiate the <code>bof.cfg</code> path with the updated CMG management IP address.</p> |
| N/A | <p><b>Symptoms:</b> The ICR is not currently supported for the IPv6 prefix delegation (MG-1329) feature.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| N/A | <p><b>Symptoms:</b> The NRF/NSSF support for TLS (limited support) (MG-2173) feature has the following limitations.</p> <ul style="list-style-type: none"> <li>• A limited set of ciphers is supported.</li> <li>• No counters/alarms have been implemented</li> <li>• No performance testing has been done.</li> <li>• TLS in outgoing connections (NRD being the client) is not supported.</li> </ul> <p>Additionally, the figure in customer documentation showing an egress gateway used for outgoing connections is under analysis and the architecture may change.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                |
| N/A | <p><b>Symptoms:</b> The Extended downlink buffering (MG-2370) feature has limited support to L2TP, GRE, MPLS, and any other Gi tunneling protocol.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A              | <p><b>Symptoms:</b> According to the Extended downlink buffering (MG-2370) feature, the max-pkt, max-pkt-size, total-buffer-capacity, and total-buffer-pkts commands must be appropriately configured during the upgrade procedure. Failing to configure the commands will set the default values in each case. For more information, refer to the <i>7750 SR MG and CMG CLI Reference Guide</i>.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A              | <p><b>Symptoms:</b> NFV3.2 VMware and NSX-T are not supported in 12.5.x and 12.7.1 CMG CBAM Templates.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                            |
| N/A              | <p><b>Symptoms:</b> SR-IOV in VMware is not supported in 12.1.1 CMG CBAM templates.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                               |
| N/A              | <p><b>Symptoms:</b> CBAM error handling retry is not supported in 12.1.1 CMG CBAM templates.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                      |
| [306436, 295980] | <p><b>Symptoms:</b> The Time of last boot timestamp for the VM may not consistently reflect the real last reboot events. The console output can be used as reference to determine the last reboot time.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                           |
| N/A              | <p><b>Symptoms:</b> The CMG templates packages for deployments with N:K redundancy do not support scale in operations for MG/LB VMs and any scale operations for DB-VM. CMG templates with versions &gt; 11.3.1 use only the new resource pool configuration model for scale out LCM Operation implementation.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                    |
| [294614]         | <p><b>Symptoms:</b> IPv4 or IPv6 reassembly is not supported with Internal-VM L3 network, so it is expected that the end-to-end MTU of the L3 network be consistent with the configured fabric MTU.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                               |
| N/A              | <p><b>Symptoms:</b> The internal fabric MTU value is configurable. However, reducing the MTU value can adversely affect the CMG call model. This functionality is only suitable for lab trials or demonstration purposes.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                         |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> Support of vMotion as part of the fully automated setting for DRS is not recommended for applications with low latency requirements and is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                              |
| N/A      | <p><b>Symptoms:</b> vMotion is supported for one MG-VM at a time. vMotion must be performed on a standby VM. If vMotion is executed on an active VM, it must be done at times when the user traffic is low (&lt;50 Mbps). vMotion under high traffic condition increases the duration of the vMotion and in combination with the FSWO timer, it may lead to unpredictable system behavior.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> If hyperthreading is enabled, the minimum vCPUs allowed per VM are listed as follows:<br/>LB-VM&gt; 12 vCPUs (minimum)<br/>MG-VM&gt; 16 vCPUs (minimum)<br/>Consult your Nokia technical support representative for the maximum number of vCPUs allowed per VM.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                        |
| [228302] | <p><b>Symptoms:</b> An MG Virtual Machine (MG-VM) may not be configured as isa-ip-reas-v.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                  |
| [269832] | <p><b>Symptoms:</b> On computers running RHEL 7.2 with OVS 2.5.0, CP traffic at high rates may cause instability at standby OAM during synchronization.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                    |

## 11.7 CNF

| Key                       | Description                                                                                                                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [435909]                  | <p><b>Symptoms:</b> Management access (SSH to Base/GRT loopback) via GRT route leaking is not supported in CNF deployments.</p> <p><b>Workaround:</b> For services (for example, X1) that may rely on GRT route leaking, use the MG-1324 feature (allow-ssh) to bypass reliance on GRT route leaking for management traffic.</p> |
| [438276/04218991, 432568] | <p><b>Symptoms:</b> The show system connections command displays an empty output.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                    |

| Key      | Description                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [430657] | <p><b>Symptoms:</b> In some circumstances, the ICR link between master and slave ePDGs can fluctuate for ePDG on CNF.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [429084] | <p><b>Symptoms:</b> In CNF the maximum number of supported VPRN per instance is 500.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                  |
| N/A      | <p><b>Symptoms:</b> File VI command is not supported in LMG.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                          |
| N/A      | <p><b>Symptoms:</b> LOAM management interface is only supported when using IPv4.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                      |

## 11.8 Combination gateways

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [345381] | <p><b>Symptoms:</b> The new PGW GTP-C node-level local overload throttling behavior, which only drops Create Session Requests (CSReq), is not recommended with the following:</p> <p>GTP-C overload reduction metrics reported in the Load Control Information (LCI) and Overload Control Information (OCI) Information Elements (IEs)</p> <p>On-the-fly changes to the <code>cpu-sampling-interval</code> command that are configured in the <code>config&gt;mobile&gt;pdn&gt;overload-control</code> context</p> <p>Furthermore, a combination of legacy and new PGW GTP-C node-level throttling mechanisms is not recommended for local-throttling levels that are configured with non-zero throttling-percent values.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> GTP-C overload reduction metrics reported in the Load Control Information (LCI) and Overload Control Information (OCI) Elements (IEs).</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| [345381] | <p><b>Symptoms:</b> On-the-fly changes to the <code>cpu-sampling-interval</code> command that are configured in the <code>config&gt;mobile&gt;pdn&gt;overload-control</code> context</p> <p>Furthermore, a combination of legacy and new PGW GTP-C node-level throttling mechanisms is not recommended for local-throttling levels that are configured with non-zero throttling-percent values.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                           |

| Key             | Description                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [253939]        | <p><b>Symptoms:</b> GTP load and overload-based throttling and reporting are not supported when multiple gateways are configured on multi-card combination gateways.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                            |
| [202887]        | <p><b>Symptoms:</b> Migrating from a two-card combination gateway configuration to a single-card combination gateway (SCCG) requires manual intervention.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                       |
| [203668]        | <p><b>Symptoms:</b> With the S4u default bearer, the S12 dedicated bearer is not supported in single-card combination configurations.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                           |
| [174321,174807] | <p><b>Symptoms:</b> A combined SGW/PGW Gateway in release 9.0: generate DDNs for blocked rating groups includes all the UE's bearers in a DDN, not just the bearer that is the target of the triggering traffic</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [221471]        | <p><b>Symptoms:</b> A combination gateway local interface cannot be configured to be used in a Diameter peer profile. The interface must be configured under the gateway for that application.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                  |
| [202101]        | <p><b>Symptoms:</b> Threshold parameters for combined SGW/PGW under <code>config&gt;mobile&gt;pdn&gt;thresholds</code> are not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                       |

## 11.9 CUPS

| Key      | Description                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [426771] | <p><b>Symptoms:</b> In CNF we support 500 VPRNs (four Interfaces + BGP peers). Exceeding this limit may cause instability and loading of the configuration may take longer.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                           |
| [429485] | <p><b>Symptoms:</b> If the <code>up-ip-allocation</code> command is enabled and a second session is allocated to the same UPF using the same static IP address, the SMF may show two active sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| [429701] | <p><b>Symptoms:</b> Bouncing/directing the session immediately to the correct MG group, which is controlled by the radius-auth-cache CLI command, the configuration does not work properly for UPF-based IP allocation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [427702] | <p><b>Symptoms:</b> In SMF configuration with 4000 Network Realms/APNs, the admin-tech support command and execution of a configuration file may take 10 to 20 minutes to complete.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [412460] | <p><b>Symptoms:</b> IP pool(s) must be configured under the config&gt;mobile&gt;profile&gt;pfc&gt;up-peer-list&gt;peer&gt;apn&gt;ip-pool-list context to see the 3GPP-IP-Address-Pool-Info AVP attribute in Access-Request RADIUS message.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| [379606] | <p><b>Symptoms:</b> Multiple AA charging groups configured on a single policy rule unit may not be used to trigger data-driven dedicated bearers.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| [346741] | <p><b>Symptoms:</b> Limited support for NB-IoT in CUPS, specific to the UP function in LB-less mode:</p> <p>NB-IoT currently supports S1-u interface, but not S11-U. Therefore, the s1u-interface-realm parameter, used for defining different interface realms for different S11-U and S1-u IP addresses, must not be configured.</p> <p>An NB-IoT-specific APN requires the SGI interface to be defined in LB mode, for example, a separate IP address pool defined in a non-MG group-specific VPRN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                            |
| N/A      | <p><b>Symptoms:</b> Continuous periodic Router Advertisement (RA) is not supported in this release of 4G CUPS if the RAs are handled by the UP function. At a successful initial PDN attach, four RAs are issued with one second interval and any RS received from the UE after the attach is always responded to with an RA. The Router Lifetime value in every RA will default to 9000 seconds.</p> <p><b>Workaround:</b> The limitation can be removed by configuring the RAs to be handled by the CP function (MG-2180). For more information, refer to section: <i>Configure UE ICMPv6 RS/RA and NS/NA Message Exchange in the Control plane and User Plane Configuration Guide</i>.</p> |
| N/A      | <p><b>Symptoms:</b> Scale and configuration restrictions apply for DHCP UE IP Address Allocation on CUPS.</p> <p><b>Workaround:</b> for information, contact your Nokia technical support representative.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Key      | Description                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [311767] | <p><b>Symptoms:</b> In CUPS deployments, the GGSN CP function does not take any action when it receives an error indication from the RNC.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                     |
| [313599] | <p><b>Symptoms:</b> Do not run a PFCP audit using the <code>tools&gt;perform&gt;mobile&gt;pdn&gt;pfcp-stale-session-audit-init</code> command at the same time as running a clear command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                    |
| [295180] | <p><b>Symptoms:</b> The mobile gateway will trigger PFCP Association State traps only for Up and Down states.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                 |
| N/A      | <p><b>Symptoms:</b> The gateway instance with CP and UP combined in a single logical instance is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                               |
| N/A      | <p><b>Symptoms:</b> Only the UP function is supported for the 7750 SR MG.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                     |
| N/A      | <p><b>Symptoms:</b> 7750 SR MG configured as CUPS UP supports 3GPP defined SGW-U, PGW-U, and GGSN-U functions for the 7750 SR MG ISM-MG-C cards only. Other hardware MSM-versions have not been evaluated.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                    |
| N/A      | <p><b>Symptoms:</b> As per 3GPP defined CUPS architecture, only GTP-C- and GTP-U-based interfaces are supported for bearer management. The GTP-C-based interfaces are supported on the CP (CMG), and the GTP-U-based interfaces are supported on the UP (CMG and 7750 SR MG).</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> 3GPP-defined CP and UP architecture for TDF/SSG (TDF-C and TDF-U) is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                       |
| [330095] | <p><b>Symptoms:</b> Configuration for 1:1 MG-VM using the resource pool and MG group.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                         |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> N:K warm redundant UP for delay tolerant applications such as Internet PDN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| [288108] | <p><b>Symptoms:</b> The network realm name associated with the APN/virtual router Instance on the UP peer in the context <code>config&gt;mobile&gt;pdn&gt;profile&gt;pfc&gt;up-peer-list&gt;peer</code> can be specified as an APN-IN and APN-OI up to 80 characters, however the APN-IN must not exceed 63 characters. This applies also to the UP function in the context <code>config&gt;mobile&gt;pdn&gt;apn</code>, where the APN name can be defined as APN-NI and APN-OI with up to 80 characters, however the APN-NI must not exceed 63 characters.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [273398] | <p><b>Symptoms:</b> The Sx reference point can only be named "default". No other name is allowed.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| N/A      | <p><b>Symptoms:</b> On S2a, CUPS is only supported for GTPv2, not for PMIP-based attaches.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| N/A      | <p><b>Symptoms:</b> GTP-U for data transport on Sx interface is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [429518] | <p><b>Symptoms:</b> The LI targets are left on the UPF and are not cleaned-up if an ISSU is performed from 23.2 to 23.5 or 23.8.</p> <p><b>Workaround:</b> The workaround is to remove and then reconfigure all the LI targets.</p>                                                                                                                                                                                                                                                                                                                                                                                     |
| N/A      | <p><b>Symptoms:</b> X2 from SX3LIF is required for packet data header information reporting (PDHIR) functionality. PDHIR is planned for a later release.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                    |

## 11.10 ePDG

| Key      | Description                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [441910] | <p><b>Symptoms:</b> WPS/MPS calls on EPDG are not treated on Par with Emergency Calls w.r.t IKE Redirection feature.</p> <p><b>Workaround :</b> No workaround available, yet.</p> |

| Key                                | Description                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [440538]                           | <p><b>Symptoms:</b> DHCP Marking for UL and DL Traffic by ePDG may differ in release 25.3 compared to R24.11.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                       |
| [438372]                           | <p><b>Symptoms:</b> When enabling the network trace feature (MG-3499) for non-SBI signaling, the ePDG is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                             |
| [430568]                           | <p><b>Symptoms:</b> The maximum number of registered UEs in N3IWF is 512 per MG group in case of VNF and 256 per MG group in case of CNF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                           |
| [428513/02899856, 428503/02916539] | <p><b>Symptoms:</b> Under High Attach Rate conditions, call trace buffers may get full and start dropping packets, the first packet required to identify IMSI for a call trace may be lost and not trace capture, and random packets may be lost on ongoing traces.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [263863]                           | <p><b>Symptoms:</b> IKE Redirect based on source IP address or by IDi realm may be counted by the counter ePDG attach failure in the show&gt;mobile&gt;pdn&gt;call-flow-stats summary command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                      |
| [248075]                           | <p><b>Symptoms:</b> For IPsec on SGI, the auth algorithm aes-xcbc and the encryption algorithm null are not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                              |
| [252076]                           | <p><b>Symptoms:</b> The IKE lockout configuration must be identical in every IPSec profile. The newest created or modified values will be applied to any pre-existing IPSec profiles.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                               |

## 11.11 GA

| Key      | Description                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [113989] | <p><b>Symptoms:</b> In a highly scaled environment, if an extremely large number of CDRs are written to local storage, the compact flash may become inaccessible.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [113563] | <p><b>Symptoms:</b> For SGW-CDRs, G-CDRs and eG-CDRs, under config&gt;mobile&gt;profile&gt;charging {epdg-charging   pgw-charging   sgw-charging}&gt;off-line&gt;partial-record-closure-trigger, the max-change-condition values outside the range of two (2) through 15 are not supported. If set to zero (0), the value of 15 will be used.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [137842] | <p><b>Symptoms:</b> In certain cases, a Ga CDR can be opened and closed in less than one second, resulting in a Service Data container reporting timeUsage 0.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                 |
| [186545] | <p><b>Symptoms:</b> In case of a bonded non-3GPP and LTE session, the charging records for the non-3GPP session may contain some additional values from the LTE session, such as MS-Timezone and User-Location-Information.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                   |

## 11.12 Gx

| Key      | Description                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [138835] | <p><b>Symptoms:</b> If more than eight Rule-Base-Name AVPs are included in one Charging-Rule-Install grouped AVP, the PGW/GGSN does not send a CCR-U message back to the PCRF to inform it about the PRBs that were not installed.</p> <p><b>Workaround:</b> Use more than one Charging-Rule-Install grouped AVP.</p> |
| N/A      | <p><b>Symptoms:</b> Time-based Gx usage monitoring for excess quota handling is not currently supported.</p> <p><b>Workaround:</b> Contact your Nokia technical support representative for information about the release in which this feature is expected to be supported.</p>                                       |

## 11.13 Gy

| Key      | Description                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [125438] | <p><b>Symptoms:</b> Online charging is only supported when the Reporting Level is set to rating group. Service ID reporting level is not supported for Online Charging.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [125520] | <p><b>Symptoms:</b> The number of rating groups that can be reported in one CCR message is limited to 16.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                |
| N/A      | <p><b>Symptoms:</b> Envelope Reporting must only be used in a laboratory environment.</p> <p><b>Workaround:</b> Contact your Nokia technical support representative for information about the release version of the production-quality feature.</p> |

## 11.14 Inter-chassis/VNF redundancy

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [317161] | <p><b>Symptoms:</b> The inter-chassis redundancy related output may not be accurate during the initial enabling of mc-complete-ue-sync.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                          |
| [137888] | <p><b>Symptoms:</b> The configure synchronization verification under the config&gt;redundancy&gt;multi-chassis&gt;mc-mobile&gt;peer command is currently disabled.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                               |
| [138071] | <p><b>Symptoms:</b> After an ICR switchover, the master gateway may send outdated time zone information in the accounting messages.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                              |
| [138072] | <p><b>Symptoms:</b> Under ICR configuration, the config&gt;mobile&gt;profile&gt;charging&gt;sgw-charging&gt;off-line&gt;partial-record-closure-trigger&gt;user-location-change command is not synchronized from the master gateway to the slave gateway. After an ICR switchover, the new master gateway may send outdated user location information in the accounting messages.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.15 IP reassembly

| Key      | Description                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [102478] | <p><b>Symptoms:</b> If IP reassembly is enabled on an interface, packets of size greater than 9K bytes are not supported on the reassembly interface.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                   |
| [102622] | <p><b>Symptoms:</b> Certain parameters in the ingress QoS policy applied on any packet destined to an interface where IP reassembly is enabled may not be honored.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                      |
| N/A      | <p><b>Symptoms:</b> The selective reassembly feature is not supported for the Home Agent. Both fragmented and non-fragmented packets destined to the Home Agent are sent to the reassembly ISAs if configured to do so.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.16 KPI-KCI counters

| Key      | Description                                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [357879] | <p><b>Symptoms:</b> The counters of the KCIBearerManagementApnCP-ISA module in UPF are not fully supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                           |
| [192988] | <p><b>Symptoms:</b> The numberOfDetachSuccessful KCI under BearerManagementApn may not increment properly in some PDP context deactivation scenarios.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.17 Lawful Interception

| Key      | Description                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> Enhancements for increased target database size and detailed target information are supported on CMG only.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                            |
| [147568] | <p><b>Symptoms:</b> The DF3 pkts tx counter under show&gt;li&gt;mobile&gt;df-peer context is not incremented if the ULIC version of the DF3 header is v0.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.18 Network Address Translation

| Key      | Description                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [317508] | <p><b>Symptoms:</b> After a CPM switchover, the system may not include the subscriber information from NAT records.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                   |
| [207712] | <p><b>Symptoms:</b> When the ISA bb MDA is provisioned in an MG slot, the CLI shows zero traffic statistics for Internal NAT ISA interfaces (_tmnx_nat-outside and _tmnx_nat-inside).</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.19 PGW/GGSN

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [405596/02366521] | <p><b>Symptoms:</b> Multiple rules referencing same AA charging group does not handle simple AA rules on default bearer with different charging parameters only.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                 |
| [429500]          | <p><b>Symptoms:</b> In case of an OCS installed policy-rule-base overriding the PCRF installed policy-rule-base, in some scenarios an additional CCR-U may be seen toward the OCS during the session establishment.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                              |
| [401766/02277754] | <p><b>Symptoms:</b> Up to 4000 user-range entries are allowed for virtual-apn-selection. In case more are added, a relevant error log is displayed in CLI output such as the following: MINOR: IMSI #2890 Reached the maximum number of allowed entries-max user-range 4000.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                     |
| [398952/02225612] | <p><b>Symptoms:</b> The qci-level command under config&gt;mobile&gt;pdn&gt;rf context is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                          |
| N/A               | <p><b>Symptoms:</b> On-the-fly changes to the enable-gtp-overload-control, s8, or validity-time commands under the config&gt;mobile&gt;pdn&gt;gtp-overload-control context, as well as to the critical, major, or minor commands under the config&gt;mobile&gt;pdn&gt;gtp-overload-control&gt;overload-reduction context, are allowed.</p> <p><b>Workaround:</b> A PDN shutdown is suggested to avoid undesired results.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [175753] | <p><b>Symptoms:</b> 3GPP2-BSID referenced by the PCRF in RAR is not supported by the PGW.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                            |
| N/A      | <p><b>Symptoms:</b> The virtual APN selection feature that supports IMEISV TAC + SVN, to override the APN received in GTP create messages, allowing reassignment to the virtual APN of choice based on selection priority and user range using the IMEI-TAC + SVN of the IMEISV content in the MEI IE will not work if the MEI IE includes IMEI and not IMIESV.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                      |
| N/A      | <p><b>Symptoms:</b> The following features must only be used in a laboratory environment. Contact your Nokia technical support representative for information about the release version of the production-quality features.</p> <ul style="list-style-type: none"> <li>• Connection over GTP-based s2a with WLAN TWAG nodes</li> <li>• Calculation of optimized GBR/MBR values toward the RAN</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [267257] | <p><b>Symptoms:</b> OAM UE ping packet sizes over 1000 bytes are not supported in the current release. Packet size is determined by the size parameter of the <code>oam ue-ping</code> command, with a default size of 56. The CLI allows larger packet sizes, but the value must be limited to 1000 or less.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                        |
| [266902] | <p><b>Symptoms:</b> OAM UE ping is not currently supported for sessions with bonded IP addresses for hybrid access.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                  |
| [271269] | <p><b>Symptoms:</b> For L2TP/PPP Regeneration, the PGW/GGSN LAC does not support sending LCP Echo Requests but will reply to them from the LNS.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                      |

## 11.20 PCMD

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447562] | <p><b>[NEW] Symptoms:</b> During a PSA reselection with configure mobile-gateway pdn 1 delay-network-timer set to 0, the PCMD object opened for the PDU Session Release procedure (E_PCMD_PROC_SMF_PDU_SESS_REL) may not be closed cleanly and is left stale. The object is subsequently detected as invalid and reclaimed by the periodic PCMD audit. The session or call processing are not impacted.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> CDATA records are reported from the GW-C/SMF, but the data volumes reported are not aligned to the PCMD reporting period, so are not reliable, for example for throughput calculations. This is because the Sx volume reporting is asynchronous and the Sx interface introduces a random delay in the reporting.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                    |

## 11.21 PFCP

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>Symptoms:</b> The following limitations apply to PFCP with Framed-Route Support:</p> <ul style="list-style-type: none"> <li>require that the RADIUS server does not assign the same Framed Route to more than one UE / PDN session</li> <li>require that the RADIUS server does not assign any overlapping Framed Routes to different UEs / PDN Sessions</li> <li>require that the UP function and the CP function must be upgraded to 11.0.R10 before the Framed Route can be used (release 11.0.R5 for LB-less)</li> <li>require that the UP function is supported in 1:1 redundancy mode</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.22 Non-IP PDN SGi tunneling

| Key | Description                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>Symptoms:</b> Non-IP PDN Type is only supported on ISM-MG-C and CMG.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key | Description                                                                                                                                                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>Symptoms:</b> Nokia recommends not having multiple sessions for a subscriber using non-IP PDN Type.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                 |
| N/A | <p><b>Symptoms:</b> Some services such as policies and external IP allocation are not supported for non-IP PDN Type.</p> <p><b>Workaround:</b> No workaround available, yet.</p>      |
| N/A | <p><b>Symptoms:</b> IP fragmentation/reassembly for non-IP data on SGi tunnels is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                          |
| N/A | <p><b>Symptoms:</b> PCO option for Link MTU is not supported for non-IP PDN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                              |
| N/A | <p><b>Symptoms:</b> The Hold-for-PDN feature for the local IP pools is not supported in this release for non-IP sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 11.23 SSG/TDF

| Key      | Description                                                                                                                                                                                                                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [281279] | <p><b>Symptoms:</b> The Acct-Session-Id sent in the feed message (Pre-CoA Acct-Start Message) of the subscriber's second subflow may be reused after multiple ISM-MG failovers.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                              |
| [259175] | <p><b>Symptoms:</b> It is recommended to limit the number of subscriber sessions to eight when installing DPI rules. If there are more than eight sessions belonging to a subscriber, the data traffic from the 9th session onward is not identified on DPI.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| N/A      | <p><b>Symptoms:</b> The functionality is for demonstration purposes and must only be used in a laboratory environment. Contact your Nokia technical support representative for information about the release version of the production-quality feature.</p> <p><b>Workaround:</b> No workaround available, yet.</p>      |

## 11.24 TWAG

| Key | Description                                                                                                                                                                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <b>Symptoms:</b> The Layer 2 VLAN Connection for WLAN APs feature is supported only for a CMG TWAG.<br><b>Workaround:</b> No workaround available, yet.                                                                                                                                                   |
| N/A | <b>Symptoms:</b> The virtual APN feature is not used by the TWAG, so virtual APN configuration commands such as the following must not be used to configure TWAG.<br>config>mobile>pdn>apn>virtual-apn<br>config>mobile>pdn>apn>virtual-apn-selection<br><b>Workaround:</b> No workaround available, yet. |

## 12 Known issues

This chapter supplies a list of known open faults in this release. If available, the workarounds are described. Known issues from past releases are added in this section with the label **[NEW]**.

### 12.1 5G

| Key             | Description                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [448169]        | <p><b>[NEW] Symptoms:</b> 5G attach fails if configuring a virtual DNN on real APN via IMSI user-range selection with TAI/LAI lists. Local UDM nf-id-list is each provisioned with one nf-profile-id bound to virtual APN via udm-selection nf-selection-profile. NRF discovery is blocked as UDM must be resolved from the local list.</p> <p><b>Workaround:</b> Configure nf-id-lists inside sba-client-services udm-client.</p> |
| [448032]        | <p><b>[NEW] Symptoms:</b> Under rare circumstances when 5G roaming call flows (between V-SMF and H-SMF) take place, the counter numberOfDedicatedBearers under KCIBearerManagementApnCPISA can get a negative value.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                   |
| [446869]        | <p><b>Symptoms:</b> The command, configure mobile-gateway profile nsmf-profile myNsmfProf sbi-srv-c-operation create http-status-code xyz application-error, can cause configuration loading failure.</p>                                                                                                                                                                                                                          |
| [444264]        | <p><b>Symptoms:</b> When configuring the preferred-locality parameter with values including special characters (for example, preferred-locality="Abc123()45"), the requester URO Query parameter is not applied.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                       |
| [444129/444100] | <p><b>Symptoms:</b> When the no max-peer-failover command is set to value 3, system instability may occur. The default value is incorrectly 3 as in following example:</p> <pre>config&gt;mobile&gt;profile&gt;npcf&gt;sbi-srv-c-operation\$ max-peer-failover - max-peer-failover &lt;value&gt; - no max-peer-failover &lt;value&gt; : [0..3] - Default: 3</pre> <p><b>Workaround:</b> Change value to 2.</p>                     |

| Key      | Description                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443767] | <p><b>Symptoms:</b> Node-selection can be removed even if group-label is still assigned/referenced to a UPF peer. This can lead to a CPM assertion after double CPM failover.</p> <p><b>Workaround:</b> Ensure that no group-label is configured to a peer before removing node-selection.</p>                      |
| [441706] | <p><b>Symptoms:</b> In the N2 handover scenario involving I-UPF insertion and removal, the serving network is incorrectly reported in the CDReq-Update and CDReq-Release messages.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                      |
| [440471] | <p><b>Symptoms:</b> If HTTP profile is used with TLS configured and used together with oauth2-profile, MG card instability may occur during manual switchover or CDB access token reconciliation.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                       |
| [440227] | <p><b>Symptoms:</b> In the Session Management Function Statistics section, the statistic Total Number of Idle UEs under the show mobile-gateway pdn statistics command shows incorrect results.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                         |
| [440461] | <p><b>Symptoms:</b> When Model-C is used for CHF communication and CHF profile is not received from NRF during discovery phase, MG card instability may occur.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                          |
| [439516] | <p><b>Symptoms:</b> For the MG-4380 feature, when interfaces on multiple services under sba-client-services are modified at the same time, session deletion rate may lead to high MG VM/POD CPU usage.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                  |
| [439954] | <p><b>Symptoms:</b> Paging retry implemented via [427920] is disabled when extended buffering is configured in namf-profile.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                            |
| [439528] | <p><b>Symptoms:</b> In some cases when a higher precedence dynamic match all PCC rule is applied to a session with an existing and better precedence match all pcc rule, then an empty PFCP Modification Request may be triggered toward the UPF/PGW-U.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [437117] | <p><b>Symptoms:</b> When SCP replies with <code>error</code> during SDM deletion with mode-fallback b is configured, fallback to model B is not applied if cache is expired.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                |
| [438642] | <p><b>Symptoms:</b> The 3gpp-sbi-message-priority header may not be used in UECM Registration Request for a WPS session if UDM has not indicated WPS ARP for the session.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                   |
| [431280] | <p><b>Symptoms:</b> In rare cases, if an inter-system handover from EPS to 5GS with indirect forwarding is cancelled, the following inter-system handover to 5GS with indirect forwarding may fail.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                         |
| [435011] | <p><b>Symptoms:</b> In case of TAU between 5G with packets buffered using extended buffering, and 4G (session in SGW/PGW), the Paging Buffer Curr packets and Paging Buffer Curr memory statistics in UPF (PGW) may show incorrect results.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [435585] | <p><b>Symptoms:</b> After Intra NF ISSU has been performed, issues may appear with AMF Event Exposure subscriptions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                        |
| [433806] | <p><b>Symptoms:</b> If CMG has both SMF and I-SMF sessions for the same UE, the <code>show&gt;mobile&gt;pdn&gt;pdn-context apn apn supi</code> command returns both sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                               |
| [434157] | <p><b>Symptoms:</b> For the 5GC-SBA server services load and overload control feature, on rare occasions the SMF may resend the 3gpp-sbi-oci header to AMF before the validity timer expires.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                               |
| [434470] | <p><b>Symptoms:</b> In case CMG has both SMF and I-SMF session for the same UE, total Number of UEs and UEs with 5G Session are counted twice on UPF PDN statistics. Additionally, QOS flows counters do not increase for ISMF sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>    |
| [431809] | <p><b>Symptoms:</b> If VUPF rejects 4G to 5G Mobility, session may not be cleared from HSMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                |

| Key      | Description                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [431812] | <p><b>Symptoms:</b> If an inter-system handover to EPS has recently been cancelled, in rare cases, the following Retrieve SM Context operation may fail.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                 |
| [432181] | <p><b>Symptoms:</b> If V-SMF insertion during N2 handover fails because UPF does not respond to PFCP Session Modification Request, session may not be cleaned up properly in V-SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                      |
| [432260] | <p><b>Symptoms:</b> In case of N2 handover with V-SMF present, if HandoverRequestAcknowledgeTransfer with QFIs in qosFlowFailedToSetupList IE is received, V-SMF will not initiate the QoS flow removal for the corresponding QFIs.</p> <p><b>Workaround:</b> No workaround available, yet.</p>      |
| [432505] | <p><b>Symptoms:</b> When PCF modifies the list of PRA Identifiers while the AM response for the creation of the subscription is pending, request for the modification of the subscription is not triggered toward AMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                   |
| [432903] | <p><b>Symptoms:</b> UDM-initiated session release or modification will not be successful after V-SMF removal.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                            |
| [433127] | <p><b>Symptoms:</b> Dedicated QoS flow creation may fail, if it is attempted during V-SMF change via N2 handover in COMPLETED phase.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                     |
| [433068] | <p><b>Symptoms:</b> If an inter-system handover from 5GS to EPS is cancelled and directly followed by a switchover of SMF to a geo-redundant standby system, in some cases the session may be removed by the SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                        |
| [432547] | <p><b>Symptoms:</b> During the re-establishment of a PDU session and while EBI Assignment is in progress, if PCF modifies the list of PRA Identifiers, the request for the modification of the subscription at the AMF is not triggered.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key             | Description                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [431819]        | <p><b>Symptoms:</b> In rare scenarios, after performing an in-service upgrade for a geographic redundant system, if there is an inter-system redirection to 5GS for a combined SGW-U+PGW-U session, the subsequent Retrieve SM Context operation for the session may fail.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                               |
| [431480]        | <p><b>Symptoms:</b> Intra NF ISSU from release 23.8.R1 to latest versions is not supported for both SMF and UPF due to error seen while Standby OAM is rebooted.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                         |
| [425666]        | <p><b>Symptoms:</b> During Xn handover where I-SMF addition is inserted, If I-UPF rejects PFCP Session Establishment/Modification Request, session is not deleted.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                       |
| [430131]        | <p><b>Symptoms:</b> Major Intra CNF ISSU from release 23.5.R2 to 23.8.R1 is not supported.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                               |
| [430161]        | <p><b>Symptoms:</b> Dedicated QoS flows are not handled correctly for any Xn Handover after I-SMF change triggered due to Xn handover.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                   |
| [430998/431331] | <p><b>Symptoms:</b> In rare occurrences, multiple ISM-MG switchover may result in missing PFCP-u tunnels. Workaround to bring back default tunnel is to configure the <code>clear&gt;mobile&gt;pdn 1&gt;bearer-context&gt;imsi default-tunnel-imsi</code> command from SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                              |
| [431217/436088] | <p><b>Symptoms:</b> Traffic may be dropped or passed unexpectedly if PCF installs a non-default qos-flow rule where flow-status is either ENABLED-DOWNLINK or ENABLED-UPLINK. With status ENABLED-DOWNLINK, both uplink and downlink traffic may be dropped. With status ENABLED-UPLINK, both uplink and downlink traffic may be passed.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [425062]        | <p><b>Symptoms:</b> Session is not cleared in I-SMF during 4G to 5G idle Mobility failure because the PSA-UPF does not respond.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                          |

| Key      | Description                                                                                                                                                                                                                                                                               |
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| [429265] | <p><b>Symptoms:</b> If I-SMF insertion during Xn handover fails because UPF does not respond to PFCP Session Establishment Request, session may not be cleaned up properly in I-SMF/PSA-SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                  |
| [430128] | <p><b>Symptoms:</b> QFIs that are missing from PathSwitchRequestTransfer IE during Xn Handover with I-SMF insertion, are not removed from PSA-SM.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                             |
| [430155] | <p><b>Symptoms:</b> PSA-UPF displays incorrect number of QoS Flows in KPI counters when using the <code>show&gt;mobile&gt;pdn&gt;qos-flow&gt;supi imsi detail</code> command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                 |
| [430482] | <p><b>Symptoms:</b> Location info is not shown in UE context output of PSA-SMF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                               |
| [430478] | <p><b>Symptoms:</b> Rule installation fails when a collision occurs between the PCF-initiated rule installation and 4G to 5G handover with I-SMF insertion in preparation phase.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                              |
| [430588] | <p><b>Symptoms:</b> UDM-initiated AMBR Modification Request received during 4G to 5G handover execution phase may be ignored.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                 |
| [430668] | <p><b>Symptoms:</b> During 4G to 5G handover with I-SMF insertion, if I-UPF does not respond to PFCP establishment request, I-SMF sends incorrect error message.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                              |
| [430808] | <p><b>Symptoms:</b> During 4G to 5G Handover with I-SMF insertion, if AMF does not send Update toward PSA-SMF, then session is not deleted in 4G.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                             |
| [430838] | <p><b>Symptoms:</b> Session not deleted from i-smf when I-UPF does not respond to IDFT creation at I-SMF removal case.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                        |
| [423388] | <p><b>Symptoms:</b> For a PDU session with multiple QoS flows, after an Xn handover with active QFIs missing from PathSwitchRequestTransfer IE, SMF does not initiate the process to have those QoS flows removed by SMF/UPF.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| [429459]          | <p><b>Symptoms:</b> In case two consecutive N2 Handovers with indirect tunnel occur, second N2 handover may fail, if indirect tunnels of the first N2 Handover are still up.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| [429936]          | <p><b>Symptoms:</b> After ICR ISSU has been performed, issues may occur with forwarding of downlink data toward eNodeB, during inter-system handover from EPS to 5GS.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| [427264] [427372] | <p><b>Symptoms:</b> For the SMF renewing UDM subscriptions (MG-3810) feature, the following apply:</p> <ul style="list-style-type: none"> <li>• subscription expiry information is missing from the <code>show&gt;mobile&gt;pdn&gt;pdn-context&gt;imsi imsi detail</code> command</li> <li>• configured values in expiry-time greater than 31536055 are not applied; this is the maximum value MG actual sends</li> <li>• under certain configuration sequence, expiry-time can be set to invalid value of 0</li> <li>• the Number of SDM Subscriptions counter is increased after renewal when UDM does not respond</li> </ul> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [421357]          | <p><b>Symptoms:</b> When connecting to NF service via the SCP, the <code>show service-stats</code> command is not updated for nf-id.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| N/A               | <p><b>Symptoms:</b> During 5G hardening task, support of multiple same services per NFProfile has been removed from release 22.5.R1. As a result, CMG supports only one NFService instance per NFProfile, for example, 1 UDM UECM and 1 UDM SDM service instance, 1 PCF policy service, and so on. Profiles are not discarded, but if more than one same service is presented, we keep only the first received.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                             |
| [404196]          | <p><b>Symptoms:</b> EPS to 5GS handovers are not rejected if DNN is in overload state.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [384689]          | <p><b>Symptoms:</b> No DNS query is sent if FQDN in UDM has more than 254 characters.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Key      | Description                                                                                                                                                                                                                                                                                                                                              |
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| [380562] | <p><b>Symptoms:</b> The PCF may remove all previously provided policy control request triggers by providing policyCtrlReqTriggers attribute set to value NULL. Removal of the triggers work with Update Response but not with Update Notify as the gateway unexpectedly rejects the message.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [380986] | <p><b>Symptoms:</b> Updating the revalidation timer with a new value before it expires may leave the original timer running. When new revalidation timer expires, the old timer will keep on running until it also expires.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                  |
| [384472] | <p><b>Symptoms:</b> Non-emergency sessions are not getting rejected when attached in emergency DNN.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                          |
| [384739] | <p><b>Symptoms:</b> PRA info is not communicated to new RAT, after inter-RAT handover.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                       |

## 12.2 All mobile gateways

| Key               | Description                                                                                                                                                                                                                                                                                                                                       |
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| [442550/04968259] | <p><b>Symptoms:</b> Page Cache usage is more than usual.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                              |
| [445581]          | <p><b>Symptoms:</b> Local MG reconcile from the active card to the standby card randomly takes longer, sometimes up to 30 minutes. The reconcile eventually completes successfully, but this affects Intra-Chassis ISSU upgrade time.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                 |
| [446754]          | <p><b>Symptoms:</b> After LTE to Wi-Fi handover if PCF reinstates rules with updated session AMBR, the the CDReq-Update message to CHF may include incorrect values for ratType,ueTimeZone, svgNode information on UUC lvi when reporting FINAL usage for updated rules.</p> <p><b>Workaround:</b> Do not provide RAT_TY_CH trigger from PCF.</p> |
| [443585]          | <p><b>Symptoms:</b> s5ToS2bHandoverFailure and 2bToS5HandoverFailurecounter may not be pegged during 4GtoePDG or ePDGto4G handover failure scenarios.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                 |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| [441880] | <p><b>Symptoms:</b> Session is shown as priority session in show mobile-gateway pdn pdn-context imsi &lt;imsi&gt; detail after non3GPP (Wi-Fi) to 3GPP handover, even though the ARP of the Session changes from WPS to non-WPS during handover.</p> <p><b>Workaround:</b> No Workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                               |
| [436976] | <p><b>Symptoms:</b> In LB-less mode, the user can configure only either the uplink-next-hop-ipv4 or the uplink-next-hop-ipv6 parameter, and either the downlink-next-hop-ipv4 or the downlink-next-hop-ipv6 parameter, under the config&gt;mobile&gt;profile&gt;policy-options&gt;traffic-redirect-profile&gt;mg-group&gt;traffic-redirect-target context. In case both IPv4 and IPv6 uplink-next-hop and downlink-next-hop types are configured, the threshold-target functionality may not work properly.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                    |
| [440129] | <p><b>Symptoms:</b> Under rare circumstances, KPI files stop to be generated after cleanup of the disk space (once it is 100% full).</p> <p><b>Workaround:</b> Perform a CPM switchover after disk space is cleaned up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                |
| [438719] | <p><b>Symptoms:</b> If MME rejects PCF-initiated QoS change, SMF sends Session Rule failure report to PCF. If the PCF re-applies the same QoS in response, another update is sent to the MME. This may lead into loop of failed QoS update attempts.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                           |
| [439045] | <p><b>Symptoms:</b> In rare scenarios when GTP context has been lost internally, PCF-initiated QoS update may trigger repeated failures.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| N/A      | <p><b>Symptoms:</b> In classic CLI, configuring the write-algorithm parameter as custom under the configure system security management-interface classic-cli write-algorithm custom command, and then configuring any commands containing an option for password or a secret that will use hash/hash2/custom hashing configuration preference, results to an incorrect saved configuration file after executing the admin&gt;save command, if the router configuration mode is classic or mixed mode. Therefore, executing such a saved configuration file fails.</p> <p><b>Workaround:</b> Use "hash" or "hash2" instead of "custom".</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                        |
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| [438872] | <p><b>Symptoms:</b> For Dedicated Bearer/QoS Flow traffic to be marked with WPS DSCP after Default Bearer or QoS Flow status has been modified from non-WPS to WPS, or vice versa, the Dedicated Bearer/QoS Flow ARP must also be modified.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                            |
| [438677] | <p><b>Symptoms:</b> During a handover from 3gpp to non-3gpp (ePDG) the Delete Bearer Request may not include the configured WPS message priority, for a PGW or combined SGW and PGW session.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                           |
| [436711] | <p><b>Symptoms:</b> Under rare circumstances, if the user on the fly removes and adds more than once the traffic-redirect-target command under the config&gt;mobile&gt;profile&gt;policy-options&gt;traffic-redirect-profile&gt;mg-group context, the session level steering may fail.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [437590] | <p><b>Symptoms:</b> In rare cases, the control to stop or start steering based on the expected stop or start steering threshold is not effective.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                      |
| [274705] | <p><b>Symptoms:</b> Modifying a dynamic PCC per RAT type configuration on the fly may not take effect on existing UEs at the next RAT Change.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                          |
| [433189] | <p><b>Symptoms:</b> After a ULCL UPF is inserted, a ModifyTaskRequest to change the LI destination may not take effect. The traffic is still sent to the old destination.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                              |
| [436611] | <p><b>Symptoms:</b> In deployment having ICR and centralized-ip-pool feature enabled, after geo-redundancy force-switchover, new master does not update routes correctly, which may impact down-link traffic.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                          |
| [435370] | <p><b>Symptoms:</b> Data-driven dedicated bearers in 4G, as well as dedicated QoS flows in 5G, cannot be created without charging or usage monitoring.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                 |
| [435865] | <p><b>Symptoms:</b> Some Hosts/Realms may not synchronize properly to slave GW in ICR deployments.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                     |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| [436457] | <p><b>Symptoms:</b> Instability may occur when Diameter GX peer replies with 3006 while number of hosts increases to maximum.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [436517] | <p><b>Symptoms:</b> Deleting the transit-sub option in classic CLI when the protocol option is configured under the service&gt;vprn&gt;ip-local-pool and router&gt;ip-local-pool contexts cannot be applied in MD-CLI.</p> <p><b>Workaround:</b> first de-configure the protocol option in MD-CLI (a protocol command does not exist in classic CLI) and then proceed with the deletion of the transit-sub option. In MD-CLI, as a workaround, delete the protocol option without deleting the complete transit-sub option; this is also accepted when committing a change.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [436796] | <p><b>Symptoms:</b> Emergency calls of IMSI/MSISDNs that belong in the Canary test user range are also handled by the CanaryPod/card. This is a known risk in release 24.7.R1 and is planned to be resolved in future releases.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                             |
| [433808] | <p><b>Symptoms:</b> If the clear command based on rat-type or user-range is issued, emergency sessions matching the rat-type or user-range are deleted.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [434404] | <p><b>Symptoms:</b> Dedicated bearers of SGW only sessions cannot be cleared with the clear command based on RAT type or user range and bearer option.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [434286] | <p><b>Symptoms:</b> Changing the rating group for a dynamic policy (PCRF/RAR) may cause interruption on collecting the transport bytes and packets for the changed policy.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| [434403] | <p><b>Symptoms:</b> SGW only sessions may not be cleared with the clear command based on rat-type or user-range if APN option is used.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| N/A      | <p><b>Symptoms:</b> The error appears if the following command is executed from classic-CLI command: mobile-gateway 1 role primary.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
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| [429665] | <p><b>Symptoms:</b> If two or more ADC rules sharing the same AA charging-group are configured under the <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;(dedicated-bearers)</code> context, removing them from the configuration may not have any effect on existing sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                    |
| [429929] | <p><b>Symptoms:</b> Under certain conditions, when MG card delays in the response to CPM LI queries, CLI LI commands/tech-supp may get stuck and CPM switchover is required to restore.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                        |
| [429996] | <p><b>Symptoms:</b> In rare circumstance when ICR reconcile occurs along with local reconcile from active MG card to standby MG card, standby MG may go out of sync (this applies to only when cam-optimization feature is enabled on gateway).</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                |
| [428087] | <p><b>Symptoms:</b> If the Support single Diameter host in S6b (limited support) (MG-2298) feature is enabled, while Gx and Gy interfaces have also single Diameter enabled, a performance degradation up to 25-30% is noticed.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                |
| [436796] | <p><b>Symptoms:</b> Emergency calls of IMSI/MSISDNs that belong in the Canary test user range are also handled by the CanaryPod/card. This is a known risk in release 24.7.R1 and is planned to be resolved in future releases.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                |
| [416789] | <p><b>Symptoms:</b> A small increase in MSCP CPU may be noticed in 22.8.R1 due to increased functionality.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                     |
| [354177] | <p><b>Symptoms:</b> When the traffic for a bearer is locally broken out, the ICE type of the CC records sent by the SGW for this bearer has the value <code>colocated_SAE_GWs</code>.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                          |
| [376342] | <p><b>Symptoms:</b> In a RADIUS group that has the <code>via-up-function</code> command enabled, if loopback is removed and then the RADIUS group is deleted the corresponding PFCP-u tunnel is not deleted.</p> <p><b>Workaround:</b> Re-create the same RADIUS group and delete the tunnel with the <code>no via-up-function</code> command in the <code>config&gt;mobile&gt;profile&gt;radius-group</code> context.</p> |

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| [354340/MG-2540] | <p><b>Symptoms:</b> When the <code>config&gt;mobile&gt;pdn&gt;gy#single-diameter-host</code> command is enabled, the PCRF-specified Primary-Event-Charging-Function FQDN will not work.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [353521]         | <p><b>Symptoms:</b> Enabling the <code>transport-byte-collection</code> command for an IMSI clears packets stats in bearer-context detail.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [440297]         | <p><b>Symptoms:</b> After QoS (ARP) is modified for a dedicated bearer, the PCRF-initiated delete of dedicated bearer may not work properly.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [353609]         | <p><b>Symptoms:</b> On-the-fly change of <code>diameter-peer-group-list</code> under the <code>config&gt;mobile&gt;pdn&gt;apn&gt;charging</code> context requires a PDN shutdown/no shutdown for change to take effect.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [353910/MG-2540] | <p><b>Symptoms:</b> When the <code>config&gt;mobile&gt;pdn&gt;gy#single-diameter-host</code> is enabled, a graceful shutdown does not work as expected and some Gy sessions may not be cleared. Two recommendations are proposed:</p> <p>The command <code>clear mobile-gateway pdn gw-id diameter-sessions peer peer-ip application-type gy graceful</code> can be configured twice with a few seconds difference to clear all sessions.</p> <p>When the suspend flag is set for a diameter peer, the <code>clear mobile-gateway pdn gw-id diameter-sessions peer peer-ip application-type gy graceful</code> command can be configured after a few minutes.</p> <p>When the <code>config&gt;mobile&gt;pdn&gt;gy#single-diameter-host</code> is enabled and the <code>graceful/non-graceful shutdown</code> command is configured, a failover to a secondary-diameter-peer does not take place, if such configuration exists.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [316532]         | <p><b>Symptoms:</b> When the PDN is in a suspend state and new PDP contexts (GTPv1) are rejected in the CPM, <code>VS.txMsgSent</code> statistics for the GTPv2 <code>CreateSessionRsp</code> are incorrectly incremented.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [299793]         | <p><b>Symptoms:</b> UL APN Rate Limiting is only supported when DL APN Rate Limiting is also enabled.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| [238704] | <p><b>Symptoms:</b> The mc-redirect traps (tmnxMcMonitorMclcrAlarm) will not be seen on CMG even though mc-redirect may have been used for certain interfaces.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| [266975] | <p><b>Symptoms:</b> If the Pi reference point and another reference point share a loopback interface with ipv6 configured, the other reference point will not work because Pi (MIPv4) does not support ipv6 addresses.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                         |
| [268110] | <p><b>Symptoms:</b> The config&gt;mobile&gt;profile&gt;policy-options&gt;ap-action-profile command can be executed while the AP action profile is in use. The operator must therefore de-reference any AP action profile referenced in the Gx profile before deleting it.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                      |
| [265509] | <p><b>Symptoms:</b> In a few and rare corner cases, the maximum duration that will be applied for buffering the downlink packets will be 1000 seconds, instead of the locally configured value. This could occur in high latency communication scenarios, where paging may take longer because the UE is using PSM or eDRX and extended buffering is activated in the SGW.</p> <p>The following AA command may not work properly when the config&gt;mobile&gt;pdn&gt;allow-multiple-apn-pdn command is enabled:<br/>tools&gt;dump&gt;aa&gt; group&gt;flow-record-search aa-sub</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [253927] | <p><b>Symptoms:</b> Changes for a PMIP profile (using the config&gt;mobile&gt;profile&gt;pmipv6 command) are effective only after issuing the shutdown and the no shutdown commands on the gateway.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                            |
| [207712] | <p><b>Symptoms:</b> The show&gt;router <i>router-instance</i>&gt;interface <i>interface-name</i> statistics command shows zero traffic statistics for Internal NAT ISA interfaces (_tmnx_nat-outside and _tmnx_nat-inside) when the ISA bb MDA is provisioned in an MG slot.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                   |
| [209849] | <p><b>Symptoms:</b> An attach will be rejected by the ePDG if the UE requires PDN type ipv6 while the subscribed PDN type of v4_or_v6 is returned in the APN Configuration on the Swm.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                         |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [220884] | <p><b>Symptoms:</b> When a UE attempts to attach after IP pool exhaustion, the IP address not available statistics for the <code>show&gt;mobile&gt;pdn&gt;statistics attach-failure-statistics</code> command are not incremented. The <code>show&gt;mobile&gt;pdn&gt;apn attach-failure-statistics</code> command may be used to obtain the failure count.</p> <p><b>Workaround:</b> No workaround available, yet.</p>    |
| [266960] | <p><b>Symptoms:</b> When the configuring <code>ip-reas</code> command on the MG-VM, benign trace messages may be displayed.</p> <p><b>Workaround:</b> Configure <code>ip-reas</code> command on the LB VM.</p>                                                                                                                                                                                                             |
| [169627] | <p><b>Symptoms:</b> On very rare occasions, issuing a shutdown command on the gateway may cause the mobile gateway group to remain in the "In Transition" state. The workaround is to clear the active card of the group. The workaround must only be used if the state does not transition to "down" within two minutes after executing the shutdown command.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 12.3 Application assurance

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444938, 445345] | <p><b>Symptoms:</b> Header enrichment does not work when the SNI appears in the second or any subsequent fragment of a fragmented ClientHello. Enrichment is only supported when the SNI is present in the first fragment because AA must update the <code>handshakeLenOffset</code> and <code>extensionsLenOffset</code> fields in the initial ClientHello segment, and these updates depend on the enrichment decision—which itself depends on the SNI.</p> <p>If AA waits until the SNI is extracted from a later fragment, and then enriches only the final fragment, the earlier length fields will not match the modified message. The server detects these inconsistencies and fails the TLS handshake. Therefore, when the SNI is not in the first fragment, AA cannot safely perform enrichment, and the SNI from later fragments is ignored for policy decisions.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [411819]         | <p><b>Symptoms:</b> Offload must not be used as an action for the command <code>config&gt;app-assure&gt;group&gt;policy&gt;aqp&gt;entry&gt;action</code>. Offloaded traffic may be dropped. (CUPS only)</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> HTTPS redirection cannot be performed for some popular websites where the browser has stricter security policies. This mechanism is called HTTP Strict Transport Security (HSTS). When AA attempts to redirect such a site, the browser blocks AA redirect request and no content is displayed to the user (not even the warning to accept an unverified certificate).</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [243319] | <p><b>Symptoms:</b> If ICMPv4 Fragmentation Needed or ICMPv6 Packet Too Big packets are issued and traverse the AA TCP Optimizer, the corresponding TCP sessions will be stuck/stalled until closed by the client, server, or the TCP Optimizer.</p> <p><b>Workaround:</b> Use the TCP MSS adjust feature or appropriately architect the network to avoid fragmentation.</p>                                                                           |

## 12.4 Call trace

| Key      | Description                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [376472] | <p><b>Symptoms:</b> Call insight metadata may show incorrect values for rule ID, precedence, and RG if data matching rule has a Policy Rule Unit with multiple AA charging groups configured.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                 |
| [368868] | <p><b>Symptoms:</b> The Diameter call trace ue does not set the IP DSCP and IP TTL value correctly.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                           |
| [334379] | <p><b>Symptoms:</b> The Diameter call trace all-ues does not set the IP DSCP and IP TTL value correctly.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                      |
| N/A      | <p><b>Symptoms:</b> The reset-user-traffic-stats parameter in the call trace debug command only resets UP statistics. The parameter cannot close running PCAP files and consequently cannot open a new one for a specific UE.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 12.5 CMG

| Key      | Description                                                                                                                                                                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447901] | <b>[NEW] Symptoms:</b> The SNMP for virtual-apn-rules accepts invalid values<br><b>Workaround:</b> No workaround available, yet                                                                                                                                                                                                             |
| [448236] | <b>[NEW] Symptoms:</b> The option no match action reject for virtual APN rules is not taking effect.<br><b>Workaround:</b> No workaround available, yet                                                                                                                                                                                     |
| [448243] | <b>[NEW] Symptoms:</b> The “Failed to find ratype-list Profile” error is observed during virtual-apn-rules when rat-type-list is modified. This error has no functional impact but cannot be suppressed.<br><b>Workaround:</b> No workaround available, yet                                                                                 |
| [448272] | <b>[NEW] Symptoms:</b> In some cases, the virtual-apn-rules modification does not take effect properly. Modification of certain filters in the virtual-apn-rules is not propagated to the MG card as expected.<br><b>Workaround:</b> Delete the rule and then apply the rule with the required changes, instead of modifying it.            |
| [448275] | <b>[NEW] Symptoms:</b> Removal or deletion of the rat-type-list, ue-location-list, or plmn-list from the system may be blocked when used in a virtual APN rule and a bulk configuration sync occurs during gateway shut/no-shut, mg-group reboot (both cards), or new mg-group addition.<br><b>Workaround:</b> No workaround available, yet |
| N/A      | Modifying another CaaS/Platform is not supported from release 25.x.R1.                                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444463] | <p>From release 25.7R2 and onwards, the use-locality parameter under the nrf-disc does not work.</p> <p><b>Workaround:</b></p> <p>Follow the steps to enable the preferred locality as NRF query parameters for N7/N40.</p> <ol style="list-style-type: none"> <li>1. Configure the preferred-locality parameter under the nrf-query-profile.</li> <li>2. Configure the nf-selection-profile to use the query profile under nrf-based.</li> <li>3. Select the configured nf-selection-profile under N7 or N40 interface.</li> </ol> <p><b>Example</b></p> <p>a. query profile</p> <pre>*A:Dut-C&gt;config&gt;mobile&gt;profile&gt;nf-selection# /configure mobile-gateway profile nrf-query-profile "query-profile1" *A:Dut-C&gt;config&gt;mobile&gt;profile&gt;nrf-query# info     preferred-locality "test_locality"</pre> <p>b. nf-selection profile</p> <pre>*A:Dut-C# configure mobile-gateway profile nf-selection-profile "nf-selection-profile1"     selection-options         nrf-based             query-profile " query-profile1"         exit     exit</pre> <p>c. use the nf-selection profile under N7 and N40.</p> <pre>*A:Dut-C&gt;config&gt;mobile&gt;pdn&gt;apn&gt;pcf-selection# info     nf-selection-profile "nf-selection-profile1" &lt;&lt;N7  *A:Dut-C&gt;config&gt;mobile&gt;pdn&gt;charging# info *A:Dut-C&gt;config&gt;mobile&gt;profile&gt;charging&gt;smf# info     chf-selection         chf-profile 1         nf-selection-profile nf-selection-profile1 profile1    &lt;&lt;N40</pre> |
| [443151] | <p><b>Symptoms:</b> In release 25.7.R2, enabling the logrotate container under CMF CNF operator pod may cause Helm rendering issues.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [440042] | <p><b>Symptoms:</b> Under rare condition, after MG switchover, new standby card that is on old version may experience instability. Standby card becomes stable after upgrade.</p> <p><b>Workaround:</b> Upgrade the standby immediately after MG switchover.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> The auto-evacuation LCM operation does not work with CBAM21 (CBNI-6905: Alarm VIM_CONNECTIVITY_OPENSTACK_KEYSTONE probably blocks FM to get VNFc related alarms).</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                       |
| N/A      | <p><b>Symptoms:</b> Under certain scenarios, after an MG-VM switchover event, a statistics object in a session may not move to a fast polling tier and may remain in a normal polling tier until a subsequent event occurs.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [299275] | <p><b>Symptoms:</b> IPv6 for CSF is not supported with CMG Templates packages 10.3.1 and later.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                             |

## 12.6 CMG appliance

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>[NEW] Symptoms:</b> After the NEDR software installation, the agent incorrectly attempts to bind the active AIM, instead of the corresponding service IP address, to the CNEDR server.</p> <p><b>Workaround:</b> Add the IP address of the central NEDR server to breth0, over the existing service IP address of the server:</p> <pre>sudo ip route add 100.127.22.94/32 dev breth0 src &lt;service ip address&gt;</pre> <p>where 100.127.22.94/32 is the IP address of the central NEDR server.</p> |

## 12.7 CNF

| Key               | Description                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442550/04968259] | <p><b>[NEW] Symptoms:</b> Page Cache usage is more than usual.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                              |
| [440474/04553044] | <p><b>Symptoms:</b> Ghost IPs in output of the show mobile-gateway pdn pdn-context command.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [440587] | <p><b>Symptoms:</b> During a sudden spike in the NAT fragments, CMG may require longer time to purge the NAT fragment lists that are expired, causing PBUF congestion and BFD flaps.</p> <p><b>Workaround:</b> To prevent traffic impact, increase BFD timers from 100 msec to 300 msec.</p> |
| N/A      | <p><b>Symptoms:</b> CMG CNF cannot be deployed on vCPU indexes above 255.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                        |
| [440405] | <p><b>Symptoms:</b> DNAI Anchor rule removal is not supported and may leave the session in an indeterminate state.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                               |

[440064]

**Symptoms:** In NCS24.11, the tenants are created by default with image validation enabled. This setting requests by the application to fetch images by using their digest instead of the tag. To continue using the image tags, the tenants must be created with image validation disabled ("tenantSignatureValidation": false); See example:

```
{
  "allowedSecurityRules": [
    "allow-privilege-escalation",
    "capabilities",
    "forbidden-sysctls",
    "host-fileSYSTEM",
    "host-namespaces",
    "host-network-ports",
    "privileged-containers",
    "proc-mount",
    "read-only-root-fileSYSTEM",
    "seccomp",
    "selinux",
    "users",
    "volumes"
  ],
  "disallowedSecurityRules": [],
  "id": "c78e63fb-e241-41c6-ae6b-b843158aa2ed",
  "name": "opa-nrd",
  "namespaces": [
    {
      "limitRange": {},
      "name": "nrd",
      "resourceQuota": {
        "hard": {
          "limits.cpu": "32",
          "limits.memory": "34Gi"
        }
      },
      "uid": "e9831410-85d6-46cb-8fc0-28d542557nrd"
    },
    {
      "limitRange": {},
      "name": "nrd-db",
      "resourceQuota": {
        "hard": {
          "limits.cpu": "6",
          "limits.memory": "16Gi"
        }
      },
      "uid": "c1b66b22-508d-4417-bedb-ad9b3e24nrd"
    }
  ],
  "networkPolicyLabel": "BLOCK_ALL_INGRESS",
  "resourceQuota": {
    "hard": {
      "limits.cpu": "50",
```

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <pre>         "limits.memory": "50Gi"       }     },     "securityPolicy": "OPA",     "tenantAdminNamespaceName": "opa-nrd-n",     "tenantSignatureValidation": false   } </pre> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| [439409] | <p><b>Symptoms:</b> Under certain conditions, after MG switchover, the VS.maxNumberOfPdnSessions counter under the KCI BearerManagementPerQciCP-ISA table is incorrectly pegged.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [436016] | <p><b>Symptoms:</b> During Intra NF ISSU toward release 24.7, MG switchovers are not supported after the CPM switchover and before the MG group is upgraded. This applies for both CNF and VNF Variants.</p> <p>Preventive action</p> <p>After the CPM switchover toward the upgraded CPM, reboot (clear) the standby card(s) that remain(s) in the source build. This action prepares the MG group for successful potential switchovers.</p> <p>Healing action</p> <p>In case of an unexpected MG group switchover, sessions cannot be served successfully.</p> <p><b>Workaround:</b> To heal this issue, an extra manual MG switchover must be executed.</p> |
| [436048] | <p><b>Symptoms:</b> Control and data fabric KPI-KCI counters are not supported in CNF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| [432733] | <p><b>Symptoms:</b> Prometheus Metrics is not available for the module KPIServiceNnrFOAuth2.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| [433335] | <p><b>Symptoms:</b> In rare scenario during Intra NF ISSU procedure, after the admin redundancy cpm-switchover now command is executed, the CDB connection may be lost. After the upgrade of the new standby CPM, the CDB connection must be restored.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                             |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> CN-A environment with Intel NIC, and CMG N:K deployment with DB-Proxy pod running over Multus interface.</p> <p><b>Workaround:</b> Follow the additional steps:</p> <ol style="list-style-type: none"> <li>1. Run <code>dpdk-devbind.py -s</code> on corresponding worker nodes, to get PCI addresses of intel vnics, mapped to <code>dpdk compatible vfio-pci driver</code>.</li> <li>2. The unbind one of the interfaces on those worker nodes from <code>vfio-pci driver</code> and bind it back to <code>ixgbevf (kernel network driver)</code> via command <code>dpdk-devbind.py --bind=ixgbevf &lt;pci address&gt;</code>. For example, if the <code>0000:00:06.0</code> is one pci address of the vNICs on the worker node, command will be <code>dpdk-devbind.py --bind=ixgbevf 0000:00:06.0</code></li> <li>3. Restart <code>sriov-dp daemonset</code> via command <code>kubectl rollout restart ds &lt;kube-sriov-device-plugin-id&gt; -n kube-system</code>, to bring up new <code>sriov-dp pods</code> under <code>kube-system namespace</code>. Here replace <code>kube-sriov-device-plugin-id</code> with a corresponding <code>sriov-daemonset id</code> for your cluster. Thus, with the <code>kubectl rollout restart ds</code> command, a new logical resource (<code>intel_sriov_netdevice</code>) is created on the corresponding worker nodes. This new logical resource (<code>intel_sriov_netdevice</code>) is created as part of <code>sriovdp-config</code> configuration map, which maps interfaces with a logical resource based on vendor ID, product ID, and device driver.</li> <li>4. Validate a new logical resource creation under <code>Allocated resources</code> section, by running command <code>kubectl describe node &lt;worker-node&gt;</code>.</li> </ol> <p>Finally, use this <code>intel.com/intel_sriov_netdevice</code> logical resource inside the <code>dbproxy multus values.yaml</code> to bring up Multus <code>net1</code> interface on DB-Proxy pod.</p> |
| [419945] | <p><b>Symptoms:</b> The <code>cmg-ts</code> utility used for collecting DB Proxy technical support is a VM-only utility. It does not work when running in the DB CNF.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| N/A      | <p><b>Symptoms:</b> The <code>vLan</code> parameters in <code>values.yaml</code> must be configured for NIC sharing feature with NCS22 and must not overlap with VLANs assigned to NCS. NCS cluster must be deployed with Provisioning VLAN option with a value other than zero.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> CMG deployment may fail due to the fact that NCS scheduler is NUMA unaware. This may happen when CMG is to use both NUMAs for MG pods. Analysis of this case is ongoing, and instructions are going to be delivered soon.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [387418] | <p><b>Symptoms:</b> In CNF deployments, CPU utilization of the <code>show&gt;system&gt;cpu</code> command output, is incorrectly displayed at 100%.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                           |

## 12.8 CUPS

| Key               | Description                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443736/05009613] | <p><b>Symptoms:</b> Under rare condition when UE sends the DHCPv6 Renew instead of sending the DHCPv6 XID Solicit request to install prefix delegated during reattach, the UPF does not install the PD correctly.</p>                                                                                                       |
| [438726/04279095] | <p><b>Symptoms:</b> CMG as UPF may unexpectedly include a Created PDR in a PFCP message that is sent toward the CPF as a response to a request that does not include any Create PDR IEs.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                        |
| [440032]          | <p><b>Symptoms:</b> Under rare scenarios, MG card on slave SMF may experience instability. However, this does not have any service impact.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                      |
| [435045/03686596] | <p><b>Symptoms:</b> In rare cases, UPF may take longer time to handle SX signaling.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                             |
| [434101]          | <p><b>Symptoms:</b> The <code>tools&gt;dump&gt;micronets</code> command does not autocomplete or display the available values for the <code>local-pool</code>, <code>VPRN</code>, or <code>network-realm</code> options.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                        |
| [424457/02731413] | <p><b>Symptoms:</b> Randomly dynamic traffic redirect does not take place when using the <code>redirect-filter-id</code> command under the <code>config&gt;mobile&gt;profile&gt;dcca</code> context for dynamic installation of the auto-redirect rule in CUPS.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [421052] | <p><b>Symptoms:</b> The <code>tools&gt;dump&gt;nat</code> commands may experience some extended completion delay.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                             |
| [374378] | <p><b>Symptoms:</b> During attach on L2TP APN in CUPS, Framed-IP-Address in accounting start is set to 0.0.0.0, and the PDP-Address AVP is missing in CCR-I on Gy interface.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                  |
| [374381] | <p><b>Symptoms:</b> RADIUS packets encapsulated in GTP in default PFCP-u tunnel are not captured in call trace.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                               |
| [331346] | <p><b>Symptoms:</b> When the PFCP overload control feature is enabled, the SMF rejects Create Session Request messages for MME initiated eMPS SGW sessions.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                   |
| [326889] | <p><b>Symptoms:</b> In N:K redundancy, the PCMD configuration may not be pushed to all cards or VMs.</p> <p><b>Workaround:</b> Nokia recommends applying the PCMD configuration last, or using the following workaround:</p> <pre>config&gt;mobile&gt;pdn gw-id no pcmd config&gt;mobile&gt;pdn gw-id pcmd profile profile-name</pre>     |
| [328152] | <p><b>Symptoms:</b> If a link layer address change occurs, an unsolicited neighbor advertisement is not sent. Having BFD configured for the IPv6 interface using <code>dynamic-sap</code> can help with MAC updates via the Neighbor Discovery.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                               |
| [330096] | <p><b>Symptoms:</b> Group create/delete operations on the same group while an MG-VM reboot is in progress could cause system instability.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                     |
| [303204] | <p><b>Symptoms:</b> In certain scenarios when the slave UP function node becomes the master node and there are Sx Modify Request messages directed toward the UP Function, the transmission may timeout at the CP function due to the retry-count being set to 1 (retry-count=1).</p> <p><b>Workaround:</b> Set the retry-count to 3.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [288065] | <p><b>Symptoms:</b> Enabling and disabling the bearer binding is not working in CUPS. This could lead to UP data flowing through the wrong bearer during the time when segmented TFT filters are being communicated to the UE.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                    |
| [296757] | <p><b>Symptoms:</b> KPI counters on the SxNodeSessionProcedures table are not fully implemented.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                  |
| [296712] | <p><b>Symptoms:</b> KPI-KCI statistics on UP function may not work for some counters.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                             |
| [296420] | <p><b>Symptoms:</b> The UPF deletes a PFCP session if the CP function responds with cause 65 (session context not found) in the error indication PFCP session report response. In all other cases, the UP function waits for the CP function to take some action.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 12.9 ePDG

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [448191] | <p><b>[NEW] Symptoms:</b> ePDG/N3IWF ESP traffic using GCM encryption/cipher algorithm does not scale beyond 8 datapath workers, and throughput may drop with more workers.</p> <p><b>Workaround:</b> No workaround available yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [445834] | <p><b>Symptoms:</b> WPS calls updated to MPS are considered as WPS call after a MSCP switchover. The calls still get the Priority status and DSCP Marking as per profile.</p> <p><b>Workaround:</b> No workaround available yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [441131] | <p><b>Symptoms:</b> For a MPS call, if the ARP of the default session does not match to any of the ARPs configured in WPS Profile, the UL Traffic of Default session may be marked with the DSCP value from the Default QCI Policy assigned to the PDN while for Dedicated Bearers, and the UL Traffic may be marked with the DSCP value configured in the WPS Profile. This applies to sessions that are identified as MPS by the HPA_INFO parameter sent by UE as well as the sessions identified as MPS by the MPS_priority AVP sent by AAA to EPDG over Diameter Messages for SWm Interface.</p> <p><b>Workaround:</b> No workaround available yet.</p> |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [44164]           | <p><b>Symptoms:</b> The SW-u does not clear the counts for number of total count (DH Group Stats) in SW-u reference point statistics are not cleared when using</p> <ul style="list-style-type: none"> <li>clear mobile-gateway pdn &lt;gw_id&gt; ref-point-stats</li> <li>swu clear mobile-gateway pdn &lt;gw_id&gt; ref-point-stats</li> </ul> <p><b>Workaround:</b> No workaround available yet.</p>                                           |
| [441726]          | <p><b>Symptoms:</b> In ePDG, for the IPsec Client Lockout feature if an MG card switchover occurs while a UE is locked out, in case of a new attach attempt for this locked-out UE, signaling occurs in SWm interface toward the AAA server.</p> <p><b>Workaround:</b> No workaround available yet.</p>                                                                                                                                           |
| [439188]          | <p><b>Symptoms:</b> Delete Bearer Request and Delete Bearer Response do not carry the MP value set as per WPS profile for WPS calls during Re-Attach and handover from 3GPP to non-3GPP (ePDG).</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                       |
| [431418/03271661] | <p><b>Symptoms:</b> Under certain conditions, ePDG with ICR may show discrepancies in counters between active and standby node, due to counter updates coming from different state on the call flow. As a result, the active node counter increases before the procedure is complete while standby node is only updated after call flow is completed.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                 |
| [436564]          | <p><b>Symptoms:</b> The value of the config&gt;mobile&gt;profile&gt;s2b-error-mapping&gt;s2b-cause-code&gt;swu-send-info command is not transferred to saved file after executing the admin save command. If the value is false, the value is set as true. In case of admin save and admin reboot commands, the user must manually re-enter the values for the swu-send-info command.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [337672]          | <p><b>Symptoms:</b> After several MG card switchovers on the ePDG, the ePDG may not throttle attaches that must be throttled due to a PGW signaled overload to the ePDG.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                              |
| [227005]          | <p><b>Symptoms:</b> Threshold limit alarms are not supported for the ePDG.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                            |

## 12.10 IP reassembly

| Key      | Description                                                                                                                                                                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [447603] | <b>[NEW] Symptoms:</b> Timeout is observed on 10% of GTP-u fragmented packets in UL direction on UPF configured on XR-8000 platform.<br><b>Workaround:</b> Use proper MTU setting on the S1u/N3 interface path between the gNB and UPF to avoid fragmentation on UL traffic.                                                                |
| [351435] | <b>Symptoms:</b> When the gtpu-source-port-dist command is enabled, the fragmented user-plane data with TCP/UDP headers use a different UDP source port than the non-fragmented user-plane data. This may cause user-plane IP packets to use different routes to reach the destination.<br><b>Workaround:</b> No workaround available, yet. |
| [288827] | <b>Symptoms:</b> The IP-reassembly MDA for L2TP packets must be configured on one of the IOM cards or ISM-MG-C cards.<br><b>Workaround:</b> No workaround available, yet.                                                                                                                                                                   |

## 12.11 ISSU/ISSD

| Key      | Description                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [448121] | <b>[NEW] Symptoms:</b> ISSD towards 26.7 is not supported for 26.7.R1 release.<br><b>Workaround:</b> No workaround available, yet.                                                                      |
| [444844] | <b>Symptoms:</b> Rollback after CPM upgrade may cause CPM instability (for example, session loss).<br>This affects the path: 26.3.R1 to 25.11.R1-R5<br><b>Workaround:</b> No workaround available, yet. |
| [440439] | <b>Symptoms:</b> In certain conditions, 24.11.R3 to 25.3.R1 ISSU upgrade may not work.<br><b>Workaround:</b> No workaround available, yet.                                                              |

## 12.12 KPI-KCI counters

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443243] | <p><b>Symptoms:</b> VS.scpFallbackDirComma under KPI ScpPeerServiceStats is not increased when SMF sends a registration message to SCP and a fallback is triggered as SCP replies with an error. Same behavior is observed for discovery. This applies to the following counters:</p> <p><b>Workaround:</b> No workaround available yet.</p>                                                                                                                                                                                                         |
| [443530] | <p><b>Symptoms:</b> SCP fallbacks counter and VS.scpFallbackDirComma under KPI ScpPeerServiceStats are not increased when SMF sends a Delete Request to SCP and a fallback is triggered.</p> <p><b>Workaround:</b> No workaround available yet.</p>                                                                                                                                                                                                                                                                                                  |
| [443958] | <p><b>Symptoms:</b> The counter VS.numberOfPfcPpeers/numberOfPfcPpeers "Number of Sx peers" and VS.pfcPUpIpv4Peers/pfcPUpIpv4Peers "Number of IPv4 user plane VNF peers" show different values due to the measurement point. In each case, the first counter is per MG Group. When aggregating this counter, the total value is the number of the UPF peers multiplied by the number of MG groups.</p> <p><b>Workaround:</b> Use the second counter for an actual reference to the number of the UPF peers connected to SMF/CP (per IP version).</p> |
| [442524] | <p><b>Symptoms:</b> In an EPS fallback scenario where UPF fails to respond to the PFCP Session Modification Request, the counter smfEPSFallbackProceduresFailure is pegged twice under KPI = BearerManagementPGW.</p>                                                                                                                                                                                                                                                                                                                                |
| [442872] | <p><b>Symptoms:</b> In an EPS fallback scenario for emergency session where UPF fails to respond to the PFCP Session Modification Request, the counter smfEmergencyFallbackProceduresFailure is not pegged under KPI=BearerManagementPGW.</p>                                                                                                                                                                                                                                                                                                        |
| [443042] | <p><b>Symptoms:</b> In an Emergency EPS fallback scenario where no Create Bearer Response is received from the MME during voice bearer activation, both the success and failure counters are pegged under KPI table BearerManagementPGW.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                 |
| [439280] | <p><b>Symptoms:</b> The SCP Request Rx and SCP Response Tx counters are not pegged. These counters must be pegged when a notification is received from a NF producer via SCP.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                            |

| Key               | Description                                                                                                                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [439851]          | <p><b>Symptoms:</b> The SCP fallbacks counter is not pegged in scenarios related to NRF Discovery and Management services.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                               |
| [433762/03529532] | <p><b>Symptoms:</b> KPI counter maxNumberOfActiveEutranBearers is much higher than maxNumberOfBearers on SMF at some scenarios.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                          |
| [433460]          | <p><b>Symptoms:</b> The nfInstanceId used in the Prometheus Labels for KPIServiceNsmfPduSsnProducerSmf module is incorrect for I-SMF/V-SMF when there is an I-SMF/V-SMF insert/Removal due to N2 handover, AMF nfID is displayed instead of peer SMF Id (PSA-SMF/H-SMF).</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [428863]          | <p><b>Symptoms:</b> The KPI counter VS.smfEPSFallbackProceduresFailure for the number of failed EPS fallback procedures in SMF does not work properly.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                   |
| [389856]          | <p><b>Symptoms:</b> The reservedIPv4Addresses/reservedIPv6Addresses in KPI-KCI may be incorrect after disabling round-robin under the config&gt;router&gt;ip-local-pool context.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                         |

## 12.13 Lawful Interception

| Key               | Description                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [440899]          | <p><b>Symptoms:</b> In some cases, the bearerActivationUnsuccessful IRI from the SGW may have failedBearerActivationReason reported as Request Rejected instead of the specifically received cause code.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                              |
| [428604]          | <p><b>Symptoms:</b> In a CMG CNF where SNMPv3 is used, the engineID must be set to a non-default value configuring the config&gt;system&gt;snmp command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                              |
| [411489/02455722] | <p><b>Symptoms:</b> Under certain scenario, following error may appear when trying to configure a new ADMF peer: MINOR: IMSI #2827 Invalid Peer idx – The peer ID 1 is already provisioned, please select another peer ID, such as 3.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [415428] | <p><b>Symptoms:</b> When an LI_X1 DeactivateAllTasksRequest is sent to the SMF and a large number of tasks exist, it responds with AcknowledgedAndCompleted in the OK field. This is sent even though the DeactivateAllTasks may not be complete.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [286595] | <p><b>Symptoms:</b> The ULI is not present in the output of the <code>show&gt;li&gt;mobile&gt;target detail</code> command when the geographic location type is SAI or RAI.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                       |
| [307499] | <p><b>Symptoms:</b> Per-target CLI statistics for x2/x3 messages (output of the <code>show&gt;li&gt;mobile&gt;target stats</code> command) are not displayed correctly when targets with duplicate IMEI are intercepted.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                          |

## 12.14 LTMT

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>Symptoms:</b> VNF running in containers</p> <p>The LTMT VNF software runs in containers using bridged networking managed by Docker. The containers run isolated, use internal IP addresses, and do not communicate directly with the outside network. The kernel of the LTMT VM performs IP and port address translations between the containers and the external network. Therefore, the LTMT VM IP address appears in use. The containers perceived the external IP address because only the destination address is changed and not the incoming source address. When going out, the source address of the container is replaced with the address of the LTMT VM. If the LTMT container restarts when receiving a high load of UDP traffic coming from the CMM or CMG, the address translation affects the incoming source address. As a result, the traffic may appear to come from the internal bridge interface. This behavior does not affect the PCMD records from the CMG, but the CMM is identified based on the source IP address. This causes the messages to drop. When this issue occurs, it can be observed using the <code>ltmt-conntrack</code> utility. Running the <code>ltmt-conntrack list</code> command shows the connection tracking entries from the VM <code>nf_conntrack</code> table. The source IP address cannot identify CMM when the following entries exist</p> <pre>udp      17 29 src=172.20.0.1 dst=172.20.0.4 sport=52724 dport=7797 [UNREPLIED] \           src=172.20.0.4 dst=172.20.0.1 sport=7797 dport=52724 \           mark=0 secctx=system_u:object_r:unlabeled_t:s0 use=1 instead of the following: udp      17 12 src=192.168.22.1 dst=192.168.22.11 sport=36311 dport=7797 [UNREPLIED] \           src=192.168.22.11 dst=192.168.22.1 sport=7797 dport=36311 \           mark=0 secctx=system_u:object_r:unlabeled_t:s0 use=1</pre> <p><b>Workaround:</b> use the <code>ltmt-conntrack delete</code> command. To delete all entries of destination port 7797, the <code>ltmt-conntrack</code> command calls the <code>conntrack</code> user space tool (<a href="https://manpages.debian.org/jessie/conntrack/conntrack.8.en.html">https://manpages.debian.org/jessie/conntrack/conntrack.8.en.html</a>). Using the <code>any</code> command can pass any identified parameter to <code>conntrack</code>. To list all TCP <code>conntrack</code> entries, use the <code>ltmt-conntrack any-L-p tcp</code> command.</p> |

## 12.15 MD-CLI

| Key      | Description                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [448186] | <p><b>Symptoms:</b> When priority rule <code>imeisv-tacsvn</code> misses the optional value <code>imeisv-tacsvn-end-ue</code>, the rule can't be matched.</p> <p><b>Workaround:</b> Include the <code>imeisv-tacsvn-end-ue</code> optional value in priority rule <code>imeisv-tacsvn</code>.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [437312] | <p><b>Symptoms:</b> Validation (via NETCONF/MD-CLI) succeeds when configuring IPs that are not reserved under <code>router&gt;ip-local-pool&gt;reserved-ipv4/6</code> and <code>service&gt;vprn&gt;ip-local-pool&gt;reserved-ipv4/6</code> but commit fails. The validations that disallow any IPs not included in the IP pools to be reserved, are missing during validation action.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                            |
| [439934] | <p><b>Symptoms:</b> The <code>peer-group</code> option under <code>config&gt;mob&gt;pdn&gt;s6b</code> command is missing in MD-CLI.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| [439948] | <p><b>Symptoms:</b> The <code>multi-homing diameter interface</code> option under <code>configure mobile-gateway pdn swm</code> command is configured in one transaction with the <code>diameter interface</code> option under <code>configure mobile gateway pdn swm</code> in MD-CLI.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                          |
| [440295] | <p><b>Symptoms:</b> Deletion of the <code>config-requests</code> option under <code>config&gt;mobile&gt;pdn&gt;apn&gt;ppp-regeneration&gt;lcp-confreq-requests</code> command, with <code>mru</code> and <code>auth-type</code> options still configured in MD-CLI, is not updated in classic CLI. Also deleting the <code>mru</code> and <code>auth-type</code> options in MD-CLI removes the <code>lcp-confreq-attributes</code> totally from the classic CLI whereas, in classic CLI, the <code>config-request</code> option remains visible.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [439909] | <p><b>Symptoms:</b> When modifying the default values of the <code>configure mobile-gateway profile mipv4-profile</code> command, validation (via NETCONF/MD-CLI) succeeds, but commit fails. The validations that disallow any value changes in default profile, are missing during validation action.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                          |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [438736] | <p><b>Symptoms:</b> If the gateway is configured in mixed mode (Classic and MD-CLI are enabled) and the mobile-gateway configuration within the MD-CLI (configure delete mobile-gateway) must be deleted, the transaction fails because the ip-address-prefix-list cannot be deleted.</p> <p><b>Workaround:</b> Commit first a transaction by deleting the remote-ip-list references for all the ip-address-prefix-lists;<br/>Follow the steps:</p> <ol style="list-style-type: none"> <li>1. remove this reference within policy-rule-unit and commit</li> <li>2. repeat for every policy-rule-unit remote-ip-list reference</li> <li>3. commit this transaction</li> </ol> <p>perform a new transaction to complete the deletion of the mobile-gateway<br/>For example, the following configuration may cause this behavior:</p> <pre>configure mobile-gateway profile list ip-address-prefix-list ipAddressPrefixList1 configure mobile-gateway policy-rule-unit "pru1 flow- description 10 match remote-ip-list ipAddressPrefixList1</pre> <p>The following commands (replicated per applicable policy-rule-unit) can be configured to remove the reference to ip-address-prefix-list ipAddressPrefixList1, assuming that the PDN and APNs are already configured with the value of the admin-state as disable:</p> <pre>configure mobile-gateway profile policy-options policy-rule- unit pru1 flow-description 10 match delete remote-ip-list validate commit configure delete mobile-gateway validate commit</pre> |
| [438014] | <p><b>Symptoms:</b> Setting the udm-uecm-profile under the config&gt;mobile&gt;pdn&gt;sba-client-services&gt;udm-client context requires initiating the udm-uecm-profile under the config&gt;mobile&gt;profile&gt;udm-uecm-profile context. Deleting in MD-CLI the udm-uecm-profile under the profile context when reference has already been deleted under the config&gt;mobile&gt;pdn&gt;sba-client-services&gt;udm-client context, fails because it still acts as referenced.</p> <p><b>Workaround:</b> Delete the initiated udm-uecm-profile from classic CLI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| [438376] | <p><b>Symptoms:</b> The pcscf-server-group-v4/6 options under the conf&gt;mobile&gt;pdn&gt;apn&gt;pco-option&gt;ue-location&gt;tai-lai-list context is not supported in 24.11 in MD-CLI.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [436137] | <p><b>Symptoms:</b> Profile references to ePDG group must be deleted and committed first to allow separate deletion of the configure mobile-gateway context.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                  |
| [436456] | <p><b>Symptoms:</b> MCI allows to configure a non-existing APN to the via-up-function parameter, leading to an error afterwards. Make sure that the APN configured under the dhcp-server-group <i>group-name</i> via-up-function apn <i>apn-name</i> command exists under the mobile-gateway pdn apn context.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [437161] | <p><b>Symptoms:</b> When the attempts to configure the characteristic option under the action-rule-unit context exceed the limit of 32 entries, the Maximum Number of Allowed Entries error is not displayed during validation. The error occurs during commit.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                               |

## 12.16 PCMD

| Key              | Description                                                                                                                                                                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [378007]         | <p><b>Symptoms:</b> In case of initial attach with piggy-backed dedicated bearer creation, the PCMD record for the MME-initiated Create Default Bearer procedure is not properly generated.</p> <p><b>Workaround:</b> No workaround available, yet.</p>            |
| [388078]         | <p><b>Symptoms:</b> In case of 4G to 3G handover in Cups, the PCMD record for SGSN-initiated Update PDP Context procedure is not properly generated.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                   |
| [331407, 344394] | <p><b>Symptoms:</b> If IP reachability to a PCMD server is resolved using BGP or IGP protocols, in specific cases intermittent system instability may occur.</p> <p><b>Workaround:</b> Nokia recommends using static-routes to reach the PCMD server.</p>          |
| [322268]         | <p><b>Symptoms:</b> The network-initiated Create Bearer to MME procedure is not reported upon dynamic application detection, which is triggered by the ADC rules pushed by the PCRF on initial attach.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [306105] | <p><b>Symptoms:</b> The network-initiated Create Bearer to MME procedure is not reported during the session creation, which is triggered by the configured rules and rule bases, under the static local PCC with different QCI/ARP than the default bearer.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| N/A      | <p><b>Symptoms:</b> The ChargingIndication flag is not set correctly in some cases of network triggered service request procedures and in procedures related to PCRF interaction in which no charging is applied in the session.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| N/A      | <p><b>Symptoms:</b> The GCID IE is not set correctly in the SGW for some cases of Bearer deactivation procedures.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| N/A      | <p><b>Symptoms:</b> The SendingNodeType IE is set to the type of the UE session/bearer for which the PCMD procedure is reported. A UE can have multiple types of sessions with different types, such as the following:<br/>SGW only session<br/>PGW only session<br/>combined SGW and PGW session – SCCG<br/>combined TWAG and SCCG session<br/>ePDG only</p> <p>Apart from the session/bearer-based procedures, there are also UE-level procedures that affect not only a single session/bearer but affect all the UE sessions that are handled on the SGW or combination gateway (from MME), such as RAB without bearer context, Suspend and Resume notifications without LBI, and Indirect or Direct Tunnel creation procedures.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 12.17 PGW/GGSN

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [439913] | <p><b>Symptoms:</b> Editing the description under configure&gt;mobile-gateway&gt;profile&gt;mipv4 default profile results in incorrect output. The description is printed, but mipv4 and exit are not printed before and after the description. This may lead to incorrect configuration error if a configuration snapshot is saved and then executed. Nokia recommends keeping the default mipv4 description.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

| Key                       | Description                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [434627]                  | <p><b>Symptoms:</b> Enabling Usage Monitoring and transport byte collection for the same policy may cause reporting inaccuracy.</p> <p><b>Workaround:</b> Avoid using these together.</p>                                                                                                                            |
| [431242]                  | <p><b>Symptoms:</b> When Gy DCCAv2 is in use, if dedicated bearers are in use, session deletion may occur during Geo-redundancy switchover.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                              |
| [411430]                  | <p><b>Symptoms:</b> For the Gx Assume Positive (retry capabilities) (limited support) (MG-3424) feature, _GGSN does not apply the QoS information from the QoS profile configured under APN, when the session goes to Assume positive during the attach.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [370567/00841379, 327210] | <p><b>Symptoms:</b> In case of DNS snooping, IPv4 only session and when both IPv4 and IPv6 DNS servers are configured within the matching criteria of a DNS PRU, then only DNS responses do not get matched to the correct RG but they end up in RG0.</p> <p><b>Workaround:</b> No workaround available, yet.</p>    |
| [385192]                  | <p><b>Symptoms:</b> If steering is configured only for one PDN type of a dual-stack UE, steering statistics incorrectly increment for both PDN types.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                    |
| [354340]                  | <p><b>Symptoms:</b> PCRF-specified Primary-Event-Charging-Function FQDN is not working when single-diameter-host is enabled.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                             |
| [341055]                  | <p><b>Symptoms:</b> When installing multiple static any/any rules with different precedence, only the rule with the highest precedence is displayed in the show&gt;mobile&gt;pdn&gt;bearer imsi detail command.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                          |
| [336497]                  | <p><b>Symptoms:</b> The action-rule-unit aa-notify-action is not visible with diff option. A commit can show the change.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                 |
| [317821]                  | <p><b>Symptoms:</b> The Gx Access-Network-Charging-Address AVP is not sent sometimes in the CCR-U message during inter-SGSN handovers when the Gn transport changes between IPv4 and IPv6.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                               |

| Key              | Description                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [313919]         | <p><b>Symptoms:</b> In multiple-UE sessions, the AMBR configuration must be identical for all UEs in HSS.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                             |
| [312992]         | <p><b>Symptoms:</b> With Multi-UE-sessions, network-initiated deletion of sessions may not be completed.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                              |
| [298647/ 298926] | <p><b>Symptoms:</b> The show&gt;mobile&gt;pdn&gt;bearer-context command displays inaccurate entries when identical policy rules exist in a multiple policy rule base.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [264132]         | <p><b>Symptoms:</b> If a Quality of Service IE is included in the Update PDP context response message sent from the SGSN, the GGSN ignores it.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                        |

## 12.18 SGW

| Key               | Description                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [389077]          | <p><b>Symptoms:</b> When a UE performs an IDLE TAU with SGW relocation in CUPS setup, if PGW sends DL traffic immediately after sending the MBResp to SGW, SGW drops the packets and does not trigger a DDN toward the MME. After a while traffic is sent again and it passes normally.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                 |
| [381119/01692079] | <p><b>Symptoms:</b> If during a RAU/TAU/HO procedure, the Linked EPS Bearer ID IE is not included either in the Create Session Request from the MME, or in the Modify Bearer Response from the PGW, the SGW sends back a Create Session Response with successful cause, but with mandatory IEs missing.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |

## 12.19 Single card combination gateway

| Key      | Description                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [174454] | <p><b>Symptoms:</b> In an SCCG configuration, the record KPIBearerTrafficMSMSGW will have all the counters set with a 0 value. The UL/DL traffic is counted as aggregate into the KPIBearerTrafficMSMPGW record.</p> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [193260] | <p><b>Symptoms:</b> In an SCCG configuration, receiving a downlink packet hashed to a non-LTE bonded session may cause an idle to active transition for the idle LTE session.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                    |

## 12.20 System

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [436782] | <p><b>Symptoms:</b> The following contexts using the prefix/full-id user-range command may fail when the start-id/end-id options have a value less than or equal to 9.</p> <pre>config&gt;mobile&gt;profile&gt;charging&gt;pgw&gt;online# config&gt;mobile&gt;pdn&gt;gx&gt;pcrf# config&gt;mobile&gt;pdn&gt;gy&gt;ocs-selection# config&gt;mobile&gt;pdn&gt;apn&gt;pcf-selection# config&gt;mobile&gt;pdn&gt;apn&gt;pco# config&gt;mobile&gt;pdn&gt;apn&gt;pcrf#</pre> <p><b>Workaround:</b> No workaround available, yet.</p> |
| [64581]  | <p>When the password-aging option is enabled, the reference time is the time of the last boot and not the current time. Password expiry is also reset on every reboot.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                                                                                                                                                                                                                                                                             |

## 12.21 TWAG

| Key      | Description                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | <p><b>Symptoms:</b> When logging into a hardware-based 7750 MG via CLI console, the following message appears on the console:</p> <p>The license is expired or missing. Please update your license as soon as possible.</p> <p><b>Workaround:</b> Nokia recommends ignoring this message because it is harmless, and it will be removed in the next release.</p> |
| [284929] | <p><b>Symptoms:</b> Traffic loss may be observed in some cases, after an inter-WLAN AP data triggered handover is performed. Depending on the PDN type, traffic is recovered if a DHCP or Neighbor Solicitation/Router Solicitation message is sent by the UE.</p> <p><b>Workaround:</b> No workaround available, yet.</p>                                       |

## 13 Usage notes

This chapter supplies information supplements or clarifies information in the user guides. Usage notes from past releases are added in this section with the label **[NEW]**.

### 13.1 3GPP release 13 standards

| Key                | Description                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [355493/1-7915794] | When the <code>rel-13</code> command is enabled under <code>config&gt;li&gt;mobile-gateway&gt;3gpp-release</code> , <code>uLITimestamp</code> is populated as 8 bytes. The last 4 bytes are filled always with zero. |
| N/A                | Baseline specification compliance is supported to 3GPP Release 13 December 2016, as described in the R9.0.R5 Release Notes. Statements of Compliance can be provided upon request.                                   |

### 13.2 5G

| Key             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442646]        | The <code>default-value-arp-pci-disabled</code> command under the <code>config&gt;mobile&gt;pdn</code> context controls the hardcoded value of <code>pre-emptionCapability</code> . This value is used in case SMF cannot fetch the value from UDM.                                                                                                                                                                                                                                                                                                                                                               |
| [430893]        | The server-side service instance <code>Nbsf_Management</code> is not supported on the SMF under the service realm. The client-side service instance <code>Nnef_PdfManagement</code> is not supported on the SMF or the UPF under the service realm.                                                                                                                                                                                                                                                                                                                                                               |
| [429828]        | Communication model D must not be enabled in the <code>namf-comm</code> service if AMF and SCP do not support AMF reselection based on NF set ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| [386185/418857] | 5G to EPS/EPS to 5G Handover in connected mode for PGW-C/SMF + PGW-U/UPF does not work without S11/S1u configuration. In 23.0.R2 release, CMG supports indirect forwarding only via S1-u/N3 and not via S5-u/N9. If the S11/S1u configuration is missing, SMF cannot select the <code>&lt;network-instance&gt;</code> to the FAR and UPF fails to handle the request. Negative response from UPF leads into session termination by the SMF. Issue can be avoided by adding a S1-u ref-point to the UPF configuration and the <code>&lt;s1-u interface realm&gt;</code> referenced under S11 ref-point to the SMF. |

| Key      | Description                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [390766] | The supported port range of Nsmf_PDUSession service instance is 80 443 8080.                                                                                                                                                       |
| [424028] | The supported port range of Nbsf_Management service instance is 80 443 8080.                                                                                                                                                       |
| N/A      | For the SMG event exposure (MG-1990) feature, the maximum limit of any UE subscriptions is eight.                                                                                                                                  |
| N/A      | For the PFCP Node ID and PFCP peer entity handling (MG-2594) feature, when the enable-partial-failure command is configured under node-id context, the command takes effect individually for all peer-entities under this node-id. |

### 13.3 All mobile gateways

| Key                                                 | Description                                                                                                                                                                                                                                                       |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446566/05636665]                                   | <b>[NEW]</b> In some cases, when RADIUS is used as NAT logging method, the Interim Update messages can also be generated.                                                                                                                                         |
| [445701/05293627]                                   | The resolution implemented for DTS 445701 makes the configuration of either of the epc-node or home-plmn-list command mandatory. If neither parameter is configured, the CMG drops all incoming requests.                                                         |
| [436778/03904301, 438402/04227868, 440222/04550356] | Event Log 99 may occasionally include logs reporting the following: Statistics bucket is not empty when the session is initialized. In case this log appears rarely and concerns a small number of bytes, it can be ignored because it does not cause any issues. |
| N/A                                                 | For MCC values that do not have an explicit MNC length standardized, the imsi-plmn-list command must be configured to indicate the actual MNC length (2 or 3 digit) to the system.                                                                                |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [433174] | <p>CPM/OAM may get busy printing extensive outputs to the SSH session (using "environment no more" option). As a result, other operations requiring the networking task, enter in contention for these resources and delay responses. As a workaround, use this output to redirect to a file, and then retrieve that file for outside system reading.</p> <p><b>Example:</b></p> <pre>admin display-config detail &gt; cf1:/output-to-a-file.txt Use only Local-url Options: # admin display-config detail &gt; - &gt; &lt;file-url&gt; &lt;file-url&gt; : &lt;local-url&gt; &lt;remote-url&gt; local-url - [&lt;cflash-id&gt;/]&lt;file-path&gt; 200 chars max, including cflash-id directory length 99 chars max each remote-url - {ftp tftp}://[&lt;login&gt;:&lt;pswd&gt;@] &lt;remote-locn&gt;/[&lt;file-path&gt;] 255 chars max, &lt;file-path&gt; 199 chars max remote-locn - { &lt;hostname&gt;   &lt;ipv4-address&gt;   "["&lt;ipv6- address&gt;"]" }[:&lt;port&gt;] ipv4-address - a.b.c.d ipv6-address - x:x:x:x:x:x:x[-interface] x:x:x:x:x:x:d.d.d.d[-interface] x - [0..FFFF]H d - [0..255]D interface - 32 chars max, for link local addresses port - [0..65535] cflash-id - cf1: cf1-A: cf1-B: cf2: cf2-A: cf2-B: cf3: cf3- A: cf3-B:</pre> |
| N/A      | <p>For the Non-terminating policy rules (MG-2233) feature, the following system limitations apply.</p> <p>When enabled, the eight million rules are divided into one million for non-terminating rules and seven million for the terminating rules (MG-VM).</p> <p><b>Example</b></p> <ul style="list-style-type: none"> <li>• average of five non-terminating rules per session, in 200k sessions</li> <li>• average of 30 non-terminating rules per session, in 33k sessions</li> </ul> <p>When disabled, the limit of the terminating policy rules per MG-VM is eight million.</p> <p>Although there is no session level limit on how many non-terminating policy rules can be installed in a session, a specific packet can only match with up to five non-terminating rules.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| N/A      | <p>For the Non-terminating policy rules (MG-2233) feature, the capacity is impacted by 5 to 25 percent that depends on the number of the active non-terminating policy rules and the number of the sessions that use the non-terminating policy rules.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [433427] | To upgrade from release 23.8R1/R2/R3 to 23.11 release, or from 23.8R1/R2/R3 to 23.8R4, the hold-timer-sec value command must be configured manually before the upgrade if the hold-timer command is also configured.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A      | The Router/VRPN interface configured under any mobile-gateway reference-point must be configured with at least one IP address. This IP address must not be modified or removed when the PDN (under the config>mobile-gateway>pdn context) is in no shutdown state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [431084] | <p>The following configuration commands can be found by issuing the tree flat command in MD-CLI, but these must not be used with any model-driven interface (NETCONF/MD-CLI/gNMI) in release 23.8R1.</p> <ul style="list-style-type: none"> <li>• configure mobile-gateway pdn s2b dual-stack-cplane-pref</li> <li>• configure mobile-gateway pdn s2b dual-stack-uplane-pref</li> <li>• configure mobile-gateway pdn s2b signaling gtp-u</li> <li>• configure mobile-gateway pdn s2b signaling gtp-u profile</li> <li>• configure mobile-gateway pdn s5 dual-stack-cplane-pref</li> <li>• configure mobile-gateway pdn s5 dual-stack-uplane-pref</li> <li>• configure mobile-gateway pdn s5 send-end-marker</li> <li>• configure mobile-gateway pdn s5 signaling gtp-u</li> <li>• configure mobile-gateway pdn s5 signaling gtp-u gtpu-keep-alive</li> <li>• configure mobile-gateway pdn s5 signaling gtp-u profile</li> </ul> |
| [428712] | Alarm for upcoming license expiration is not visible at "show system alarms". The CMG user can use the show>system>license command to retrieve the information about the expiration of the deployed license.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| [382756] | In the event of a double MG failure on Slave node, customers must make sure to do a recovery procedure in the Slave node before doing any further ICR switchover to make this node Master. Recovery procedure is to perform local MG switchover on the specific group where double MG failure happened.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| N/A      | If the operator does not intend to configure GTP-U path management failure actions, a new GTP profile with keep-alive interval set to 0 must be configured for S1-u and N3 interfaces to disable GTP-U path management for them. Otherwise, performance impact may be observed due to the vast number of that kind of peers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [402422/02301716] | Switch over state" field name of show>mobile-gateway system command, is changed into graceful switchover as this field is updated only when the admin redundancy mg-switchover command is executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| N/A               | "Switch over state" field of show mobile-gateway system command output is updated only when MG-VM controlled switchover is triggered using admin redundancy mg-switchover command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| N/A               | With the algorithm ( <u>Improve IP allocation mechanism (MG-2786)</u> ) and with an approximately equal attach/detach rate amongst the IP-pools, the system stays balanced. If the UEs (found on a specific IP-pool) are all detached, or the IP-Pools are of unequal size, the load-sharing algorithm tries to use all available IP-pools, and not to load balance the IP-pools. Such a load-balancing algorithm requires an algorithm that keeps a per IP-pool occupancy rate, and then, order the IP-pools based on the stored occupancy rate. This enhancement is currently not planned. However, the load-sharing algorithm can efficiently perform balancing in most cases while improving the current mechanism.                                                                          |
| [378338]          | Automatic re-balancing – as per <u>Improve IP allocation mechanism (MG-2786)</u> feature – of UE IP subnets across CP MG groups during system growth or ICR ISSU has been disabled. Systems that are subject to such MG growth (adding new CP MG groups to existing systems) may end up with non-balanced UE IP subnets across those MG groups. Systems experiencing non-balanced UE IP subnets across MG groups may need to either block or to drain existing UE IP prefixes or add new prefixes to achieve balance across MG groups. Systems that require MG growth and do not use ICR ISSU to upgrade, can add new MG groups during the upgrade procedure. The system reboot required by the upgrade procedure ensures that UE IP subnets are balanced across all new and existing MG groups. |
| N/A               | <b>Workaround:</b> Nokia recommends that before committing the policy changes with the config>mobile>profile>policy-options#commit command, operators must confirm that there are no on-going police changes already taking place by running the tools dump mobile-gateway gw-id policy-on-the-fly-update-stats command and checking the 'Policy Update End time'. This is particularly relevant for Systems with a high number of UEs and/or active policy rules.                                                                                                                                                                                                                                                                                                                               |
| [361790]          | PDP/bearer contexts may be removed up to five minutes earlier than the configured value of the session-timeout command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Key              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [353764]         | When the ECMP with OSPF is enabled, the ECMP routes are displayed in the routing table; however, they are installed incorrectly in the FIB tables. This may impact routing and traffic. This is applicable only to MG deployments. ECMP inter-working with static route and BGP are not impacted.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A              | For the OAM Switchover Enhancement in case of Lost Access to Storage Nodes (MG-2332) feature, the following apply: <ul style="list-style-type: none"> <li>An OAM switchover is triggered only if the secondary OAM is available.</li> <li>In the current release there is no buffering of the OAM charging data. User data which was written to disk is lost.</li> </ul> Switchover when detecting storage failure is implemented only for VirtIO base devices.                                                                                                                                                                                                                                                                              |
| [337204]         | The <code>config&gt;mobile&gt;profile&gt;sig-rate-limit-profile&gt;max-message-rate</code> command is per Diameter client. If the <code>config&gt;mobile&gt;pdn&gt;gx&gt;single-diameter-host</code> or <code>config&gt;mobile&gt;pdn&gt;gy&gt;single-diameter-host</code> command is enabled, the <code>max-message-rate</code> refers to the total rate; that is, the maximum number of allowed transactions per second on a Diameter client for all the MG groups because the mobile gateway is gathering all the Diameter messages into one diameter client. If <code>single-diameter-host</code> is not enabled, the <code>max-message-rate</code> is configured per MG group because each MG group is running its own Diameter client. |
| [333867]         | The <code>dynamic-rule-to-internal-rule-map rules</code> command must be applied to dedicated bearers only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [323525, 322315] | If the default interface is not configured under PDN or a specific GTP interface, the override profile will not take effect for this specific interface. Therefore, it is mandatory to configure a default maxGTP interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| N/A              | For DSCP-based bearer binding, the <i>dscp-value</i> parameter of the <code>config&gt;mobile&gt;profile&gt;policy-options&gt;policy-rule-unit&gt;flow-descriptor&gt;match&gt;tos-tc</code> command corresponds to a DSCP value in decimal format (range 0 to 63). In GTP-c, this value is mapped under the Traffic Flow Template to its corresponding Type of Service value in hexadecimal format.                                                                                                                                                                                                                                                                                                                                           |
| [311696]         | The GW/Group/VM field has been added to the <code>show&gt;mobile&gt;pdn&gt;pdn-context detail</code> command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| N/A              | When a DPR is received, the first connection attempt (CER) is performed on the interval specified for the <code>dpr-timeout</code> command. If a TCP connection is established but no CEA is received, then on expiration of the transaction timer, a new TCP/CER connection is attempted after the length of time equal to the configured DPR timeout.                                                                                                                                                                                                                                                                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | When using IP local pools, the mobile gateway is designed to be most efficient at allocating UE IP addresses with a high number of UEs. The absolute minimum is a /30 prefix per MG group. Furthermore, the system is designed to work most efficiently when each MG group is configured with at least 2000 addresses (equivalent to a /21 prefix) per MG group using ISM-MG-C cards. Operators must consider these parameters when increasing the number of MG groups in a node. |
| [306997] | Type NAI is not supported with the stats keyword in the <code>show&gt;li&gt;mobile&gt;target</code> command.                                                                                                                                                                                                                                                                                                                                                                      |
| [306183] | If a v4 framed route is received when establishing a v46 PDN session, the v4 framed route is installed and the v46 PDN session is established. Previously, a v4 PDN session was established, instead of the requested v46 PDN session.                                                                                                                                                                                                                                            |
| N/A      | The <code>tools&gt;perform&gt;mobile-gateway&gt;pdn&gt;force-clear&gt;path</code> command must only be used under direction of your Nokia technical support representative.                                                                                                                                                                                                                                                                                                       |
| [299075] | Only MDA 1 is supported for ISM-MG memory and CPU statistics in the <code>show&gt;mobile&gt;ism-mg</code> context.                                                                                                                                                                                                                                                                                                                                                                |
| N/A      | To detach a large number of UEs, Nokia recommends using per APN-based clear commands (for example: <code>clear mobile-gateway pdn 1 bearer-context apn</code> ).<br>Ensure that the configuration does not cross 256/sec throughout all gateways because the value on the CLI is based on the MG group. Therefore, you must set this value according to the total number of the groups present in the system.                                                                     |
| N/A      | All the policy rules in one policy rule base must have the same data-driven-dedicated-bearer value on the action-rule-unit.                                                                                                                                                                                                                                                                                                                                                       |
| N/A      | Policy rules and policy rule bases that are configured under the <code>config&gt;mobile&gt;pdn&gt;apn&gt;static-local-pcc&gt;dedicated-bearers&gt;default-bearer-qci arp</code> command cannot have the same QCI and ARP combination as the entry itself.                                                                                                                                                                                                                         |
| N/A      | The session deletion rate configured using the <code>config&gt;mobile&gt;pdn&gt;session-deletion-rate</code> command is not an enforced rate for session deletion. It is treated by the system as the maximum rate to be applied. The same applies for the explicit rate set by the <code>clear&gt;mobile&gt;bearer-context</code> commands.                                                                                                                                      |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | Enabling the <code>disable-learned-dest-realm</code> command will disable dynamic learning of destination realms thereafter. For existing sessions that have already learned a destination realm, the realm that was last learned is used from then on.                                                                                                                                                                                                                                                                                                                               |
| N/A      | The same IP interface must not be used for the same reference-point or by the default signaling interface configured in multiple PDN instances.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| N/A      | In any policy-rule-base, all associated policy-rule-units with the <code>aa-advanced-functions</code> command enabled must have the same flow-gate status and charging properties configured.                                                                                                                                                                                                                                                                                                                                                                                         |
| N/A      | All charging groups defined under AA (using the <code>config&gt;aa&gt;group&gt;policy</code> command) must be part of an active policy rule (using the <code>config&gt;mobile&gt;profile&gt;policy-options&gt;policy-rule</code> command). Under certain conditions, unexpected packet drops, or charging may occur if AA detects traffic for a charging group that is not part of an active policy rule.                                                                                                                                                                             |
| N/A      | The PDN type of a real APN and virtual APN may differ, and only the PDN type of the final APN is checked during an attach.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A      | Receiving and processing a GTP Error Indication is supported on S1-u, S4, S5, S8, S2a, S2b, S11-u, N3, N9, and Gn/Gp (GTPv1) interfaces. Support for other interfaces will be added in future releases.                                                                                                                                                                                                                                                                                                                                                                               |
| N/A      | A policy-rule-base that contains both AA policy rules and 5-tuple policy rules must also contain a policy-rule with the "auto-redirect" flow-gate-status for Gy-based HTTP redirection to work correctly.                                                                                                                                                                                                                                                                                                                                                                             |
| [241369] | Modification of default SWw tunnel profile parameters for TWAG is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [240819] | APCO in an Update Bearer Request is encoded under bearer context.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [243533] | The <code>retry-count</code> and <code>retry-timeout</code> commands in the <code>config&gt;mobile&gt;profile&gt;radius</code> context require a shutdown and no shutdown action in the radius-group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer. As of release 9.0.R5, the <code>retry-count</code> and <code>retry-timeout</code> commands have been replaced by the <code>auth-retry-count</code> , <code>auth-retry-timeout</code> , <code>acct-retry-count</code> , and <code>acct-retry-timeout</code> commands. |
| [227504] | The reporting level for a rating group installed to an active session cannot be changed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [227504] | For the Service ID or Sponsor ID reporting level, a billing method of only online charging is not supported. Offline charging must be enabled for PCC rules with these reporting levels.                                                                                                                                                                                                                                    |
| N/A      | Configuring the suppress-dotzero-broadcast-address command does not take effect for IPv4 pools with masks from /25 to /30.                                                                                                                                                                                                                                                                                                  |
| N/A      | Starting in release 7.0.R1, the only supported combined SGW/PGW configuration is with both functions running on the same gateway instance. Consequently, MG groups run both functions. Running each function on separate groups is no longer supported.                                                                                                                                                                     |
| N/A      | It is recommended that all charging groups defined under AA (using the <code>config&gt;aa&gt;group&gt;policy</code> command) be part of an active policy rule (using the <code>config&gt;mobile&gt;profile&gt;policy-options&gt;policy-rule</code> command). Under certain conditions, unexpected packet drops, or charging may occur if AA detects traffic for a charging group that is not part of an active policy rule. |
| N/A      | If dual-stack is available for the user-plane encapsulation, IPv6 is used by default.                                                                                                                                                                                                                                                                                                                                       |
| N/A      | If it is necessary to shut down the gateway function, appropriate actions must be taken when restoring the function. This is recommended to prevent signaling packets on the interfaces toward the UE from reaching the gateway until all the necessary interfaces facing the network are operational.                                                                                                                      |
| N/A      | On an SGW, it is advisable to block access to S4/S11/S12 interfaces by turning down the service/physical interface or by applying a filter until all charging interfaces are verified to be ready and the S5/S8 peers can be reached.                                                                                                                                                                                       |
| N/A      | A PGW/GGSN must not be reachable on S5/S8/Gn/Gp/S2a/S2b interfaces, until the charging servers and the AAA and PCRF functions are ready.                                                                                                                                                                                                                                                                                    |
| N/A      | The Reference point field displayed with the <code>show&gt;mobile&gt;pdn&gt;ref-point-stats {s1u   s11   s5   s8   gn   gp   s2a   s2b   s4   s12}</code> command shows a number of peer instances, that is, $x + 1$ , where $x$ is the number of mobile gateway groups and 1 accounts for the active CPM.                                                                                                                  |
| N/A      | After a switchover between the active and standby ISM-MG, the DDN damp timer is reset. A subsequent DDN from the SGW will be forwarded to the MME. If the MME confirms that the UE is not reachable, further DDNs will be dampened during the DDN damp timer.                                                                                                                                                               |

| Key      | Description                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | The <code>show&gt;mobile&gt;pdn&gt;statistics summary</code> command displays the utilization of all mobile gateway groups assigned to the SGW. It summarizes the numbers from the various active cards, including the number of eNodeB, MME, PGW, and Rf peers.                                                                  |
| N/A      | When using a very large number of eNodeBs, it is recommended to disable path management on the S1-u interface to save resources. To disable path management, the GTP profile associated with the S1-u interface must have the keep-alive interval configured to 0.                                                                |
| N/A      | Charging (offline and/or online) is required for correct reporting of KPI bearer traffic statistics on either a per-APN or per-interface basis.                                                                                                                                                                                   |
| N/A      | An APN named "unknown" must be configured on the PGW/GGSN to handle sessions/PDP requests that are destined to APNs that are not specifically configured. The "unknown" APN is not configured by default, but it is a recommended configuration. It may be left in the shutdown state.                                            |
| N/A      | If dual-stack is available for the user-plane encapsulation, IPv6 is used by default.                                                                                                                                                                                                                                             |
| N/A      | The maximum number of filter descriptions that can be signaled when establishing a dedicated bearer or secondary PDP context is 15. Therefore, any policy rule unit that is expected to be used for creating such a bearer or context must not have more than 15 flow descriptions.                                               |
| N/A      | A policy rule base that contains an AA policy rule requires a 5-tuple policy rule to be configured for bearer binding. An AA policy rule has a policy rule unit matching on an AA charging group. A 5-tuple policy rule has a policy rule unit with one or more flow descriptions matching the network and/or transport criteria. |
| N/A      | An SDF Next-Hop selection can be configured to redirect traffic to several next-hops, in which case redirected traffic will be load-balanced among the next-hops per PDN connection. However, there is no failover mechanism, so connectivity to the next-hop must have its own redundancy.                                       |
| N/A      | Adding the same physical interface under multiple Pi reference points is not supported.                                                                                                                                                                                                                                           |
| [147756] | The following is the behavior for handling two instances of Create Pdp Context Request back-to-back for the same APN.                                                                                                                                                                                                             |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | If the second Create Pdp Context Request is from a different SGSN peer, a Delete Pdp Context Request will be sent to the SGSN peer that sent the first Create Pdp Context Request.                                                                                                                                                                                                                                                                                         |
| N/A      | If both SGSN peer and control TEID are the same for the two Create Pdp Context Requests, no Delete Pdp Context Request will be sent.                                                                                                                                                                                                                                                                                                                                       |
| N/A      | If the same SGSN peer is used but both the control TEID and NSAPI are different, a Delete Pdp Context Request will be sent to the SGSN.                                                                                                                                                                                                                                                                                                                                    |
| N/A      | When using a PCRF to install a PCC over the Gx interface, the configured ARP values must match the following mapping table: <ul style="list-style-type: none"> <li>• Pre-Rel-8 (Gn) EPS ARP (Gx)</li> <li>• 1</li> <li>• 6</li> <li>• 11</li> </ul>                                                                                                                                                                                                                        |
| N/A      | A statically configured policy rule with a flow-gate status set to HTTP redirect can only be activated from the PCRF by referencing a statically configured policy rule base that contains this policy rule. Furthermore, if it is necessary to allow traffic to destinations (such as redirect server IP address, DNS servers, and so on) while the session is in the redirect state, those policy rules must also be specified as part of the redirect policy rule base. |
| N/A      | The charging method configured under the <code>config&gt;mobile&gt;profile&gt;policy-options&gt;charging-rule-unit&gt;charging-method</code> command is now considered and can be used to enable offline and/or online charging.                                                                                                                                                                                                                                           |
| [225551] | The pause chrg hold timer start and pause chrg hold timer stop counters have been moved from the PDN Gateway Statistics section to the Serving Gateway Statistics section under <code>show&gt;mobile&gt;pdn&gt;statistics</code> .                                                                                                                                                                                                                                         |
| [269162] | When subscription-id-override is enabled under <code>config&gt;mobile&gt;pdn&gt;s6b</code> , the following commands with the MSISDN option are not supported. <ul style="list-style-type: none"> <li>• <code>show&gt;mobile&gt;pdn&gt;ue-context msisdn</code></li> <li>• <code>clear&gt;mobile&gt;pdn&gt;bearer-context msisdn</code></li> </ul>                                                                                                                          |
| [259533] | The reject-foreign-subscribers command is not supported under the <code>config&gt;mobile&gt;pdn&gt;apn</code> command.                                                                                                                                                                                                                                                                                                                                                     |
| N/A      | The configured SSH DSCP policy applies to all applications that use SSH on the gateway.                                                                                                                                                                                                                                                                                                                                                                                    |

| Key | Description                                                                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | If the DSCP configuration is modified after a TCP connection to a peer is brought up, the TOS (DSCP) of packets sent after the configuration change will reflect the modified value. There is no need to bring down the TCP connection to the peer. |
| N/A | You cannot specify the FQDN group and remote IP options for the same flow description.                                                                                                                                                              |
| N/A | When the <code>aa-advanced-functions</code> command is enabled, IP-flows resolved for the FQDN group are diverted to AA.                                                                                                                            |

## 13.4 Application assurance

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | In all platform types (7750MG, CMG, CMG-a, CNF, and so on) the ISA AA MDAs must be configured in the second MDA complex of MG cards or VMs.                                                                                                                                                                                                                                                  |
| N/A | The bandwidth capacity for an AA subscriber is equal to the full capacity of the ISA-AA card, provided there is a realistic diversity of 5-tuple flows. The 5-tuple flow limits remain at 200 Mb/s per flow (in both directions).                                                                                                                                                            |
| N/A | The <code>isa-aa.tim</code> image is available in the same directory as other <code>.tim</code> images. The image contains the AA software used on MS-ISA and the protocol list loaded by the CPM. The AA software can be upgraded independently of the 7750 SR MG software within a major release of the 7750 SR MG.                                                                        |
| N/A | Operators using applications maintained by Nokia for charging or control must update both protocol signatures and the AA policy definition on a regular basis. New and updated protocols are available in the <code>isa-aa.tim</code> file while the AA policy update is provided through technical support. For more details, see <a href="#">AA signatures upgrade procedure for CMG</a> . |
| N/A | An AA policy rule requires an active 5-tuple rule for bearer binding. If no such 5-tuple rule is explicitly activated using either PCRF or static configuration, the system will automatically enable a 5-tuple any-any rule that allows all user traffic to be diverted to the AA engine.                                                                                                   |

## 13.5 Call trace

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [305565] | The all option has been removed from the <code>config&gt;mobile&gt;profile&gt;call-insight&gt;ue&gt;user-traffic</code> command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| N/A      | OAM UE ping packets cannot be captured by call trace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| N/A      | User-plane call trace in CMG systems is supported for general availability since release 8.0.R8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| N/A      | <p>When user traffic is enabled using the <code>user-traffic</code> command in the context <code>config&gt;mobile&gt;profile&gt;call-insight&gt;ue</code>, user-plane packets are captured. The packets are captured once (for both ingress and egress interfaces) as they pass through the gateway and are captured without the tunneling and transport headers (valid for SR 7750 hardware). New arguments have been added in the <code>user-traffic</code> command and user-plane packets can be captured with tunnel and transport headers in LB-VMs (valid for CMG and Cloud ePDG).</p> <ul style="list-style-type: none"> <li>User traffic can be enabled for up to 15 UEs.</li> <li>PCC metadata is recommended to be enabled for up to 5 UEs</li> </ul> |

## 13.6 Charging

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [429418] | Charging profiles under the <code>config&gt;mobile&gt;profile&gt;charging</code> context may now be deleted on the fly. When a charging profile is deleted, existing sessions using the profile will continue as is. For new session establishments, if the referred charging profile does not exist, then the session will not be charged.                                                                                                                                                                                          |
| [430710] | The reference of a non-existing PGW/SGW/SMF-charging profile is allowed under the <code>config&gt;mobile&gt;pdn&gt;charging&gt;charging-profile {home   roam   visiting}</code> or <code>config&gt;mobile&gt;pdn&gt;apn&gt;charging&gt;charging-profile {home   roam   visiting}</code> contexts. This may be considered as a misconfiguration but there is no longer any restriction at CLI to prevent this. For new session establishments, if the referred charging profile does not exist, then the session will not be charged. |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>The following charging information triggers that are configured in the <code>config&gt;mobile&gt;profile&gt;charging&gt;pgw-charging&gt;off-line&gt;charging-info-trigger</code> context are supported for the Rf interface only and must not be used for Ga charging.</p> <ul style="list-style-type: none"> <li>• <code>gy-final</code></li> <li>• <code>gy-forced-reauthorization</code></li> <li>• <code>gy-qht</code></li> <li>• <code>gy-quota-exhausted</code></li> <li>• <code>gy-rating-condition-change</code></li> <li>• <code>gy-tariff-time-change</code></li> <li>• <code>gy-threshold-vqt</code></li> <li>• <code>gy-validity-time</code></li> <li>• <code>ePDG</code></li> </ul> |

### 13.7 CMG appliance

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p><b>[NEW]</b> After the replacement of <code>svr-1</code>, OAM-A comes up with a delay of approximately 15 minutes. During this period, OAM-B remains the active unit, and logging in to <code>svr-1</code> is not possible.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| N/A | <p>To upgrade from an older to a newer release, see section <a href="#">CMG-a upgrade paths</a> for the supported paths.</p> <p>However, when upgrading to release 26.3.R1 from an earlier release, TPM encryption is not enabled by default.</p> <p>To enable TPM encryption in release 26.3.R1, you must activate it during clean installation only, applying one of the two methods:</p> <ul style="list-style-type: none"> <li>• performing a full software re-installation following the installation instructions, or</li> <li>• executing a server replacement procedure on each server</li> </ul> <p>For the server replacement method, it is mandatory to ensure that Secure Boot and TPM is enabled in the BIOS prior to executing the server replacement command. Failure to do so will result in the installation failing.</p> <p>This functionality applies only to Gen11 and Gen12 hardware. For Gen10 platforms and AirFrame systems, TPM is not supported and, therefore, TPM encryption cannot be enabled on those systems.</p> |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [442114/04523207] | The management information base (MIB) used for the communication between the CMG-a SNMP agent and the SNMP manager is the <code>TIMETRA-AIRFRAME-MIB.mib</code> file and the enterprise object identifier (OID) is 6527 assigned to Nokia (TiMetra) by Internet Assigned Numbers Authority (IANA). In the AirFrame deployments, when the MP9 firmware package is used (BMC version 3.94.00), the <code>TIMETRA-AIRFRAME-MIB.mib</code> file defines the SNMP trap OID to be 111. Any previous firmware package comes with an MIB that defines the SNMP trap OID to be 97.          |
| N/A               | <p>Due to the following vulnerabilities that are identified in HP iLO:</p> <ul style="list-style-type: none"> <li>• TLS Version 1.0 Protocol Detection</li> <li>• TLS Version 1.1 Protocol Deprecated</li> <li>• SSL Medium Strength Cipher Suites Supported (SWEET32)</li> <li>• SSH Server CBC Mode Ciphers Enabled</li> </ul> <p>the operator must set the values in HP iLO under Security &gt; Encryption as follows:</p> <ul style="list-style-type: none"> <li>• set Security state to <b>High Security</b></li> <li>• leave TLS 1.0 and TLS 1.1 <b>unchecked</b></li> </ul> |
| [434097/03580555] | For the CMG-a update to RHEL 9.2 (MG-4124) feature, in RHEL 9, the OpenSSL library does not support the DES privacy protocol. The protocol is marked as insecure, so DES support has been removed.                                                                                                                                                                                                                                                                                                                                                                                 |
| [431571/03252889] | In CMG-a6 deployments, where HPE DL325 Gen10+ AMD EPYC servers are used, the average CPU utilization for certain MG-VMs may be higher than the average CPU utilization for all other MG-VMs. The reason for this is that some application processes may need to communicate with application processes that are located on a different group of CPU cores. In this case, the MG-VM performance is affected.                                                                                                                                                                        |
| N/A               | <p>To access DB-VM on CMG-a RM19 N:K deployments, the user can use the SSH from either the hypervisor-1 or hypervisor-2 with the <code>ssh cmga@10.10.0.103</code> command.</p> <p><b>Note:</b> If server 2 is replaced, access to the DB-VM is not possible from server 2.</p>                                                                                                                                                                                                                                                                                                    |
| N/A               | When a server has been replaced on CMG-a, the software rollback procedure does not work. Only the software upgrade or fresh installation can be performed. CLI output has been updated to display an error in case the user attempts to rollback an appliance with a replaced server.                                                                                                                                                                                                                                                                                              |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | CSF on all CMG-a appliances (RM18/RM19) has been reconfigured to use the SR-IOV connectivity. For that reason, two CSF networks are now configured in each VM with new VLAN IDs that must be configured on WBX/7850 switches as well. For more details, refer to the <a href="#">CMG-a software upgrade</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| N/A | Due to unexpected connectivity issues that may occur with WBX directly connected to IPMI ports, you must configure the IPMI connectivity on CMG-a8 through an external switch. Connectivity between external switch and WBX must be at least 10GE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| N/A | Contact your Nokia technical support representative for any Layer 2 functionalities to be enabled in CMG-a including but not limited to VPLS, rVPLS, VRRP, and LAG.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| N/A | On the RM18 and RM19 CMG-a, you must use either RM19 Server PP4.2 (BMC 3.60.10), PP4.6 (3.60.15) or PP4.8 (3.61.01).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A | <p>The CMG-a host SSH ciphers and MACs have been updated to minimize vulnerability of SSH access. The following lists the updates:</p> <p>ssh2-enum-algos:</p> <ul style="list-style-type: none"> <li>  encryption_algorithms (7)</li> <li>  <a href="#">chacha20-poly1305@openssh.com</a></li> <li>  aes128-ctr</li> <li>  aes192-ctr</li> <li>  aes256-ctr</li> <li>  aes128-cbc</li> <li>  aes192-cbc</li> <li>  aes256-cbc</li> <li>  mac_algorithms (6)</li> <li>  <a href="#">hmac-sha2-512-etm@openssh.com</a></li> <li>  <a href="#">hmac-sha2-256-etm@openssh.com</a></li> <li>  <a href="#">umac-128-etm@openssh.com</a></li> <li>  hmac-sha2-512</li> <li>  hmac-sha2-256</li> <li>  hmac-ripemd160  hmac-ripemd160</li> </ul> <p>The SSH client must support at least one cipher and MAC algorithm listed above, otherwise an SSH connection is not possible after a 10.0.R10 software upgrade.</p> |
| N/A | For CMG-a8 release 10.0 and 9.0.R10 and later, it is recommended to use Nuage 7850 VSG-VSA-5.2.2-27 software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| N/A | CPU hyperthreading on the CMG-a2/a8 is always enabled and cannot be disabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Key | Description                                                                                                                                                                                                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | To enable AirFrame server fault and performance management on the CMG-a2, the server's IPMI management port must be connected via an RJ45 Cat5 Ethernet cable to an external switch/router with Layer 2 connectivity to the configured AIM interface, as described in the <i>CMG Appliance Installation and System Administration Guide R11.0.R1</i> . |

## 13.8 CMG

| Key             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [443872]        | A plain text hashed admin-password that is configured under default Day 0, may cause security vulnerabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| [434487]        | <p>The value of track-mobile cannot be modified when this interface is referenced by any other application.</p> <ul style="list-style-type: none"> <li>• <code>config&gt;router interface no track-mobile</code></li> <li>• <code>config&gt;router interface track-mobile</code></li> <li>• <code>config&gt;service ies interface ipv6 no track-mobile</code></li> <li>• <code>config&gt;service ies interface ipv6 track-mobile</code></li> <li>• <code>config&gt;service ies interface no track-mobile</code></li> <li>• <code>config&gt;service ies interface track-mobile</code></li> <li>• <code>config&gt;service vprn interface no track-mobile</code></li> </ul> <p><code>config&gt;service vprn interface track-mobile</code></p> |
| [435073/347037] | As per standard, SGW cannot change its UP F-TEID during connected mode handover or Service Request. 3GPP allows the F-TEID to change during idle mode mobility, that is, TAU/RAU procedures, but to support seamless handover between s4 and s11, the mobile gateway must use the same IP address for CP, and the same IP for UP on CUPS and non-CUPS deployment.                                                                                                                                                                                                                                                                                                                                                                          |
| [431365]        | In CMG, when employing Cloud database, max-preparation-time must be set higher than 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [371305] | <p>Security hardening changes applied to the SSH daemon configuration parameters for the DB Proxy VM. The updated parameters and values are:</p> <ul style="list-style-type: none"> <li>• Compression no</li> <li>• LogLevel VERBOSE</li> <li>• MaxSessions 2</li> <li>• AllowAgentForwarding no</li> </ul> <p>The MaxSession parameter change affects the Ansible configuration which uses SSH multiplexing. To avoid problems, either temporarily set the MaxSessions parameter to 10 or set the SSH client configuration to ControlMaster no.</p> <p><b>Note:</b> The "-o ControlMaster=no" must be added to attribute <code>ansible_ssh_extra_args</code> in both <code>ansible/executeDbActions.yaml</code> and <code>ansible/verifyDbStatus.yaml</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| N/A      | <p>For fast and reliable resource allocation, Nokia recommends setting the hyper-threading parameter to 2 (that is, ht=2).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A      | <p>The Openstack CMG CBAM templates can be used with minor adaptations for CMG deployment in VMware VIO. The main differences in the VMware VIO templates are:</p> <ul style="list-style-type: none"> <li>• 1 CSF/DSF network</li> <li>• no CSF/DSF VLAN IDs</li> <li>• no SR-IOV port type</li> <li>• no vf2vf extension</li> <li>• IDE bus type in OAM CF1/CF2 disks</li> <li>• OAM/DB volume types</li> </ul> <p>Volume type resources and IDE bus type disks definitions are specific to VIO deployment to prevent relevant issues when disks are not recognized by the system. The above adaptations are supported by the Spinner tool. The root packages included in Spinner Tool are the MG+ ones. To generate LB mode templates, the LB-Mode Package Conversion parameter must be set to True. The VMware Integrated Openstack parameter supports two options:</p> <ul style="list-style-type: none"> <li>• DPDK (CSF/DSF networks are created by CBAM)</li> <li>• DPDK (CSF/DSF networks are created externally)</li> </ul> <p>In the first one, CBAM can create CSF/DSF network entities during instantiation.</p> <p>In the second one, the user is required to have created these networks manually in VIO and then just define them UUIDs in CBAM side. Also, VIO restricts using only /32 addresses in externalIpAddressList extension. Stack creation fails if the 0.0.0.0/0 address is used. If multiple external IP addresses are required, you must separate them with commas, for example: 1.1.1.1/32, 2.2.2.2/32.</p> |

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [391044/02084623] | When statistics are collected from an NFM-P via SNMP for a large number of APNs, the <code>config&gt;mobile&gt;pdn&gt;apn&gt;dedicated-bearer-idle-timeout-profile default</code> command must be set under all APNs to avoid delays in CLI.                                                                                                                                                                                                                     |
| N/A               | In CMG NK deployments, the users must be cautious with the software images for DB-VM and CMG-VMs. Both images must be from the same release. If images from different releases are selected in CBAM GUI (for example, UUID in Openstack, or the OVA file consists of disks from different images), CBAM cannot detect this mismatch and continues the software upgrade LCM operation.                                                                            |
| [368285]          | MG-VM controlled switchover must not be used in parallel with commands that can lead to bulk/massive operations, such as <code>clear</code> commands, <code>admin tech-support/tools monitor/perform</code> , and policy configuration changes.                                                                                                                                                                                                                  |
| [347037]          | The CMG platform does not support configuration of both ends of the S11/S4 interfaces with the same IP addresses. To support seamless handover between S4 and S11, it is recommended that the MME/S4-SGSN end uses different IP addresses, while the gateway (SGW) end uses the same IP address.                                                                                                                                                                 |
| [348761]          | The configuration of the <code>local-ecmp-egress</code> command is not visible when an <code>admin&gt;save&gt;detail</code> is configured before an <code>admin&gt; reboot</code> . As a workaround, the <code>admin&gt;save</code> command can be used instead of <code>admin&gt;save&gt;detail</code> .                                                                                                                                                        |
| N/A               | For deployment using VMware NFV3.2 only, a known issue in the VMware vCloud Director in releases 9.7 and 10.0 results in the CPU and memory reservations not being preserved on newly deployed VMs, as specified in the CMG templates. Consequently, the CMG-VMs (which have latency sensitivity set too high) do not power on. To correct this issue, the network operator must change these settings manually after deployment, and before powering on the VM. |
| N/A               | Contact your Nokia technical support representative for any Layer 2 functionalities to be enabled in CMG, including but not limited to VPLS, rVPLS, VRRP, and LAG.                                                                                                                                                                                                                                                                                               |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>Version 12.1.1 CMG CBAM Templates consist of the following two sets of VMWare Templates for N:K Redundancy:</p> <ul style="list-style-type: none"> <li>Package A set supports Grantful and Grantless Modes with single EPCs and they are named as cMgCbamVmxnet3MgRedirectNkRedundancyPackageA_v12.1.1 and cMgCbamVmxnet3NkRedundancyPackageA_v12.1.1</li> <li>Package B set support only Grantless Mode with multiple ECPs and they are named as cMgCbamVmxnet3MgRedirectNkRedundancyPackageB_v12.1.1 and cMgCbamVmxnet3NkRedundancyPackageB_v12.1.1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| N/A | <p>In N:K Redundancy, after the instantiation of the CDB VM with CBAM templates, unless the IP address of the external interface belongs to the subnet as the respective ip used by the CMG, then the CDB-CMG communication does not work properly due to invalid routing table in the CDB-VM.</p> <p>This is because by default, the "default" route is served by the mgmt interface (in our case eth0). Hence, a route must be added in the cdb vm to instruct the traffic from/toward the CMG ip to be served by the external interfaces (bond0), otherwise communication does not work.</p> <p>Example of such routes:</p> <pre>sudo ip route add IP_OF_CMG dev bond0 via GATEWAY_OF_EXTERNAL_IP_OF_CMG sudo ip route add IP_OF_CMG dev bond0</pre> <p><b>Example</b></p> <pre>sudo ip route add 10.0.0.192 dev bond0 via 192.168.22.1</pre> <p>To make this route persistent across reboots, you need to create the relative route file:</p> <pre>sudo vi /etc/sysconfig/network-scripts/route-bond0 10.0.0.192 dev bond0 via 192.168.22.1</pre> |
| N/A | <p>In dynamic licensing with CLM, if one OAM VM is offline/down in CMG side for any reason and healing is needed, customer has to check through show bof the licenseURL file name that is currently saved. Then, they must copy the &lt;NF_NAME&gt;_CMG_temporary_license.dat contents to the already existing license file that was used in OAM which is currently down. This is important because for the missing/deleted OAMs CMG CBAM Templates cannot update BOF so if we do not manually update this with wildcard license content, healed OAM comes up with expired license and is in down state (since CMG looks by default for BOF in cf1 which includes old license name and machine UUID which has changed). Furthermore, the &lt;NF_NAME&gt;_CMG_temporary_license.dat license is configured to be used by the other OAM during healing.</p>                                                                                                                                                                                              |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | CMG Template packages for N:K redundancy supports two instantiation levels. The minimum one that starts with three number of instances for the mg_server and the medium one, which is also the default and starts with four number of instances for the mg_server. Three and four mg_groups are created during CMG instantiation for the minimum and medium instantiation levels which means that the users must fill the ipPoolsConfig.json file starting from mg_group_4 and mg_group_5 respectively if they want to use the fully automated scale out operations in CUPS and 5G deployments with NSP.                                                                                                                                                                                                                                                    |
| [299915] | For optimum performance with OVS DPDK, the Rx and Tx queue size must be increased to the maximum supported value of 1024 on every compute by modifying the tx_queue_size and rx_queue_size parameters in the /etc/nova/nova.conf file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| N/A      | When an OAM-VM switchover takes place, the Time of Last Boot field is updated for both OAM-VMs (active and standby).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| N/A      | <p>The <i>7750 SR MG and CMG Configuration Guide</i> provides configuration guidelines for Openstack/KVM-based LB-less deployments. These guidelines do not apply for VMware-based LB-less deployments. Consult your local support organization for information about VMware-based LB-less deployments in the current release.</p> <p><b>Note:</b> You must strictly follow the recommended deployment model. Any addition or removal of routing features that deviate from the recommendation is not supported. Please consult the next level of support when other features are required.</p>                                                                                                                                                                                                                                                             |
| N/A      | <p>For VMXNET3 vNICs, Nokia recommends performing the following:</p> <ul style="list-style-type: none"> <li>• Configure split Rx and Tx completion system contexts threads only when there is adequate number of free cores, for example, one per system contexts thread for all vNICs.</li> <li>• Pin the system contexts threads, by setting the value of sched.cpu.latencySensitivity.sysContexts to the following:</li> <li>• number of threads per vNIC x number of vNICs</li> </ul> <p>For example, for 2 vNICs, use the value 6, if there are 3 system context threads per vNIC, or 4 if there are 2. However, when the number of available cores left to accommodate the system context threads is smaller than this value, then it is recommended to not use pinning. The host scheduler spreads the threads optimally to the available cores.</p> |
| N/A      | The CPU reservation configured for the MG-VM cores must take into account the accurate CPU clock rate (with any decimal digits). This is important to ensure that no other tasks take resources from the cores allocated to the MG-VM. Pinning the sysContexts is also affected if the value configured does not correspond exactly to 100.0% of the CPU cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | CMG uses a single Tx/Rx queue. The system uses an internal proprietary method to efficiently distribute the packets for further processing.                                                                                                                                                                                                                                                              |
| [267673] | When using the Multiple NIC Tasks feature with dual data fabrics, make sure that all data fabric ports are serviced by the same NIC Task, to avoid possible system instability.                                                                                                                                                                                                                          |
| [246908] | Unless the boot options file has been saved, the <code>admin&gt;redundancy&gt;synchronize boot-env</code> and <code>admin&gt;redundancy&gt;synchronize config</code> commands synchronize the default boot options file instead of the SMBIOS attributes. To avoid instabilities, run the <code>bof save</code> command immediately after the VNF instantiation to properly synchronize the two OAM VMs. |
| N/A      | When VF-to-VF communication is enabled using SMBIOS ( <code>vf2vf=1</code> ), flat networks are required for the data switch fabric and external networks.                                                                                                                                                                                                                                               |
| [228418] | On the CMG platform, selective reassembly is always enabled. See also the restrictions in Functional limitations.                                                                                                                                                                                                                                                                                        |

## 13.9 CNF

| Key               | Description                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [444849]          | Intra-CNF Canary testing for SMF is not supported for path 25.3.R8 to 25.7.R7.                                                                                                                                                                                                                                                                                                                                                   |
| [441602/04786672] | In CMG CNF deployments, when the enhanced statistics polling operation ( <code>mzmStatsPoll</code> ) is enabled, system instability may occur. The reason for the instability is that the data structure used to allocate quota runs out of resources, that is, storage space.                                                                                                                                                   |
| N/A               | After installation, the value of the secondary configuration in <code>bof</code> is <code>/conf/mg.cfg</code> ; therefore, executing the <code>admin&gt;redundancy&gt;synchronize&gt;config</code> command fails.<br>As a workaround, you must modify it manually.                                                                                                                                                               |
| [402191]          | To cover consecutive reboots on a LMG container, the number of the TMD files kept by LMG has been increased from 2 to 25. All TMD files are incorporated into the tech-support file when it is collected. The file may be large and take long time to be collected. In such cases, manual intervention is required to back up and clean up the large TMD files. Contact your Nokia technical support representative for support. |

| Key | Description                                                                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | CMG CNF uses only node NTP configuration, time reference will be used from infrastructure on each POD (LOAM, LMG, LLB); therefore, CMG CLI NTP configuration is not required to be added. |

## 13.10 CUPS

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [430464] | The timeout-msec option was removed from the <code>config&gt;mobile&gt;profile&gt;pfcpc&gt;pfcpc-profile&gt;message-retransmit</code> command from current release the single timeout configuration item covers all time units.                                                                                                                                                                                                                                                                                                                                                    |
| [392300] | The interface-realm and APN must be configured with different names. The 3GPP Interface Type IE (IE Type 160) is required to identify the correct reference-point-type if more than one reference-point-types use the same "interface-realm" names.                                                                                                                                                                                                                                                                                                                                |
| [372812] | A flow description is mandatory within a PRU at the CP/SMF. Based on it, the SMF decides whether the rule is a match-all or a specific filtering rule. Nokia recommends that the same policy is configured on both the SMF and the UPF.                                                                                                                                                                                                                                                                                                                                            |
| [352386] | If the UPF receives a Quota Holding Time (QHT) of 0 for a URR, it is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [348203] | PFCP messages may be large, often exceeding 2 KB, particularly in the CP-to-UP direction. Therefore, Nokia recommends that customers ensure that possible fragmentation in Sx/N4 is correctly handled, either by setting a large enough MTU on Sx/N4, or by enabling ip-reassembly for the Sx/N4 reference point.                                                                                                                                                                                                                                                                  |
| [326453] | <p>In the case of an MG group scale-in in the UP Function, it is recommended to remove the IP pool linked to this MG group from the CP Function PFCP configuration in the following contexts:</p> <ul style="list-style-type: none"> <li><code>config&gt;mobile&gt;profile&gt;pfcpc&gt;up-peer-list&gt;peer&gt;apn&gt;ip-pool-list</code></li> <li><code>config&gt;mobile&gt;profile&gt;pfcpc&gt;up-peer-list&gt;peer&gt;network-realm</code></li> </ul> <p>This is required to minimize the possibility of sending Sx messages to an IP address that the CMG-U cannot handle.</p> |
| N/A      | The peak throughput statistics will not be reported by the SMF/CP function from release 11.0.R1 onwards. The counters are included in the XML with a zero value.                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [312582] | In CUPS systems, policy rules associated to the same rating group must be configured with the same stat-rule-unit.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [314059] | The SMF config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool-distribution command requires more than one active MG group in the SMF or more than 256 active UEs to load balance UE IP addresses from different IP pools across multiple MG groups in the UPF. If the SMF contains a single active MG group, load balancing of UE IP addresses from different IP pools will be done in groups of 256 IP addresses.                                                                                                       |
| [314874] | QCI policy configuration must be identical for all QCIs that share the same DSCP.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| N/A      | The output of show commands at the bearer-context and pdn-context level reflects the local values of the parameters existing in any node. For a given PDN session, these parameters may be displayed differently in the SMF when compared to the corresponding values in the UPF node, depending on the relevance of the parameters and how these parameters are used in each node type. Some examples of parameters that may be displayed differently in SMF versus UPF are bearer-id, policy-rule names, and charging information. |
| [294227] | In UPF, the show>mobile>pdn>ue-context command displays the number of PDN sessions, and not the number of UEs.                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 13.11 ePDG

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [348879] | Pseudo random jitter that reaches up to 50% of the rekey margin is added to the ESP-SA/IKE-SA rekey time range. This determines that the rekey time can be up to 50% more than the 10% (rekey-margin) from the ESP-SA/IKE-SA lifetime. For example, if the ESP-SA/IKE-SA lifetime is set to 120 seconds and the rekey-margin is set to 10% (default), the resulting rekey time range is between 108 seconds (ESP-SA/IKE-SA lifetime less rekey margin) and 114 seconds (ESP-SA/IKE-SA lifetime less 6 seconds/50% of the rekey margin). |

## 13.12 Ga

| Key      | Description                                                                                                                                                                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [109424] | Nokia recommends configuring the node ID to be used in CDRs, in the <code>config&gt;mobile&gt;pdn&gt;charging&gt;node-id</code> context. Since each ISM-MG card generates its own Local Record Sequence Numbers (LRSN), the node ID along with the ISM-MG card number will be populated in each CDR to uniquely identify the CDRs generated by this gateway. |

## 13.13 Gx

| Key | Description                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | Result codes 4010, 4012, 5003, and 5030 must be specifically configured as continue or retry-and-continue to enter Gy Assume Positive mode, even if the 4xxx/5xxx result codes are configured as continue or retry-and-continue. For more details, refer to the <i>7750 SR MG and CMG Configuration Guide</i> . |

## 13.14 Hardware

| Key | Description                                                                                                                                                                          |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | Contact your Nokia technical support representative for information about supported capacity and scale aspects of the ISA and ISA2 modules for IP reassembly and other applications. |
| N/A | IP reassembly MDAs can be defined either in the second complex of an MG IOM, or in any complex of non-MG IOMs.                                                                       |
| N/A | The ISM-MG-C assembly does not contain removable ISA modules. Hot swap or field replacement of the entire ISM-MG-C assembly is supported.                                            |
| N/A | The first generation of ISA and ISA2 MDAs for IP reassembly can coexist in the same node but not in the same IP reassembly LAG.                                                      |
| N/A | Currently, in OpenStack, it is not possible to have three additional disks cf1, cf2, and cf4 as compact flash disks. Only cf1 and cf2 are supported (apart from the boot disk cf3).  |
| N/A | Only Nokia-sourced Compact Flash devices are supported for the 7750 SR MG.                                                                                                           |

| Key | Description                                                                                                                                                                                                                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | If ZFS compression is enabled in the Compact Flash that is used to store the <code>bof.cfg</code> file, the booting procedure may fail because a compressed <code>bof.cfg</code> file is not readable during boot. This is particularly relevant to CMG systems in which, by default, the <code>bof.cfg</code> file is loaded from CF1. |

## 13.15 ICR

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [424280] | When using ICR, Diameter peers must be created and activated first in slave and then to master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [382267] | The command to enable or disable Master Traffic Detection <code>master-traffic-detection {enable   disable}</code> , and the command to configure the ICR Traffic Detection Polling Interval <code>traffic-detection-poll-timer &lt;seconds&gt;</code> , appear under the <code>config&gt;redundancy&gt;multi-chassis&gt;peer&gt;mc-mobile</code> CLI context. Also, the new fields are shown in <code>show&gt;redundancy&gt;multi-chassis&gt;mc-mobile peer ip-address</code> command output but the feature is not supported in release 12.0.R9. The existing ICR behavior is not affected. |
| N/A      | When using ICR, shutting down a service or a mobile gateway needs to be performed on the slave first, then on the master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| N/A      | Contact your Nokia technical support representative for methods of procedure for modifying the software or hardware configuration of a mobile gateway in an ICR setup.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| N/A      | The lawful interception peers and targets must be configured on both the master and slave nodes of an ICR gateway.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| N/A      | The <code>config&gt;redundancy&gt;multi-chassis peer&gt;shutdown</code> command is not supported. To perform an ICR switchover, use any of the following commands on the current master switch: <ul style="list-style-type: none"> <li><code>admin&gt;redundancy&gt;mc-mobile-switchover</code></li> <li><code>admin&gt;reboot</code></li> </ul>                                                                                                                                                                                                                                              |
| [208265] | Initiating a manual ICR switchover with the <code>now</code> option specified will impact traffic for about 10 seconds, even when <code>mc-redirect</code> is enabled.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [163342] | Upon an ICR state going to Hot, the upstream routers must be validated to ensure that the proper routes and metrics have been advertised before any additional actions are taken that would rely on ICR to recover.                                                                                                                                                                                                                                                                                                                                                                           |

| Key      | Description                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| [178871] | When using ICR, adding new groups must be performed on the slave first and then on the master with no ICR switchover. |
| N/A      | PGW ICR is not supported when using multiple S2b interfaces.                                                          |

## 13.16 KPI-KCI counters

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>In release 22.0.R1, and for consistency purposes, the measObjLdn index has changed in the SBI services KPI statistics, as follows:</p> <p>In the following modules, the SvcRealmId has been added:</p> <ul style="list-style-type: none"> <li>KPIServiceNsmfPduSsnProducerAmf</li> <li>KPIServiceNamfCommunication</li> <li>KPIServiceNamfEventExposure</li> <li>KPIServiceNnrfOAuth2</li> <li>KPIServiceNnrfNfManagement</li> <li>KPIServiceNnrfDiscovery</li> <li>KPIServiceNchfConvergedCharging</li> </ul> <p>In the following modules, the SvcInstance has been added:</p> <ul style="list-style-type: none"> <li>KPIServiceNudmUecm</li> <li>KPIServiceNudmSdm</li> <li>KPIServiceNpcfSmPolicyControl</li> </ul> <p>Examples of the new measObjLdn (the difference is indicated in bold):</p> <ul style="list-style-type: none"> <li>measObjLdn="KPI=ServiceNsmfPduSsnProducerAmf,GroupName=Nsmf,SvcRealmId=1,<b>SvcInstance</b>=pdu1,NfInstanceId=..."&gt;</li> <li>measObjLdn="KPI=ServiceNamfCommunication,GroupName=Namf,SvcRealmId=0,<b>SvcInstance</b>=comm1,NfInstanceId=..."&gt;</li> <li>measObjLdn="KPI=ServiceNamfEventExposure,GroupName=Namf,SvcRealmId=1,<b>SvcInstance</b>=Tnamf,NfInstanceId=..."&gt;</li> <li>measObjLdn="KPI=ServiceNnrfOAuth2,GroupName=Nnrf,SvcRealmId=1,SvcInstance=Tnrf,address=1.2.3.4,port=8080"&gt;</li> <li>measObjLdn="KPI=ServiceNnrfNfManagement,GroupName=Nnrf,SvcRealmId=1,<b>SvcInstance</b>=nfm1,address=11.127.5.5,port=8080"&gt;</li> <li>measObjLdn="KPI=ServiceNnrfDiscovery,GroupName=Nnrf,SvcRealmId=1,<b>SvcInstance</b>=disc1,address=11.127.5.5,port=8080"&gt;</li> <li>measObjLdn="KPI=ServiceNchfConvergedCharging,GroupName=Nchf,<b>SvcRealmId</b>=1,<b>SvcInstance</b>=mySrvIsNm2,NfInstanceId=...,IpAddr=0.0.0.0,port=0"&gt;</li> <li>measObjLdn="KPI=ServiceNudmUecm,GroupName=Nudm,SvcRealmId=1,<b>SvcInstance</b>=uecm1,NfInstanceId=0,IpAddr=10.10.7.79,port=0"&gt;</li> <li>measObjLdn="KPI=ServiceNudmSdm,GroupName=Nudm,SvcRealmId=1,<b>SvcInstance</b>=sdm1,NfInstanceId=0,IpAddr=10.10.7.74,port=0"&gt;</li> <li>measObjLdn="KPI=ServiceNpcfSmPolicyControl,GroupName=Npcf,SvcRealmId=1,<b>SvcInstance</b>=policy1,NfInstanceId=0,IpAddr=10.10.7.75,port=7890"&gt;</li> </ul> <p>This change appears in the KPI-KCI counters spreadsheet, in column INDEX of this module, part of the Customer Documentation set.</p> |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | In 22.0.R1 release, two new MESSAGE values are supported in the KPIPfcNodeProcedureCauseCode, for counting the following Session Set Delete Procedure messages: <ul style="list-style-type: none"> <li>msgType=pfcSessSetDelReq</li> <li>msgType=pfcSessSetDelRsp</li> </ul>                                                                                                                                                                                   |
| N/A | From release 22.0.R1, KPISessionManagementPgwApnCpm is also generated on the UP gateway.                                                                                                                                                                                                                                                                                                                                                                       |
| N/A | KCIBearerManagementVprnPoolCP-ISA is not generated on the UP gateway, where IP-pools are only supported as 'remote'. Per MG group IP-pool statistics are not supported for remote pools.                                                                                                                                                                                                                                                                       |
| N/A | The KPIReferencePointS12 and KPIReferencePointS1U, which contain one counter for bearer contexts not found, are never generated in the XMLs.                                                                                                                                                                                                                                                                                                                   |
| N/A | When all statistics under a specific path (list or container in YANG) are zero, these statistics are not reported. If only one counter under a specific path is non-zero, all the other counters in this path are reported with zero.                                                                                                                                                                                                                          |
| N/A | The following error GTP packet counters per 5G reference point are not supported and are always reported as zero: <ul style="list-style-type: none"> <li>VS.errorGtpPktsN3</li> <li>VS.errorSgiPktsN6</li> <li>VS.errorGtpPktsN9a</li> <li>VS.errorGtpPktsN9c</li> </ul>                                                                                                                                                                                       |
| N/A | VS.msgSent and VS.msgReceived in KPIGtpMsgFailureCauseCodeCP-ISA and KPIPfcSessionProcedureCauseCode modules are now counting both successful messages and messages with failures cause values. [MG-2514]                                                                                                                                                                                                                                                      |
| N/A | The LB-VM CPU statistics in the KPI-KCI (2178.2) (MG-2492) feature adds support to report resource usage statistics for the LB-VMs. The group=0 portion in the INDEX is used to denote a specific measurement for an LB-VM. If there are post processing formulas averaging over all measurements to obtain an average CPU/memory utilization of all MG-VMs, the group=0 measurement must be excluded from the averaging scope because it refers to the LB-VM. |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | For KPI-KCI XML generation in release 11.0.R5, the zero value counters behavior is consistent across all counter modules. Prior to this change, some counter modules were written in the XML file for a specific collection period even though all counter values were zero. In release 11.0.R5 and onwards, the behavior of all counter modules is such that if all counters have zero values, the table is not written to the XML file to save space. This behavior now applies to all counter modules. This change is introduced with the feature KPI-KCI Statistics for Control and User Plane (MG-2178). |
| N/A | Per APN per QCI drop stats are now reported in a new KPIBearerTrafficApnQci table. The old KPIApnQciDropPackets table has been deprecated and is no longer available.                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### 13.17 Lawful Interception

| Key                                | Description                                                                                                                                                                                                                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A                                | If 5G LI peers are to be reached via a VPRN interface, only local-port 80, 8080, or 443 must be used. This restriction does not apply to 5G LI peers reachable via the base routing instance.                                                                                                                 |
| [431794]                           | When the LI configuration field under the <code>config&gt;li&gt;mobile&gt;custom-correlation-id-format</code> is set to <code>enable</code> , the correlation ID used in LI_X2 and LI_X3 messages is still 8 octets in length, but the first 4 octets will be set to zeros. This field only affects 5G calls. |
| [427867, 427402/02897189/02834401] | Starting from release 22.8, changes have been introduced to enforce security in the certificate validation process by including host IP address validation. In previous version, the IP address validation was not enforced despite its state to be mandatory.                                                |
| N/A                                | To get 5G peer information, the <code>tmnxMobLiDynamicPeerTable</code> and <code>tmnxMobLiDynamicPeerStatusTable</code> MIB objects are used. For 4G peer information, the <code>tmnxMobLiDfPeerEntry</code> and <code>tmnxMobLiDfPeerStatusEntry</code> MIB objects are used.                                |

## 13.18 LTMT

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>ICR — BGP monitoring as ICR switchover criteria for partial failure trigger. For LTMT Workflows based ICR - BGP monitoring and ICR switchover, the fast detection and action within few seconds is critical.</p> <p>In this case adjust LTMT configuration for high parallelism like this:</p> <pre>workflowThreadsNum=12 connectionPoolWaitPromptTime=3 connectionPoolMinSize=10 connectionPoolMaxSize=20 connectionPoolCheckInterval=240 connectionPoolIdleTimeout=720 connectionPoolLoginRetryInterval=300 connectionPoolMaxReuseCount=50 connectionPoolMaxReuseHour=1</pre> <p>For even more workflows running in parallel, multiply the values in rows:</p> <pre>workflowThreadsNum=12 connectionPoolMinSize=10 connectionPoolMaxSize=20</pre>                                                  |
| N/A | <p>Remove private keys from the ltmt-user pod</p> <p>The ltmt-user container image shipped with private keys for TLS and SSH. The keys can be divided into two groups: sample keys that are not in use in the runtime and keys that will be in use by the runtime container.</p> <p>All keys have been removed from the container image.</p> <p>As some keys are still required, they are created automatically at runtime when the container starts up. The automatically created keys are:</p> <ul style="list-style-type: none"> <li>• /etc/ltmt/privkey.pem</li> <li>• /etc/ltmt/server.pem</li> <li>• /opt/ssh/ssh_host_rsa_key</li> <li>• /opt/ssh/ssh_host_ecdsa_key</li> <li>• /opt/ssh/ssh_host_ed25519_key</li> </ul> <p>The keys are re-generated every time the container is restarted.</p> |

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>MG management connection — LTMT executes CLI commands on the mobile gateway periodically using MG management router access and credentials (MG CLI). LTMT can consume up to 3 SSH sessions in parallel at times. Configure the mobile gateway to have enough SSH the mobile gateway required.</p> <p>Configure the mobile gateway to allow enough SSH sessions by using the following command: <code>config&gt;system&gt;login-control&gt;ssh&gt;inbound-max-sessions number-of-sessions</code></p> <p>number-of-sessions: 0 to 50, default = 5</p> <p><b>Note:</b> Nokia recommends using the maximum value, that is, 50.</p> <p>The CMG user that is used by the LTMT must have classic CLI as default and not MD-CLI. Currently the LTMT does not support MD-CLI.</p> |
| N/A | <p>Caution during MG maintenance — When performing maintenance, ensure all [TS] RULE(s) are DEACTIVATED (that is, when the TS indicator pilot light circle is RED), otherwise the system triggers a TS file creation many times on the DUT.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| N/A | <p>Call Trace — LTMT automatically creates <code>pcmdvaidCT</code> and <code>pcmdvaidCTendless</code> call insight profiles that use the LTMT destination settings configured in CMG. If you modify the LTMT IP address, the IP address of the <code>live-output</code> parameter in the <code>pcmdvaidCT</code> and <code>pcmdvaidCTendless</code> profiles must also be updated.</p>                                                                                                                                                                                                                                                                                                                                                                                      |

## 13.19 MD-CLI

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [441735] | <p>If configuration changes are implemented using both classic and MD-CLI, and some configuration is not committed, a validation error may occur if the commands have dependencies. Executing the update command in MD-CLI clears the error.</p>                                                                                                                                                                                                                                                                                 |
| [424656] | <p>Under the <code>config&gt;mobile-gateway&gt;profile&gt;dcca</code> context via MCI, the following mutually exclusive commands are required to be handled in separate transactions to configure/modify the existing values.</p> <pre>   tmnxMobProfDccaFhSessContTimer   tmnxMobProfDccaSessRestoreTimer     tmnxMobProfDccaCcrInitOnFirstPkt   tmnxMobProfDccaCcriOnFirstOnlnRg     tmnxMobProfDccaFirstPktBehavior   tmnxMobProfDccaOneTimeRed     tmnxMobProfDccaSupportedVersion   tmnxMobProfDccaSupportedVendor   </pre> |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [436080] | <p>The following commands the under <code>configure&gt;mobile-gateway&gt;profile&gt;dcca-profile</code> context are not planned to be implemented in MCI. The same commands will be deprecated in classic CLI in future releases. Nokia recommends using the equivalent commands through CCFH profile configuration.</p> <pre>configure&gt;mobile-gateway&gt;profile&gt;dcca-profile +-- cc-failover-handling +-- cc-session-failover +-- ccfh-fh-action +-- failure-handling     +-- msc-result-code         +-- service-delay-timer     +-- result-code         +-- behavior     +-- result-code-3xxx     +-- result-code-4xxx     +-- result-code-5xxx +-- fh-session-continue-timer +-- fh-volume-limit +-- session-restore-retry-timer</pre> |
| [436924] | <p>Modifying the context via MCI under the <code>configure mobile-gateway profile charging pgw-charging-profile/sgw-charging-profile 0</code> command, is not allowed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 13.20 NETCONF

| Key | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | <p>Release 24.7.R1 changes the delivery method of the YANG module files on the compact flash. For release prior to 24.7.R1, the YANG modules were delivered as individual YANG files in the <code>cflash/support/YANG</code> directory. From release 24.7.R1 and onwards, the YANG files are delivered with the software image as part of a single <code>yang.tim</code> file, which is the same method used to deliver other integral components of the operating system. This file is signed, which increases the integrity of the YANG modules delivered from the node and decreases synchronization times.</p> <p>If the model-driven management is enabled, use the <code>admin redundancy synchronize boot-environment</code> command to synchronize the <code>yang.tim</code> file associated with each image between the primary and standby CPM. If the schema path is configured to be a local directory, the preceding command recursively copies all YANG files from the local directory to the standby CPM.</p> <p>If the model-driven management is not enabled on the device and disk space is limited, the <code>yang.tim</code> file can be deleted from the flash card. If the file is missing, the system still successfully completes the synchronization (using the <code>admin redundancy synchronize boot-env</code> command) but displays a warning message and reports that it failed to copy the file in the system logs. Nokia recommends storing the <code>yang.tim</code> file with the other software image components on the compact flash.</p> |

## 13.21 PGW/GGSN/HA

| Key | Description                                                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | Operator can install several any/any rules per session, with different precedence and rating group. Traffic matches to the any/any rule with lower precedence value.                                   |
| N/A | Create/Update PDP Responses from the GGSN may be missing alternate addresses when the Gn-c and Gn-u are using mixed IP types - one uses IPv4 and the other uses IPv6; this follows the 3GPP standards. |
| N/A | This Virtual APN Selection with IMEISV TAC + SVN enhancement will not work if MEI IE includes IMEI and not IMIESV.                                                                                     |

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [283572] | RADIUS authentication and accounting supports retry timeout values of up to 255 seconds for the RADIUS profile configuration in the <code>config&gt;mobile&gt;profile&gt;radius&gt;auth-retry-timeout</code> context or <code>acct-retry-timeout</code> command. However, if the retry timeout is set to 255 seconds, and the retry count is greater than 1 (set using the <code>config&gt;mobile&gt;profile&gt;radius&gt;auth-retry-count</code> or <code>config&gt;mobile&gt;profile&gt;radius&gt;acct-retry-count</code> command), then the system may allocate memory for extended periods. It is therefore recommended to avoid setting a retry count greater than 1 when the retry timeout is 255 seconds. |
| N/A      | When using the PGW non-IP SGI tunneling feature, the PDN connections will be associated with the Application Server by the APN. A single APN will support either non-IP PDN type or IP PDN types but not both.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| N/A      | The PGW/GGSN is compliant with the 3GPP Technical Specification (TS) 29.212. The Feature-List-ID AVP is as per table 5.4.1.1 of the TS. Depending on the PCRF capability, it is recommended to upgrade the PCRF, either before or at the same time as the PGW/GGSN, to match the version of the TS.                                                                                                                                                                                                                                                                                                                                                                                                              |
| N/A      | The PGW/GGSN is compliant with the 3GPP Technical Specification 32.299. The default Service-Context-Id AVP in Diameter charging packets is as per section 7.1.12 of the TS. If required, it can be adapted on the PGW/GGSN by configuring it in the <code>configure&gt;mobile&gt;profile&gt;dcca service-context-id</code> context. Please refer to the <i>7750 SR MG and CMG Configuration Guide</i> for more details.<br>Also note that configuring the <code>epc-node</code> command under the PDN context with proper MCC/MNC values will update the AVP as per the TS.                                                                                                                                      |

## 13.22 Rf

| Key      | Description                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [107128] | If both IPv4 and IPv6 addresses are configured for the Rf interface and the FQDNs for both IPv4 and IPv6 addresses need to be resolved, the following line is required to be added to the <code>config&gt;system&gt;dns&gt;address-pref</code> <code>ipv6-first</code> configuration command. |
| [114702] | The Serving-Node-Change-Limit partial-record-closure trigger is not applicable for Rf charging.                                                                                                                                                                                               |

## 13.23 Reference points

| Key | Description                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A | If both S5 and S8 reference point interfaces are configured and are different, there is no ambiguity in the selection of S5 and S8 interfaces.                      |
| N/A | If only one of S5 and S8 reference point interfaces are configured, then both (S5 and S8) reference point interfaces map to the same interface.                     |
| N/A | If neither of S5 and S8 reference point interfaces are configured, then both (S5 and S8) reference-point interfaces map to the default signaling interface for GTP. |
| N/A | If both S5 and S8 reference-point interfaces are configured, there is no ambiguity in the selection of S5 and S8 interfaces.                                        |
| N/A | If only one of S5 and S8 reference-point interfaces are configured, then both (S5 and S8) reference-point interfaces map to the same interface.                     |
| N/A | If neither one of S5 and S8 reference-point interfaces are configured, then both (S5 and S8) reference-point interfaces map to the default signaling interface.     |
| N/A | If both Gn and Gp reference-point interfaces are configured, there is no ambiguity in the selection of Gn and Gp interfaces.                                        |
| N/A | If only one of Gn and Gp reference-point interfaces are configured, then both (Gn and Gp) reference-point interfaces map to the same interface.                     |
| N/A | If neither one of Gn and Gp reference-point interfaces are configured, then both (Gn and Gp) reference-point interfaces map to the default signaling interface.     |
| N/A | Gn/Gp reference-point interfaces are currently NOT inherited from S5/S8 reference-point interfaces and vice versa.                                                  |
| N/A | If both S5 and S8 reference-point interfaces are configured, there is no ambiguity in the selection of S5 and S8 interfaces.                                        |

## 13.24 SNMP

| Key      | Description                                                                                   |
|----------|-----------------------------------------------------------------------------------------------|
| [267657] | In SNMP, the corporate name in sysObjectID has now been changed from Alcatel-Lucent to Nokia. |

## 13.25 System

| Key      | Description                                                                                                                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N/A      | When creating a new log file on a Compact Flash disk card, the system will check the amount of free disk space, and the amount must be greater than or equal to the lesser of 5.2 MB or 10% of the Compact Flash disk capacity.                                                                                                                                   |
| N/A      | Gateway functionality is supported only on chassis mode D.                                                                                                                                                                                                                                                                                                        |
| [123222] | The KPI-KCI counters for reference point statistics are now pegged separately in transmit and receive directions.                                                                                                                                                                                                                                                 |
| N/A      | When creating GTP/GTPv2 sessions at a high rate, it is recommended to ensure that CPU protection is not triggered. Use the <code>show&gt;system&gt;security&gt;cpu-protection&gt;violators</code> command to find out if a port, interface, or SAP violates the rate limit for traffic sent to the CPM. If they do, then CPU protection settings must be adapted. |

## 13.26 TWAG

| Key      | Description                                                                       |
|----------|-----------------------------------------------------------------------------------|
| [267657] | Any 7750 SR MG TWAG deployment is required to run on ISM-MG-C and FP3 based IMMs. |

## 14 Documentation issues

This chapter lists the documentation issues throughout this release. Each documentation issue includes the release under which the correction or update will be published. The CMG documentation set is available in Nokia Discovery Center at <https://doc.networks.nokia.com>.

The label **[NEW]** is added to the documentation issues that are introduced in current release.

### 14.1 Issues in documentation guides

| Key      | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [446843] | <p><b>[NEW]</b> In the following guides, a note is added that CMG currently supports IPCP exchange only when the LNS initiates it:</p> <ul style="list-style-type: none"><li>• <i>CMG CUPS Configuration Guide</i>, under the section: PPP regeneration — L2TP LAC</li><li>• <i>CMG Configuration Guide</i>, under the section: IPCP DNS address assignment</li></ul> <p>The update appears in release 26.7.R1.</p> |
| [446383] | <p><b>[NEW]</b> In the <i>CLI Reference Guide</i>, the value for the <code>s1u-n3-path-management-stats</code> command under the <code>configure mobile-gateway pdn kpi-kci accounting-policy</code> context is incorrectly documented as <code>pathMgmtFails</code>. The value is corrected as <code>path-mgmt-fails</code>.</p> <p>The update appears in release 26.7.R1.</p>                                     |
| [447230] | <p><b>[NEW]</b> In the <i>Application assurance Configuration Guide</i>, under <code>AA HTTP enrichment restrictions</code> section, the <code>strict-in-order</code> parameter is incorrectly documented. The parameter is corrected as <code>tcpstrict-order</code>.</p> <p>The update appears in release 26.7.R1.</p>                                                                                            |

# 15 Corrected security vulnerabilities

The *CVSS expert score* refers to either the Nokia or third-party software provider’s Expert Score, and the *Original/base score* is the score provided by the SVM scanner’s vulnerability database.

For detailed CVE information, see <https://cve.mitre.org>.

For information about CVE weakness descriptions, see <https://cwe.mitre.org/>.

For information about CVSS scoring, see <https://www.first.org/cvss/specification-document>.

## Corrected security vulnerabilities

| CVE ID                                                                                                                               | Vulnerability name | CVSS expert score | Original / base score | Third-party software component and version |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-----------------------|--------------------------------------------|
| The list of the corrected common security vulnerabilities for CMG is available on request from your Nokia Security Point of Contact. |                    |                   |                       |                                            |

## 16 References

### 16.1 Operating documentation

The CMG documentation set is available in Nokia Discovery Center at <https://doc.networks.nokia.com>.

To view or download the CMG Operating Documentation:

1. Log in to the [Discovery Center](#).
2. From the existing menus, either directly select CMG as the product or navigate as follows:
  - Product categories – Core Networks
  - Product subcategories – Mobile Packet Core
  - Products – CMG (Cloud Mobile Gateway)
3. Click **Browse**.
4. Choose the release of your interest from the drop-down menu.
5. Click on
  - the **Operating Documentation** to list the available Operating Documentation sets of major releases. Click on the **CMG (Cloud Mobile Gateway)** documentation set to view or download the CMG documentation set.
  - the **Operating Documentation** and then **Change Documentation** subcategory to view or download any publication of individual guides applicable to all current releases and any guide that must be published for a maintenance release.
  - the **Release Documentation** and the **SW Release Note** or **Compatibility** subcategories for the available CMG SW Release Notes or Compatibility Report, respectively.
  - the **Reference Documentation** to list the PDF versions of the referenced 7750 SR OS reference guides.

To view or download the 7750 SR MG Operating Documentation (up to release 24.7), follow the same steps but select the 7750 SR MG product instead. The 7750 SR MG and CMG software release is based on release 20.10.R10 of the 7750 SR OS and therefore inherits many (but not all) of the SR 20.10.R10 features, mostly without modification. The related SR OS configuration information is provided in the guides listed in this section.

The Discovery Center is listed as a product of its own in the Discovery Center portal where you can access all tool-related information. The contents are presented in the [Discovery Center overview](#).

## 16.2 License text

License text, as required by third-party software, is available in the CMG documentation set in the CMG CNF Software License Terms and CMG VNF Software License Terms.

## 16.3 Dimensioning

For dimensioning information, contact your Nokia technical support representative.

## 17 Appendices

17.1 CMG compatibility with MantaRay releases

CMG compatibility with MantaRay releases

| NetAct<br>Release/<br>CMG Release | 25R2-NM        | 25R3-NM        | 26R1-NM                   |
|-----------------------------------|----------------|----------------|---------------------------|
| 26.3                              | Not Compatible | Not Compatible | Supported<br>(FM/PM only) |

Not Compatible: Combination not verified from MantaRay or CMG RnD.

26R1-NM supports CMG YANG models from 25.11.R1 CMG software.

## 17.2 Lightweight Real Time Monitoring and Troubleshooting Tool (LTMT) compatibility

LTMT is backward compatible. Nokia recommends always using the latest (newest) LTMT version with any CMG version at customer.

LTMT 26.7 is tested with:

- the latest CMG release, that is, CMG 26.7
- the previous CMG releases, that is, 24.7, 24.11, 25.3, 25.7, 25.11, and 26.3

## 17.3 Supported solutions for LB-less architecture CMG deployments

### Supported solutions for LB-less architecture CMG deployments

| # | Solution                             | Description                                                             | Deployment infrastructure | Protocols on super/aggregate VPRN                 | Protocols on host VPRN       |
|---|--------------------------------------|-------------------------------------------------------------------------|---------------------------|---------------------------------------------------|------------------------------|
| 1 | BGP peering via Super/Aggregate VPRN | Super/Aggregate VPRN:<br>ports on LB-VM<br>Host VPRN:<br>ports on MG-VM | OpenStack                 | iBGP (preferred)<br>/eBGP<br>static-routes<br>BFD | static-routes<br>BFD         |
| 2 | Direct BGP peering via MG Host VPRN  | Super/Aggregate VPRN: N/A<br>Host VPRN:<br>ports on MG-VM               | VMware                    | N/A                                               | eBGP<br>static-routes<br>BFD |

Routing protocols other than those listed in this table are NOT tested or supported for LB-less architecture deployments. Using any other routing protocol that deviates from the above table may result in unexpected/undesired behavior.

## 17.4 Memory requirements

The required memory per MG-VM varies on the following:

- features activated
- expected traffic
- allocated UP cores

### Memory requirements

|                                          | Memory                          |
|------------------------------------------|---------------------------------|
| Per WorkThread (vCPU assigned to the WT) | ~ 2 GB RAM statically allocated |
| MG MDA                                   | ~ 25 GB                         |
| AA MDA, 1million AA subscribers          | ~16 GB                          |
| AA MDA, 2million AA subscribers          | ~ 22 GB                         |
| TCP-O, 1 million AA subscribers          | ~ 10 GB                         |
| TCP-O, 2 million AA subscribers          | ~ 20 GB                         |
| NAT on MG-VM                             | ~ 20 GB                         |
| ePDG MDA                                 | ~ 3 GB                          |
| Each 100k sessions                       | ~ 1-1.5 GB                      |
| Per WorkThread (vCPU assigned to the WT) | ~ 2 GB RAM statically allocated |

### Minimum RAM of MG-VM per CMG deployment

| ePDG  | SMF/SAEGW-C | SAEGW-U/SAEGW/UPF |
|-------|-------------|-------------------|
| 48 GB | 64 GB       | 64 GB             |

## 17.4.1 Memory configurations for CMG

### Supported RAM of CMG-VMs

| OAM-VM          | LB-VM           | MG-VM                                                                                                                                              |
|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 GB (minimum) | 24 GB (minimum) | 64 to 240 GB<br>Required memory per MG-VM varies depending on which features are activated, on the anticipated traffic, and on UP core allocation. |

### Examples

The following is a UPF or non-CUPS deployment example of recommended memory for MG-VM with 26 WTs for UP, default AA and TCPO flows (8M flows), and maximum traffic of ~1M UEs per VM. The number of UP WTs affect RAM requirements, and each WT consumes ~2 GB of RAM.

| CMG mode              | Minimum MG memory requirements | +AA    | AA+NAT | AA+NAT+TCPO |
|-----------------------|--------------------------------|--------|--------|-------------|
| LB mode/ LB-less mode | 100 GB                         | 116 GB | 132 GB | 148 GB      |

The following is an SMF deployment example of recommended memory for MG-VM with 18 vCPUs (14 control cores) and maximum traffic of ~1M UEs per VM.

| CMG mode | Minimum MG memory requirements |
|----------|--------------------------------|
| LB mode  | 64 GB                          |

## 17.5 Updating the SSH client configuration to allow connecting to CMG SSH server

The 25.X.X CMG release uses an SSH version that supports legacy host keys. When you try to establish an SSH connection using a recent SSH client version, the system responds with the following error:

```
Unable to negotiate keys with no matching host key type found. Their offer:
ssh-rsa, ssh-dss.
```

As a workaround, perform the following steps.

1. Create the following files.

- `/etc/ssh/ssh-legacy.conf`  
`.include /etc/ssl/openssl.cnf`  
`[openssl_init]`  
`alg_section = evp_properties`  
`[evp_properties]`  
  
`rh-allow-sha1-signatures = yes`
- `/etc/ssh/ssh_config`  
`Host *`  
`KexAlgorithms +diffie-hellman-group14-sha1`  
`HostKeyAlgorithms +ssh-rsa`  
`PubkeyAcceptedKeyTypes +ssh-rsa`  
  
`RequiredRSASize 1024`

2. Execute the following command.

```
export OPENSSL_CONF= /etc/ssh/ssh-legacy.conf
```

This step can be applied either to the host running the SSH client or to the LOAM pod. If these modifications are applied to the LOAM pod, the change is not persistent.

To make the change persistent, modify the `Deployment.yaml` file.

```
- /bin/sh
- -c
- >-
  echo -e ".include /etc/ssl/openssl.cnf" >> /etc/ssh/ssh-legacy.conf;
  echo -e "[openssl_init]" >> /etc/ssh/ssh-legacy.conf;
  echo -e "alg_section = evp_properties" >> /etc/ssh/ssh-legacy.conf;
  echo -e "[evp_properties]" >> /etc/ssh/ssh-legacy.conf;
  echo -e "rh-allow-sha1-signatures = yes" >> /etc/ssh/ssh-
legacy.conf;
  echo -e "Host *" >> /etc/ssh/ssh_config;
  echo -e "  KexAlgorithms +diffie-hellman-group14-sha1" >>
/etc/ssh/ssh_config;
  echo -e "  HostKeyAlgorithms +ssh-rsa" >> /etc/ssh/ssh_config;
  echo -e "  PubkeyAcceptedKeyTypes +ssh-rsa" >> /etc/ssh/ssh_config;
  echo -e "  RequiredRSASize 1024" >> /etc/ssh/ssh_config;
```

```
{{- if $loamMgmtDict }}  
  
{{- if $dualStackEnable }}
```

You can include these commands as part of the LOAM pod initContainer and set the environment variable under LOAM deployments.

## 17.6 Configuring manually the network interface card and PCIe settings

Follow the procedure to manually configure the network interface card and PCIe settings for HP DL325/DL345 and AirFrame RM19 Servers.

This procedure must be performed during clean installation of CMGa 26.3.Rx and onwards because of the feature MG-4976 (Use of TPM and secure boot for full disk encryption).

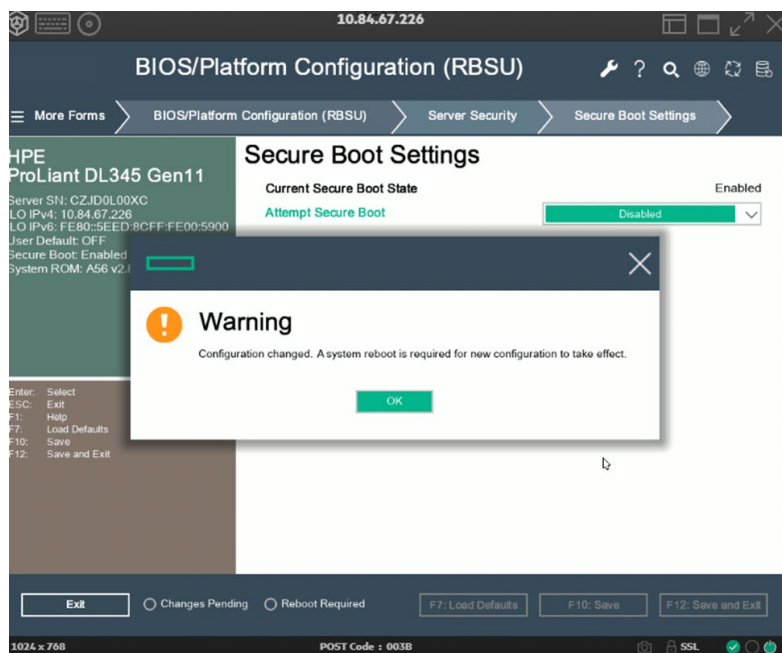
**NOTE:** This is transitional Method of Procedure (MOP) due to an issue identified on RHEL. Resolution to this issue is planned for delivery in the forthcoming releases.

In specific, the Mellanox driver parameters must be manually configured:

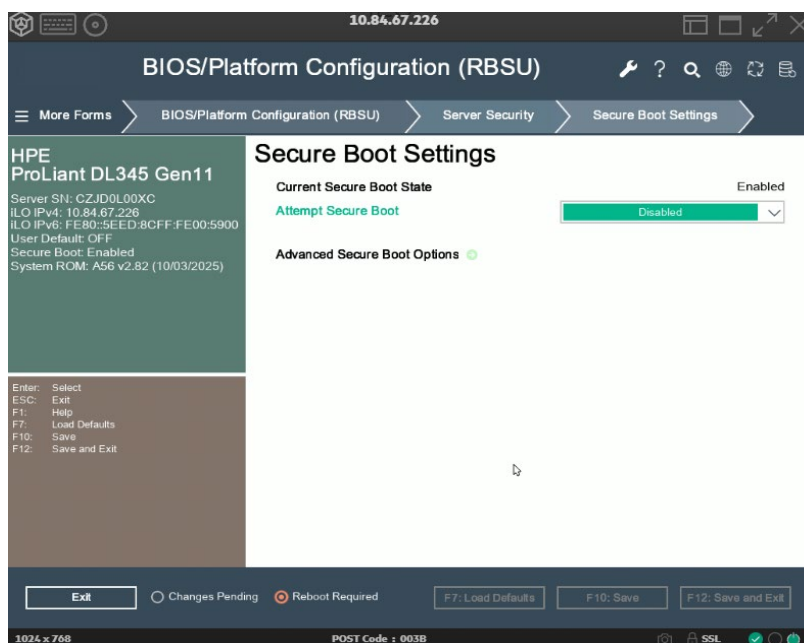
- DYNAMIC\_VF\_MSIX\_TABLE=1
- STRICT\_VF\_MSIX\_NUM=0
- NUM\_VF\_MSIX=16
- VF\_LOG\_BAR\_SIZE=3
- CQE\_COMPRESSION=1

Perform the following steps.

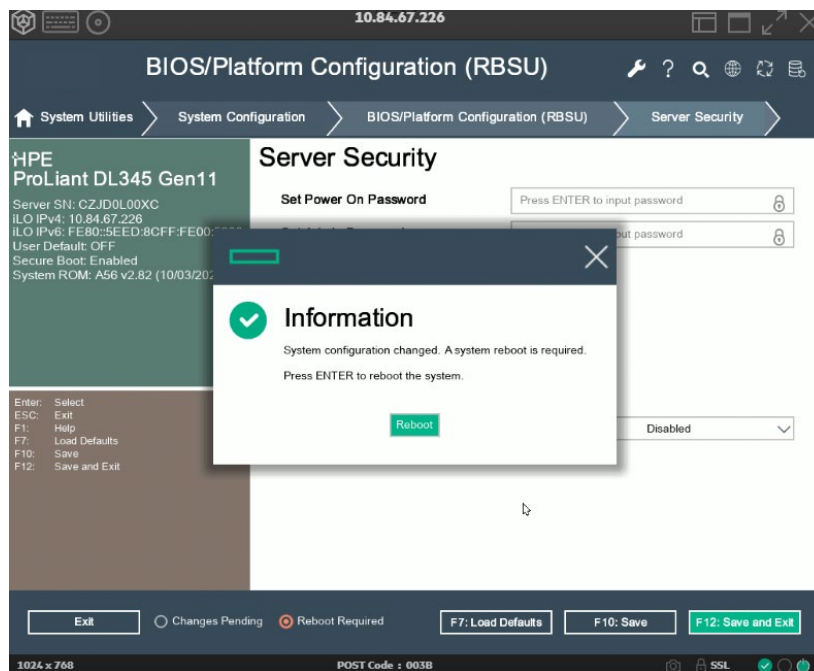
1. Disable Secure Boot.
  - a. Reboot the server.
  - b. Press **F9** for System Utilities during POST.
  - c. Go to **System Configuration**.
  - d. Open **BIOS/Platform Configuration (RBSU)**.
  - e. Go to **Server Security**.
  - f. Set **Attempt Secure Boot** to **Disabled** to turn it off.



g. Select **Exit** from the bottom menu on the left.



h. Press **F12** to reboot the server.



## 2. Enable PCI enhancement.

- a. Check that you can have access on of Mellanox Card.

```
mstconfig -d `lspci |grep Mellanox |grep -v Virtual |awk '{print $1}' |head -1` query
```

**Note:** You must perform the following steps as a **root user** only.

## 3. Configure the settings.

cd /root/postInstall/ & execute ./11-setup-mellanox.sh

## 4. Reboot the server to apply the configuration.

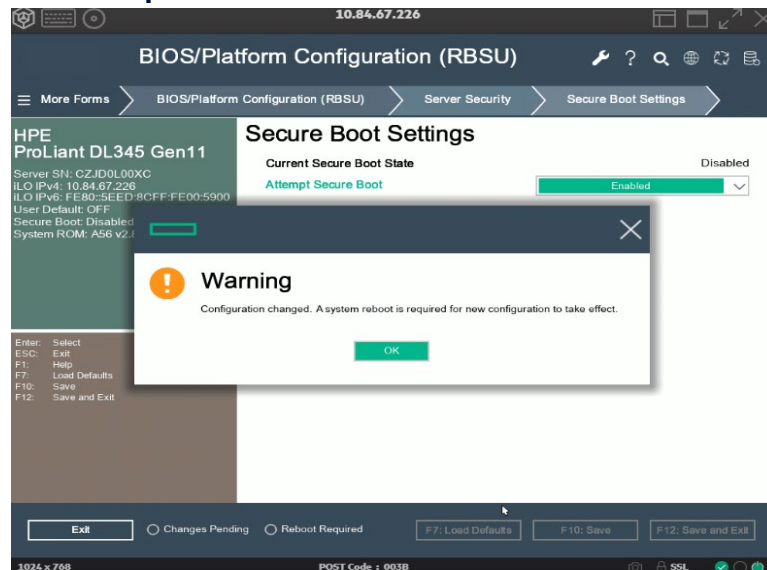
## 5. Check the values.

```
mstconfig query | grep  
"DYNAMIC_VF_MSIX_TABLE\|STRICT_VF_MSIX_NUM\|NUM_VF_MSIX\|VF_LOG_BA  
R_SIZE\|CQE_COMPRESSION"
```

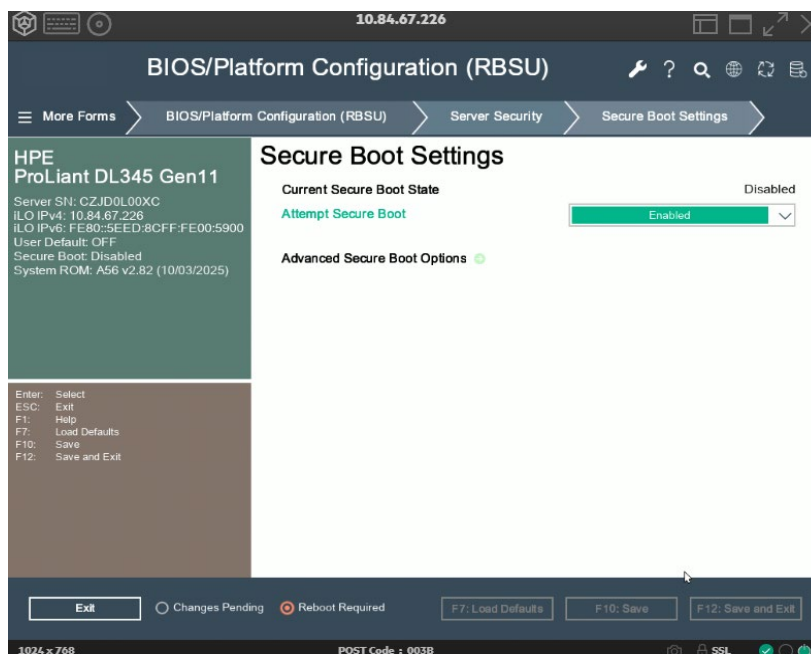
## 6. Enable Secure Boot.

- a. Reboot the server.
- b. Press **F9** for System Utilities during POST.
- c. Go to **System Configuration**.
- d. Open **BIOS/Platform Configuration (RBSU)**.

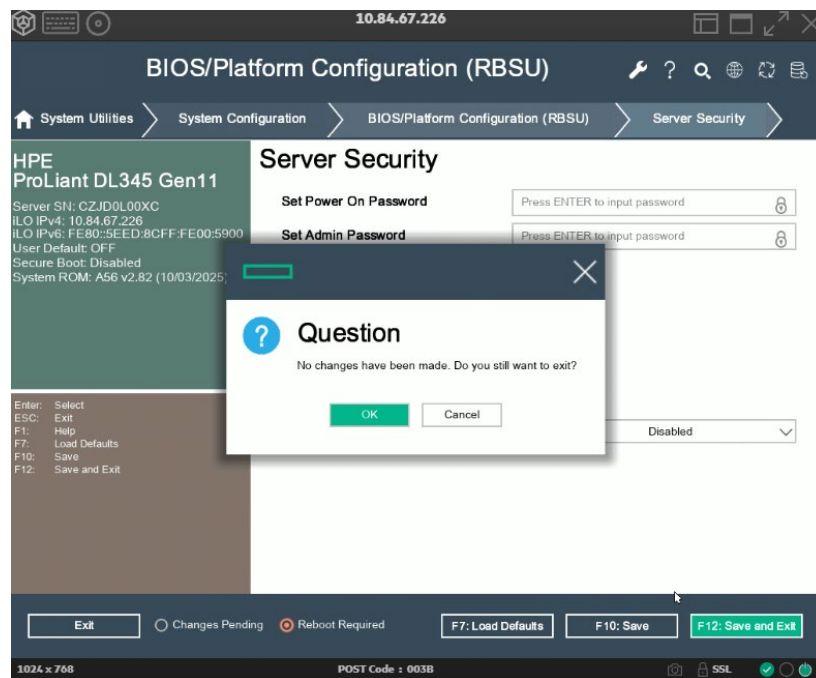
- e. Go to **Server Security**.
- f. Set **Attempt Secure Boot** to **Enabled** to turn it on.



- g. Select **Exit** from the bottom menu on the left.



- h. Press **F12** to reboot the server.
  - a. Select **Ok** to exit.



b. Press **ENTER** to reboot.

